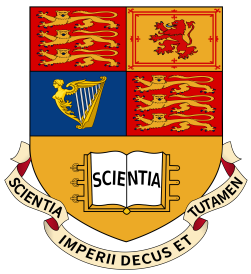
5D Supersymmetric Gauge Theories at Infinite Coupling: Brane Webs, Quivers, and Chiral Rings

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Abstract

This dissertation gives a pedagogical account of a sequence of methods for probing strongly coupled five-dimensional $\mathcal{N} = 1$ gauge theories of varying Chern–Simons level k, told through the calculations of $SU(3)_k + 5F$ for $k = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}$, and $SU(3)_k + 6F$ for $k = 0, 1, 2$. Beginning with field theory and moduli spaces, the discussion proceeds through Type IIB five-brane webs and magnetic quivers to Hilbert series, highest-weight generating functions and plethystic methods, culminating in an operator-level construction of the infinite–coupling Higgs-branch chiral ring relations for all six theories. These results extend the known highest-weight generating function counting data with explicit polynomial relations between the identified mesonic, baryonic, glueball and instanton sectors under the enhanced ultraviolet symmetry. The relations allow access to the non-reduced finite-coupling information invisible to the reduced Higgs variety; in particular, the nilpotency of the glueball is recovered in all six theories and is independent of the Chern–Simons level, while the relations further suggest that successive glueball powers are supported on increasingly constrained mesonic loci.

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1 Introduction

One of the most powerful governing principles in the study of modern fundamental physics is the idea of symmetry. It provides a systematic means of separating the physically meaningful structure from the particular variables used to describe it. Quantum field theories leverage this understanding by constructing models that respect the symmetries of spacetime, as well as further symmetries acting internally on the different species and quantum numbers of elementary particles. Gauge theory takes this principle a step further by requiring an internal symmetry to hold locally and necessitating the introduction of gauge fields, turning an abstract invariance into a dynamical interaction. From this single structural requirement, a beautiful class of interacting theories emerge.

Physicists seek a unified theory of everything, but for the time being our understanding remains organised by energy scale, a framework within which we have nevertheless been remarkably successful. Much of this rests on perturbation theory, where physical quantities are approximated in expansions of a small coupling, with renormalisation providing the machinery in which quantum corrections generated by such expansions can be absorbed into the parameters of the theory. This methodology breaks down when the relevant coupling becomes large, and perturbation theory ceases to provide reliable access to the dynamics. States that were previously heavy may become light, symmetries not manifest in the weakly coupled presentation may emerge, and an alternative formulation may make the relevant structure of the theory more transparent. One route through this difficulty is to identify features that remain well defined across such reorganisations, providing a means of relating limits that may at first appear entirely distinct.

Supersymmetry has proved particularly fruitful in pursuing this programme. The Coleman–Mandula theorem severely restricts how spacetime and internal symmetries may combine in an interacting relativistic quantum field theory, while its graded extension singles out supersymmetry as the non-trivial possibility beyond this structure. By relating bosonic and fermionic degrees of freedom, supersymmetry places strong constraints on the quantum dynamics and protects certain quantities from otherwise uncontrolled corrections. Although no experimental evidence has yet established supersymmetry as a symmetry of nature, these constraints make supersymmetric theories an exceptionally useful mathematical laboratory, allowing structure inherent to gauge theory to be followed across energy scales.

Five dimensions offer a particularly revealing arena for these ideas. Gauge theories in five dimensions are perturbatively non-renormalisable, thus no weakly coupled Lagrangian can furnish a complete description at arbitrarily high energies. In the presence of eight supercharges however, the supersymmetric restrictions preserve enough structure to make the strong-coupling regime tractable, constraining the low-energy dynamics and protecting distinguished sectors of the spectrum.

Remarkably, many such theories admit non-trivial ultraviolet fixed points, with the familiar gauge theory emerging as a deformation of an underlying superconformal theory. Five-dimensional supersymmetric gauge theories therefore offer a particularly revealing context in which to study how weakly coupled descriptions reorganise into strongly coupled physics, while retaining sufficient control to follow that transition in detail.

String theory provides a dual language in which to carry out this study. Beyond its role as a proposed model for quantum gravity, string theory has uncovered striking geometric descriptions of quantum field theories. Gauge theories may arise on branes, while masses, couplings and vacuum parameters can be represented by geometric separations and deformations of such extended objects. In five dimensions, this geometric perspective allows the transition to strong coupling to be followed through brane webs, whose worldvolume gauge theories can be traced into regions that lie beyond conventional field-theoretic control. Including those where no Lagrangian description is possible.

The interaction between string theory and quantum field theory has in turn brought increasingly sophisticated areas of mathematics into the study of fundamental physics. Geometry, topology, representation theory and algebra have become deeply connected with the study of gauge theories, particularly through their moduli spaces of vacua. Such spaces can be difficult to characterise directly, and algebraic geometry provides a natural framework for approaching this problem. By considering the algebra of all possible functions on the space, one can encode its defining information algebraically and, from this data, recover the space's geometric structure. Combined with brane realisations, this provides a highly practical language in which dual descriptions giving rise to the same geometry may be used interchangeably, according to which is most advantageous for computation.

This is the perspective adopted throughout this dissertation, we follow the theory through a succession of complementary viewpoints, chosen according to the questions they make most accessible: from field theory, to string-theoretic geometry, to quiver descriptions, and finally to the algebraic structure of the protected operator ring. Each description makes different features transparent, and their interplay makes it possible to probe strongly coupled physics beyond the reach of any single approach. This work is written as a pedagogical account for Master's students familiar with advanced quantum field theory, representation theory, string theory and supersymmetry, with the aim of conveying the necessary intuition and practical tools to pursue related investigations independently.

The chapters are arranged to build these ingredients in turn. *Chapter 2, Theories in Five Dimensions and the Moduli Space,* introduces the five-dimensional supersymmetric theories and the physical structures that survive towards strong coupling. *Chapter 3, Type IIB Brane-Webs,* recasts the same theories geometrically in string theory and follows their behaviour towards infinite coupling. *Chapter 4, Magnetic Quivers for HyperKähler Constructions,* introduces an auxiliary quiver description of the resulting moduli spaces. *Chapter 5, Chiral Rings and the Hilbert Series,* turns from the geometry to the algebra of protected operators, while *Chapter 6, Plethystic Reconstruction of Generators and Relations,* develops the method used to recover explicit algebraic information from that counting data. *Chapter 7, Infinite Coupling Relations for $SU(3)_k$,* presents the resulting operator relations for the theories studied. The final chapters then use these relations to investigate what they imply about the finite- and strong-coupling descriptions, with particular attention to the glueball sector and to questions suggested by the reconstructed ring structure.

2 Theories in Five Dimensions and the Moduli Space

Five-dimensional supersymmetric gauge theories provide a rich setting where the distinction between low-energy effective descriptions and their ultraviolet completions is unavoidable. This is already apparent from dimensional analysis; for a Yang–Mills theory in d spacetime dimensions, the mass dimension of the inverse squared coupling is $d-4$ and therefore in 5d,

$$[g_{\mathrm{YM}}^{-2}] = 1. \tag{2.1}$$

Thus, the Yang–Mills interaction is power-counting non-renormalisable, and the 5d Lagrangian should consequentially be regarded as an effective description, valid below an energy cutoff[1, 2]. To begin examining the physics beyond these energies, it is useful to identify the defining data constituting such gauge-theory prescriptions. First, we define the dynamical structure; the gauge group G, the matter representations $\{\mathcal{R}_f\}$, the amount of supersymmetry, and any allowed discrete or Chern–Simons data.[1] The theory may then be deformed by non-dynamical parameters such as real masses and gauge couplings. And lastly, the choice of vacuum (through expectation values of scalar fields) determines the low energy behaviour.

Type	**Dynamics**	**Deformations**	**Vacuum**
Data	$G, \{\mathcal{R}_f\}, \mathcal{N}, \theta_{\mathrm{disc}}/k$	$g_0^{-2}, m_f, \ldots$	$\langle\phi\rangle, \langle q\rangle, \ldots$

Table 2.1: The three types of data entering a gauge-theory presentation.

Global symmetries must also be considered, since they constrain the operator spectrum and the deformations that may be introduced. A continuous global symmetry is encoded by its conserved-current operators and may be probed by coupling them to non-dynamical background sources. In a supersymmetric theory these sources assemble into full supermultiplets, so that supersymmetry-preserving parameters such as real masses may be realised as fixed values of their scalar components.[2] If $S[\Phi;\lambda]$ denotes the action at fixed external parameters λ_a, ,

$$S[\Phi;\lambda+\delta\lambda] = S[\Phi;\lambda] + \sum_a \delta\lambda_a \int d^d x\, \mathcal{O}_a(x) + \mathcal{O}(\delta\lambda^2).$$

where $\mathcal{O}_a$ is the local operator coupled to the deformation parameter λ_a, and Φ denotes the dynamical

[1] Topological information that will be defined later in the chapter. See 2.4,2.7.

[2] Coupling conserved currents to background gauge fields is the standard source formulation of continuous global symmetries, underlying both Ward identities and the modern treatment of 't Hooft anomalies [3]. Supersymmetric mass deformations may be organised through background flavour-current multiplets [4]; for the corresponding structure in 5d SCFTs, see Ref. [5]. Details of the multiplet construction are further discussed in Sec. 2.1.2.

fields. Varying λ_a changes the theory, whereas expectation values of the dynamical fields Φ select a vacuum at fixed λ_a.

The gauge coupling dimension rules out perturbative renormalisability, however supersymmetry constrains several aspects of the theory strongly enough to extract information on the physics. Our focus is the geometry of supersymmetric vacua, and how certain massive states stay encoded in this structure as we move towards strong coupling. We therefore turn to the supersymmetric constraints themselves, both to make this control precise and to understand what they help reveal beyond the gauge-theory description.

2.1 Minimal supersymmetry in five dimensions

2.1.1 Eight supercharges and R-symmetry

The irreducible complex spinor representation S of $\mathrm{Spin}(1,4)$ has complex dimension four and is pseudoreal. It therefore does not admit an ordinary Majorana condition. [3] Minimal supersymmetry is instead described on $S \otimes \mathbb{C}^2$ with real supercharges Q^A_α, $A = 1, 2$. The additional complex space carries a non-degenerate antisymmetric form, and their combined antilinear structure squares to $+1$. One may therefore impose the symplectic-Majorana condition,

$$(Q^A)^* = B\,\epsilon^{AB} Q^B, \tag{2.2}$$

where B implements complex conjugation on S and ϵ^{AB} is the antisymmetric invariant tensor on $\mathbb{C}^2$. The condition selects an eight-dimensional real subspace, giving the eight real supercharges of the algebra conventionally denoted $5d\,\mathcal{N} = 1$. The index A is thus acted on by the automorphism group, or the *R-Symmetry*,

$$USp(2)_R \simeq SU(2)_R. \tag{2.3}$$

As $SU(2)_R$ acts non-trivially on the supercharges, covariance of supersymmetry transformations can induce an action on all fields within a supermultiplet. It therefore appears as an internal symmetry of the supersymmetric theory, tied directly to the supersymmetry algebra.

Theories with eight real supercharges form a closely related class, including $6d\,\mathcal{N} = (1,0)$, $4d\,\mathcal{N} = 2$ and $3d\,\mathcal{N} = 4$ theories. Although the algebras differ with dimension, many features familiar in the study of extended supersymmetry in lower dimensions have direct analogues in $5d\,\mathcal{N} = 1$ [1].

2.1.2 Supermultiplets

The field content of a $5d\,\mathcal{N} = 1$ gauge theory is organised into vector multiplets for the gauge sector and hypermultiplets for the matter sector. The on-shell vector multiplet can be written schematically as,

[3] When complex representations are used to describe physical real fields, the reality condition must be compatible with the symmetry action. If J denotes the corresponding antilinear map, equivariance requires $J\rho(g) = \rho(g)J$. For an irreducible complex representation V, such a map exists precisely when $V \simeq \overline{V}$. After normalisation, $J^2 = +1$ defines a real representation, while $J^2 = -1$ defines a pseudoreal one. A quick check for compact groups is the appearance of an invariant singlet in $V \otimes V$: a singlet in $\mathrm{Sym}^2 V$ signals a real type, whereas one in $\wedge^2 V$ signals a pseudoreal type.

$$\mathcal{V} = (A_\mu, \phi, \lambda^A), \tag{2.4}$$

where A_μ is the gauge field, ϕ is a real scalar, and λ^A are the symplectic-Majorana gaugini; all transforming in the adjoint of the gauge group. A hypermultiplet contains four real scalar degrees of freedom and their fermionic partners,

$$\mathcal{H} = (q^A, \psi). \tag{2.5}$$

The scalars transform as a doublet of $SU(2)_R$ and take values in a quaternionic matter representation on which the gauge and flavour symmetries act. In complex notation this is expressed by a symplectic reality condition,

$$(q^{Aa})^* = \epsilon_{AB}\Omega_{ab}q^{Bb},$$

where ϵ_{AB} and Ω_{ab} are the invariant antisymmetric tensors on the $SU(2)_R$ doublet and the quaternionic matter representation space respectively. The two pseudoreal structures therefore combine to give a genuine reality condition on q^{Aa}.

Conserved currents of the global symmetry G_F transform in the adjoint representation. To form a flavour-invariant source coupling, the corresponding background sources must transform in the dual representation. For compact G_F, the adjoint is self-dual, so the sources are again adjoint-valued.[4] Supersymmetry requires the vector source to be completed into a full background multiplet. Its real scalar component couples to flavour-charged matter, and assigning it a constant value m reproduces the ordinary supersymmetric mass terms. Therefore a background scalar fixes a mass parameter, while expectation values of dynamical scalars select supersymmetric vacua.

2.2 Supersymmetric vacua and moduli-space geometry

A supersymmetric vacuum has vanishing vacuum energy, therefore static, spatially constant scalar configurations require the scalar potential to vanish. Gauge-invariant solutions to the vacuum equations define the classical *Moduli Space* $\mathcal{M}$ [5] of supersymmetric vacua,

$$\mathcal{M} = \frac{\{(\phi, q) \mid V(\phi, q) = 0\}}{G}. \tag{2.6}$$

With a fixed background, moving through $\mathcal{M}$ changes the vacuum in which the theory is expanded, thus the possible unbroken symmetry, spectrum and mass of excitations, resulting light degrees of freedom and ultimately the low-energy effective field theory. The geometry of $\mathcal{M}$ is the intrinsic structure organising the accessible effective descriptions within the same underlying theory.

Considering the multiplet origin of such VEVs distinguishes $\mathcal{M}$ into *branches* [2, 6]. Vacua parametrised by vector-multiplet scalars form the *Coulomb branch* $\mathcal{C}$, those parametrised by hypermultiplet scalars form the *Higgs branch* $\mathcal{H}$, and compatible simultaneous expectation values give mixed branches. We now study $\mathcal{C}$ and $\mathcal{H}$ separately, focusing on the geometry that organises their vacua.

[4] Equivalently, the vector source may be regarded as a non-dynamical G_F gauge field, or globally as a background connection on a G_F-bundle; see [3].

[5] More generally, a moduli space parametrises equivalence classes of objects. Depending on the problem, these may be gauge-equivalent vacua, solutions to differential equations, bundles, instantons, complex structures, or representations.

2.2.1 The Coulomb branch and the effective prepotential

The vector-multiplet scalar is real and adjoint valued, and may be diagonalised by a gauge transformation. The diagonal generators span the Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$, so every gauge-equivalent orbit of scalar VEVs admits a representative in $\mathfrak{h}$.[6] Residual gauge transformations act on the representatives through the Weyl group W, so classically,

$$\mathcal{C}_{\mathrm{cl}} = \mathfrak{h}/W. \tag{2.7}$$

The real dimension of $\mathcal{C}_{cl}$ is therefore $\operatorname{rank} G$. To determine the corresponding mass spectrum, we decompose the adjoint-valued gauge field,

$$A_\mu = A^i_\mu H_i + \sum_{\alpha\in\Delta} A^\alpha_\mu E_\alpha. \tag{2.8}$$

Consider a generic point of the Coulomb branch $\alpha(\phi) \neq 0, \forall\alpha \in \Delta$, so that ϕ lies away from the root hyperplanes. Its stabiliser is the maximal torus, and the gauge symmetry is broken to $G \longrightarrow U(1)^r$. The masses arise from the covariant-derivative $\mathrm{Tr}(D_\mu\phi)^2$, and thus $\mathrm{Tr}[A_\mu, \phi]^2$; the Cartan components A^i_μ commute with ϕ and remain massless, while the root-space components (W-bosons) acquire masses through,

$$[\phi, E_\alpha] = \alpha(\phi)E_\alpha, \quad M_{W_\alpha} = |\alpha(\phi)|. \tag{2.9}$$

Similarly, although hypermultiplet VEVs vanish on the Coulomb branch, their fluctuations remain in the spectrum and decompose into gauge and flavour weight spaces. If w is a gauge weight, ν a flavour weight, and $m \in \mathfrak{h}_F$ the background flavour scalar, then the corresponding component has mass,

$$M_{\mathcal{H}_{w,\nu}} = |w(\phi) + \nu(m)|\,. \tag{2.10}$$

The vanishing of these masses parametrise the boundaries of Coulomb chambers. Massless W-Boson loci restore the corresponding root generators to the massless spectrum and enhances the unbroken gauge symmetry, while a combination $w(\phi)+\nu(m) = 0$ introduces an additional massless hypermultiplet component into the low-energy spectrum.

Away from such hyperplanes the massive fields may be integrated out, leaving a low-energy $U(1)^r$ gauge theory. In a generic Abelian theory, the scalar and gauge kinetic terms could carry independent field-dependent couplings; eight supercharges strongly constrain this freedom by placing the fields in common vector multiplets. Remarkably, the entire local coupling structure, and more generally the full local vector-multiplet effective Lagrangian may be encoded in a single real function of the Coulomb parameters, the *prepotential* $\mathcal{F}(\phi)$. For the uncompactified five-dimensional theory on $\mathbb{R}^{1,4}$, the prepotential is piecewise cubic, with perturbative corrections terminating at one loop [1, 2].

[6] More invariantly, for compact connected G, the one-parameter subgroup generated by any $\phi \in \mathfrak{g}$ lies in some maximal torus $T \subset G$, so $\phi \in \mathrm{Lie}(T)$. Since all maximal tori are conjugate, there exists $g \in G$ such that $\mathrm{Ad}_g\,\phi$ lies in a chosen Cartan subalgebra $\mathfrak{h}$.

Writing the flavour contribution $\nu(m) \equiv m_f$ for the f-th hypermultiplet, within a fixed chamber it takes the standard form,

$$\mathcal{F}_{\text{eff}}(\phi) = \frac{1}{2g_0^2} h_{ij}\phi^i\phi^j + \frac{k}{6} d_{ijk}\phi^i\phi^j\phi^k + \frac{1}{12}\left[\sum_{\alpha\in\Delta} |\alpha(\phi)|^3 - \sum_f \sum_{w\in\mathcal{R}_f} |w(\phi) + m_f|^3\right]. \tag{2.11}$$

The first two terms are classical: the quadratic term is fixed by the Yang–Mills coupling, while the cubic term is present when the gauge algebra admits an invariant symmetric tensor d_{ijk}, with discrete coefficient k whose topological interpretation is discussed in Sec. 2.7. The root and weight sums are the one-loop contributions of massive vector and hypermultiplets respectively. Since their masses enter through absolute values, $\mathcal{F}_{\text{eff}}$ is piecewise cubic, with its polynomial form changing across massless loci. A field-theoretic derivation of Eq. (2.11) is given in App. A.

Through the prepotential, the kinetic terms define a metric on the scalar space $\mathfrak{h}/W$, while the gauge kinetic terms define an effective coupling matrix controlling the strength and mixing of the surviving $U(1)^r$ gauge fields. By supersummetry, both are given by the Hessian,

$$G_{ij}(\phi) = (g_{\text{eff}}^{-2})_{ij}(\phi) = \frac{\partial^2\mathcal{F}}{\partial\phi^i\partial\phi^j}, \tag{2.12}$$

up to the chosen normalisation. Its structure therefore provides a powerful probe of the theory: the Hessian will enter the consistency conditions for a possible UV completion, while higher derivatives reveal further physical data, with particular importance attached to the Chern–Simons coefficient k.

2.2.2 The Higgs branch: hyperkähler geometry and the coordinate ring

In contrast to the adjoint Higgs mechanism on the Coulomb branch, the hypermultiplet expectation values lie in matter representations. Expanding the kinetic term $|D_\mu q|^2$ about such a vacuum produces the gauge-boson mass matrix,

$$(M_A^2)_{ab} \propto \mathrm{Re}\Big[\big(T_a\langle q\rangle\big)^\dagger \big(T_b\langle q\rangle\big)\Big], \tag{2.13}$$

where the generators satisfying $T_a\langle q\rangle = 0$ remain unbroken, and the rank of the unbroken symmetry may be reduced. On the Coulomb branch, conjugating the single adjoint scalar into the Cartan subalgebra reduced the identification of physical, gauge invarient vacua to a finite Weyl action. Hypermultiplet scalar VEVs however, obey coupled vacuum constraints and generally admit no comparably simple description [7]. The eight-supercharge structure instead restricts the vacuum space to a highly special class of geometries, providing an intrinsic route to its description.

On smooth Higgs-branch stratum, the massless scalar fluctuations parametrise motion between neighbouring vacua, and their dynamics are described by a two-derivative nonlinear sigma-model. Eight supercharges require its target to be hyperkähler: it admits three complex structures I, J, K satisfying $I^2 = J^2 = K^2 = IJK = -1$, with respect to each of which the metric is Kähler [8].[7]

[7] The derivation is reviewed in App. B.

To construct this target from the microscopic theory, begin with the full hypermultiplet scalar space $\mathcal{X}$, which before imposing the vacuum equations already carries a flat hyperkähler structure inherited from the canonical kinetic terms. The gauge group G acts on $\mathcal{X}$ through hyperkähler isometries, preserving the metric and the three symplectic forms ω_r.

Gauging the triholomorphic isometries generated by k_a couples the hypermultiplets to the vector multiplet through the covariant derivative $D_\mu q$. Supersymmetric completion then forces the auxiliary $SU(2)_R$ triplet Y_r^a to couple linearly to a corresponding triplet of functions $\mu_r^a(q)$ of the hypermultiplet scalars. Cancellation of the supersymmetry variations fixes their derivatives in terms of the gauge action and hyperkähler forms,

$$d\mu_r^a = \iota_{k_a}\omega_r, \qquad r = 1, 2, 3, \tag{2.14}$$

so the μ_r^a are precisely the moment maps associated with the gauge action. Geometrically, their local existence follows from $\mathcal{L}_{k_a}\omega_r = 0$ and $d\omega_r = 0$ via Cartan's identity. The supersymmetric completion and its symplectic interpretation are derived in App. C.

Non-zero hypermultiplet masses generally lift charged Higgs directions, while Fayet–Iliopoulos parameters shift the moment-map level set. We therefore study the undeformed pure Higgs locus with $m_f = 0$, vanishing Fayet–Iliopoulos parameters, and $\phi = 0$. The relevant auxiliary-field terms are,

$$\mathcal{L}_Y = \frac{1}{2g^2}\sum_{a,r}(Y_r^a)^2 + \sum_{a,r} Y_r^a \mu_r^a. \tag{2.15}$$

Eliminating Y_r^a gives,

$$V_H(q) = \frac{g^2}{2}\sum_{a,r}(\mu_r^a)^2, \tag{2.16}$$

so supersymmetric vacua satisfy $\vec{\mu} = 0$. Quotienting by gauge equivalence therefore gives,

$$\mathcal{H}_{\text{cl}} = \vec{\mu}^{-1}(0)/G \equiv \mathcal{X}///G, \tag{2.17}$$

where /// denotes the *hyperkähler quotient* of $\mathcal{X}$ by G [9].

The moment-map relation ensures that the hyperkähler structure descends to this quotient. If v is tangent to $\vec{\mu}^{-1}(0)$, then $d\mu_r^a(v) = 0$, and hence $\omega_r(k_a, v) = 0$. The restricted symplectic forms are therefore insensitive to directions along the gauge orbits and descend to the quotient. When the level set is regular and the gauge action is free, the resulting space is hyperkähler and,

$$\dim_{\mathbb{H}} \mathcal{H}_{\text{cl}} = \dim_{\mathbb{H}} \mathcal{X} - \dim G, \tag{2.18}$$

since each gauge generator removes three real directions through the moment-map equations and one through gauge equivalence. Where these conditions fail, the quotient may instead develop singular strata [10, 11].

With this construction, explicitly solving the three real moment-map equations and classifying their G-orbits is generally impractical. The usefulness of this description is that after choosing one complex structure, the same vacuum space admits an algebraic formulation.

Choose I and write $\mu_{\mathbb{R}} = \mu_I, \mu_{\mathbb{C}} = \mu_J + i\mu_K$. The hyperkähler quotient imposes both $\mu_{\mathbb{C}} = 0$ and $\mu_{\mathbb{R}} = 0$ before quotienting by the compact group G. In the present zero-FI setting, it may equivalently be represented as

$$\mathcal{H}_{\rm cl} = \frac{\mu_{\mathbb{C}}^{-1}(0) \cap \mu_{\mathbb{R}}^{-1}(0)}{G} \simeq \mu_{\mathbb{C}}^{-1}(0) /\!\!/ G_{\mathbb{C}}. \tag{2.19}$$

Within each relevant complexified orbit closure the real moment map selects the representative satisfying $\mu_{\mathbb{R}} = 0$, unique up to the compact action of G. The quotient by $G_{\mathbb{C}}$ therefore packages the real equation and gauge identification into a holomorphic quotient. Its complex geometry is encoded by the coordinate ring

$$\mathbb{C}[\mathcal{H}_{\rm cl}] \simeq \mathbb{C}[\mu_{\mathbb{C}}^{-1}(0)]^{G_{\mathbb{C}}}. \tag{2.20}$$

This provides the algebraic object through which the Higgs branch may be studied: tools from algebraic geometry can be used to extract explicit geometric information and will be developed further in Chapters 5 and 6.

This Higgs-branch construction is protected by the eight-supercharge structure: at generic finite coupling it does not vary continuously with vector-multiplet couplings and holds while the massless field content remains unchanged [6]. At the five-dimensional ultraviolet fixed point however, non-perturbative states can become massless, introducing new light degrees of freedom and allowing the Higgs branch to enlarge. To understand this mechanism, we must first develop the physics of the non-perturbative spectrum.

2.3 Central charge and the BPS bound

In $5d$ $\mathcal{N} = 1$, the super-Poincaré algebra admits extensions that include the scalar central charge Z. The algebra may be written schematically as

$$\{Q_\alpha^A, Q_\beta^B\} = \epsilon^{AB}(C\Gamma^\mu)_{\alpha\beta}P_\mu + \epsilon^{AB}C_{\alpha\beta}Z + \cdots, \tag{2.21}$$

where the omitted terms include tensorial extensions associated with extended objects, and convention-dependent factors will not be important here.

For a massive state we may work in the rest frame, $P^\mu = (M, \mathbf{0})$, and diagonalise the algebra into combinations $\mathcal{Q}_\pm$ satisfying

$$\{\mathcal{Q}_+, \mathcal{Q}_+^\dagger\} = 2(M + |Z|), \qquad \{\mathcal{Q}_-, \mathcal{Q}_-^\dagger\} = 2(M - |Z|). \tag{2.22}$$

For any normalised state $|\Psi\rangle$, Hilbert-space positivity then gives

$$0 \leq \langle\Psi|\{\mathcal{Q}_-, \mathcal{Q}_-^\dagger\}|\Psi\rangle = 2(M - |Z|) \qquad \Longrightarrow \qquad M \geq |Z|. \tag{2.23}$$

The final inequality is known as the *BPS bound* [12]. When the bound is saturated, half the supercharges annihilate the state, and the associated multiplet contains fewer states than a generic massive, or *long*, multiplet and is therefore said to be *shortened*.

Its mass is then fixed by the algebra,

$$M_{\mathrm{BPS}} = |Z|, \tag{2.24}$$

rather than by an unconstrained pole/self-energy condition. So long as the state remains in a shortened BPS multiplet, its mass is tied directly to the central charge.

The remaining question is therefore what determines Z. Crucially, it depends on the scalar vacuum expectation values and background parameters, so the Coulomb-branch coordinates introduced above directly control the protected BPS spectrum through $M_{\mathrm{BPS}} = |Z|$. As these moduli vary, BPS masses vary and may vanish at special loci. This dependence follows from the field-theoretic realisation of Z as a surface term in the supercharge algebra, following Witten–Olive[12]. Writing the conserved supercharge as,

$$Q_\alpha^A = \int_{\Sigma_4} d^4x\, \mathcal{J}_\alpha^{0A}, \qquad \Sigma_4 \simeq \mathbb{R}^4, \tag{2.25}$$

total derivatives in the local supercurrent algebra contribute to the anticommutator through [13],

$$\{Q, Q\} \supset \int_{\Sigma_4} d^4x\, \partial_m \mathcal{Z}^m = \int_{S^3_\infty} d\Sigma_m\, \mathcal{Z}^m, \qquad S^3_\infty = \partial\Sigma_4. \tag{2.26}$$

For an electrically charged state in an Abelian Coulomb vacuum, this surface term reduces to the pairing between the asymptotic vector-multiplet scalar and the electric flux,

$$Z_{\mathrm{el}} = \int_{S^3_\infty} d\Sigma_m\, \phi^I D_I^m = a^I q_I, \qquad a^I \equiv \lim_{|x|\to\infty} \phi^I(x), \qquad q_I \equiv \int_{S^3_\infty} d\Sigma_m\, D_I^m, \tag{2.27}$$

where $I = 1, \ldots, r$ labels the Cartan generators and D_I^m is the electric displacement determined by the low-energy effective Lagrangian [14]. The central charge therefore pairs the conserved electric charge with the scalar expectation value specifying the Coulomb vacuum. For a W-boson associated with a root α, $q = \alpha$, and hence,

$$Z_{W_\alpha} = \alpha(a), \qquad M_{W_\alpha} = |\alpha(a)|, \tag{2.28}$$

in agreement with the mass obtained from the Higgs mechanism in Eq. (2.9).Real hypermultiplet masses may equivalently be introduced as constant background values of flavour-vector-multiplet scalars; in the supersymmetry algebra this amounts to shifting the relevant vector-multiplet scalar by the mass parameter [13]. Thus for a flavour current j_f^μ,

$$Z_{\mathrm{fl}} = m_f s_f, \qquad s_f \equiv \int_{\Sigma_4} d^4x\, j_f^0. \tag{2.29}$$

Combining the dynamical and background contributions, the particle central charge therefore takes the form,

$$Z(a, m) = q_I a^I + s_f m_f + \cdots. \tag{2.30}$$

The tensorial extensions suppressed in Eq. (2.21) similarly constrain BPS extended objects. In five dimensions, magnetic charge for a one-form gauge field is measured by the flux through an S^2 linking the source. Since there are four spatial dimensions, the corresponding source extends along one spatial direction and is therefore string-like.

For a straight static string, the transverse space is $\mathbb{R}^3$, and its energy per unit length may be written using the Coulomb-branch metric $G_{ij} = \partial_i \partial_j \mathcal{F}$ as,

$$\begin{aligned} T &= \frac{1}{2}\int_{\mathbb{R}^3} d^3x\, G_{ij}\left(\mathbf{B}^i\cdot\mathbf{B}^j + \nabla\phi^i\cdot\nabla\phi^j\right) \\ &= \frac{1}{2}\int_{\mathbb{R}^3} d^3x\, G_{ij}(\mathbf{B}^i \mp \nabla\phi^i)\cdot(\mathbf{B}^j \mp \nabla\phi^j) \pm \int_{\mathbb{R}^3} d^3x\, \mathbf{B}^i\cdot\nabla\left(\frac{\partial\mathcal{F}}{\partial\phi^i}\right). \end{aligned} \tag{2.31}$$

The final term is a surface contribution, since $G_{ij}\nabla\phi^j = \nabla(\partial_i\mathcal{F})$. For magnetic charge n^i_m, the resulting bound is,

$$T \geq \left|n^i_m \frac{\partial\mathcal{F}}{\partial a^i}\right|, \qquad T_{\mathrm{BPS}} = \left|n^i_m \frac{\partial\mathcal{F}}{\partial a^i}\right| \tag{2.32}$$

when the first-order Bogomol'nyi equation $\mathbf{B}^i = \pm\nabla\phi^i$ is satisfied [1, 15–17].

This saturation condition also preserves part of the supersymmetry. For a straight static string, taking x^4 along the string and $x^{1,2,3}$ transverse to it, the vector-multiplet gaugino variation reduces schematically to $\delta\lambda^A \propto (\frac{1}{2}F_{ij}\Gamma^{ij} + \Gamma^i\nabla_i\phi)\epsilon^A$. Using $B_i = \frac{1}{2}\epsilon_{ijk}F_{jk}$, the Bogomol'nyi equation reduces this variation to a single projector on the supersymmetry parameter, leaving half of the supercharges unroken. The monopole string is therefore $1/2$-BPS.

Taken together, we see that once the charge sector is fixed, the Coulomb-branch vacuum and the supersymmetric deformation parameters directly control the protected BPS masses and tensions: electric particle masses depend on a^I, while magnetic-string tensions depend on the dual coordinates $\partial_i\mathcal{F}$. The loci encountered earlier are therefore naturally associated with the vanishing of these protected quantities, where the corresponding BPS object becomes massless or tensionless. In five dimensions, this same BPS structure admits an additional topological contribution not yet included in the discussion.

2.4 Instantons

Instantons were introduced in four-dimensional Yang–Mills theory as finite-action Euclidean solutions capturing quantum tunnelling beyond perturbation theory [18, 19]. With the development of five-dimensional supersymmetric gauge theories, the same four-dimensional gauge configuration acquired a new interpretation: supported entirely on a spatial slice, it describes a localised solitonic particle carrying topological charge.

2.4.1 From four-dimensional instantons to five-dimensional particles

Consider Yang–Mills theory on Euclidean $\mathbb{R}^4$. An instanton is a non-trivial finite-action solution of the self-duality equation,

$$F = \pm *_4 F, \tag{2.33}$$

with the two signs corresponding to instantons and anti-instantons. The finite-action condition constrains the gauge field at infinity: for the smooth configurations considered here, finiteness of

$$S_E \propto \int_{\mathbb{R}^4} \mathrm{Tr}(F \wedge *_4 F) \tag{2.34}$$

requires $F \to 0$ sufficiently rapidly as $|x| \to \infty$. The connection is consequently asymptotically flat and hence locally pure gauge, $A \propto g^{-1} dg$. The asymptotic field therefore determines a map on the three-sphere at infinity,

$$g : S^3_\infty \longrightarrow G. \tag{2.35}$$

This boundary map requires a consideration in the interagation over gauge-configurations in the Euclidean path integral. A continuous deformation between finite-action configurations induces a continuous deformation of their boundary maps, namely a homotopy. Possible non-homotopic maps therefore distinguish components of the configuration space which cannot be reached from one another by perturbative fluctuations. For a simple compact gauge group, these maps are classified by the third homotopy group,

$$\pi_3(G) \simeq \mathbb{Z}, \tag{2.36}$$

and are therefore labelled by an interger *winding* number. This same integer emerges directly from the curvature; the gauge-invariant four-form $\mathrm{Tr}(F \wedge F)$ is closed and by using invariance of the trace with the Bianchi identity $DF = 0$,

$$d\ \mathrm{Tr}(F \wedge F) = \mathrm{Tr}[D(F \wedge F)] = 2\,\mathrm{Tr}(DF \wedge F) = 0. \tag{2.37}$$

Locally it is therefore expressible as,

$$\mathrm{Tr}(F \wedge F) = d\omega_3(A), \tag{2.38}$$

where $\omega_3(A)$ is the Chern–Simons three-form [20]. Stokes' theorem relates its integral over $\mathbb{R}^4$ to the asymptotic S^3_∞, where substituting the pure-gauge form of the connection reduces the result to the winding of g. Choosing the trace normalisation $\int \mathrm{Tr}(F \wedge F) = 8\pi^2$, one obtains,

$$I = \frac{1}{8\pi^2} \int_{\mathbb{R}^4} \mathrm{Tr}(F \wedge F) = \mathrm{wind}(g) \in \mathbb{Z}. \tag{2.39}$$

Thus I labels the topological sector of a finite-action configuration, where the (anti-)self-dual solutions are the semiclassical saddles in the non-zero sectors. [8] Its physical significance becomes clear by writing,

$$\mathbb{R}^4_E = \mathbb{R}_\tau \times \mathbb{R}^3, \tag{2.40}$$

and interpreting $A_\mu(\tau, \mathbf{x})$ as a Euclidean-time history of three-dimensional gauge configurations $A_i(\tau, \mathbf{x})$. By the same topological argument, the vacuum configurations approached as $\tau \to \pm\infty$ are labelled by integer winding numbers $n_\pm$. An instanton is then a Euclidean history which approaches one vacuum as $\tau \to -\infty$, passes through configurations of non-zero field strength, and approaches another as $\tau \to +\infty$. Its topological charge measures the change between them,

$$I = n_+ - n_-. \tag{2.41}$$

Instantons therefore mediate tunnelling between distinct winding-labelled semiclassical vacua [19, 21]. For a unit instanton, the Euclidean action is $S_E = 8\pi^2/g^2_{\mathrm{YM}}$, so its leading semiclassical contribution to the path integral is proportional to,

$$e^{-S_E} = e^{-8\pi^2/g^2_{\mathrm{YM}}}.$$

This dependence has an essential singularity at $g_{\mathrm{YM}} = 0$ and cannot be generated at any finite order in perturbation theory. Instanton tunnelling consequently mixes the winding-labelled semiclassical vacua, with the quantum vacuum described by their superpositions [21]. The effect is therefore intrinsically non-perturbative.

The same gauge profile acquires a new interpretation in five-dimensional Lorentzian spacetime,

$$\mathbb{R}^{1,4} = \mathbb{R}_t \times \mathbb{R}^4_{\mathrm{space}}. \tag{2.42}$$

The four coordinates supporting the instanton are now all spatial, so at fixed t the configuration is localised on $\mathbb{R}^4_{\mathrm{space}}$. Its evolution therefore traces a worldline in time, and the instanton becomes a five-dimensional solitonic particle.

The corresponding topological density defines the five-dimensional conserved current,

$$J_I = \frac{1}{8\pi^2} *_5 \mathrm{Tr}(F \wedge F), \qquad d(*_5 J_I) = \frac{1}{8\pi^2} d\ \mathrm{Tr}(F \wedge F) = 0, \tag{2.43}$$

where conservation follows from the closedness established above. Its charge on a constant-time slice is

$$Q_I = \int_{\mathbb{R}^4_{\mathrm{space}}} *_5 J_I = \frac{1}{8\pi^2} \int_{\mathbb{R}^4_{\mathrm{space}}} \mathrm{Tr}(F \wedge F) = I. \tag{2.44}$$

Thus the same topological invariant that measured the change of vacuum winding in four dimensions becomes a conserved particle charge in five dimensions, generating the topological global symmetry $U(1)_I$.

[8] Equivalently, after compactifying $\mathbb{R}^4$ by its point at infinity to S^4, the map $g : S^3 \to G$ acts as the transition function between two gauge patches. For $SU(N)$, with the standard trace convention, I is the second Chern number.

2.4.2 BPS instanton mass and the inverse-coupling deformation

The particle interpretation allows the topological charge to be related directly to the energy of the soliton. For a static configuration on $\mathbb{R}^4_{\text{space}}$, the Yang–Mills energy admits the Bogomol'nyi completion [22],

$$\begin{aligned} E &= \frac{1}{g_0^2}\int_{\mathbb{R}^4} \mathrm{Tr}(F \wedge *_4 F) \\ &= \frac{1}{2g_0^2}\int_{\mathbb{R}^4} \mathrm{Tr}[(F \mp *_4 F) \wedge *_4 (F \mp *_4 F)] \pm \frac{1}{g_0^2}\int_{\mathbb{R}^4} \mathrm{Tr}(F \wedge F). \end{aligned} \tag{2.45}$$

The first term is non-negative, while the second is fixed within a sector of instanton number I. Using the normalisation of Eq. (2.39) therefore gives,

$$E \geq \frac{8\pi^2}{g_0^2}|I|. \tag{2.46}$$

The bound is saturated precisely when $F = \pm *_4 F$, the same (anti-)self-duality condition that minimises the four-dimensional Euclidean action at fixed I. In the supersymmetric theory, self-duality has an additional consequence. For a static instanton background with $F_{0i} = 0$ and constant vector-multiplet scalar, the gaugino variation reduces schematically to,

$$\delta\lambda^A \propto F_{ij}\Gamma^{ij}\epsilon^A. \tag{2.47}$$

On the spatial $\mathbb{R}^4$, self-dual and anti-self-dual two-forms act on opposite spinor chiralities. For $F = \pm *_4 F$, the variation therefore vanishes for supersymmetry parameters obeying one chirality projector, leaving four of the eight real supercharges unbroken. The instanton soliton is consequently $1/2$-BPS, so by the superalgebra analysis of Sec. 2.3 its mass is fixed by the central charge.

The corresponding contribution follows from the topological symmetry $U(1)_I$. As for the flavour symmetries before, its conserved current belongs to a current multiplet that may be coupled to a background vector multiplet. The background gauge field couples to J_I, while supersymmetry completes this interaction so that the background scalar m_0 multiplies the Yang–Mills kinetic term. Turning on m_0 therefore produces the Yang–Mills deformation itself [23]. With the present normalisation,

$$m_0 = \frac{8\pi^2}{g_0^2}. \tag{2.48}$$

Thus the inverse gauge coupling may be viewed intrinsically as the background scalar conjugate to instanton number. The particle central charge is therefore extended by the topological charge,

$$Z = q_i a^i + s_f m_f + I\, m_0 + \cdots . \tag{2.49}$$

At the Coulomb and flavour origin, $a^i = m_f = 0$, so a BPS state carrying only instanton charge has

$$M_I = |I| m_0 = \frac{8\pi^2}{g_0^2}|I|, \tag{2.50}$$

in agreement with the self-dual energy bound.

This interpretation makes the strong-coupling limit immediate. Removing the Yang–Mills deformation sends $g_0^{-2} \to 0$ and hence $m_0 \to 0$. At the Coulomb and flavour origin, BPS states carrying non-zero instanton charge may then become massless, bringing into the low-energy spectrum degrees of freedom that were intrinsically non-perturbative at finite coupling. Their appearance marks a qualitative change in the spectrum at the strong-coupling limit, raising the question of whether the resulting massless degrees of freedom define a consistent interacting ultraviolet fixed-point theory.

2.5 Ultraviolet consistency from the Coulomb branch

A consistent endpoint of this strong-coupling limit would be a renormalisation-group fixed point, where the dimensionless couplings cease to run and the theory becomes scale invariant [24]. With eight supercharges, the interacting fixed points relevant here are $5d\,\mathcal{N}=1$ superconformal field theories [1, 25].[9] One may therefore ask which conditions the low-energy gauge theory must satisfy for such a limit to be physically admissible.

The Coulomb-branch dynamics give a particularly useful diagnostic. As established in Eq. (2.12), the Hessian of the prepotential determines both the Coulomb-branch metric and the effective Abelian gauge couplings. Its first derivatives also determine the tensions of BPS magnetic strings through Eq. (2.32). Consequently, a physically admissible gauge-theory region must have non-negative kinetic terms and non-negative tensions for its physical BPS states and strings.

This observation underlies the field-theoretic consistency conditions first developed by Seiberg and subsequently systematised by Intriligator, Morrison and Seiberg [1, 2]. In the original formulation, the prepotential was required to remain convex throughout the Weyl chamber in the strong-coupling limit. Later, geometric and brane constructions showed that this can be too restrictive: the perturbative Weyl chamber need not coincide with the region described by a single gauge-theory phase. In particular, a BPS string may become tensionless at finite distance, marking a boundary beyond which the same effective description need not remain valid. The relevant positivity conditions are therefore imposed on the *physical Coulomb branch*, bounded by such loci, rather than on an arbitrarily extended perturbative chamber [26, 27].

A simple application gives a useful necessary bound on the matter content. Consider an $SU(N)$ gauge theory with N_f massless hypermultiplets in the fundamental representation. Writing the Cartan scalar in diagonal form, choose the Coulomb-branch direction [26],

$$\phi = \mathrm{diag}(a, 0, \ldots, 0, -a), \qquad a > 0. \tag{2.51}$$

Along this ray the classical cubic invariant vanishes,

$$\mathrm{Tr}\,\phi^3 = a^3 + (-a)^3 = 0, \tag{2.52}$$

so the Chern–Simons contribution to the prepotential (2.11) drops out. Taking the strong coupling limit then isolates the vector and hypermultiplet contributions. Other Coulomb directions retain the Chern–Simons dependence and can impose further constraints.

[9] A five-dimensional gauge theory need not UV-complete to an intrinsically five-dimensional fixed point. Some instead arise from six-dimensional SCFTs compactified on a circle, while others admit no decoupled quantum-field-theoretic ultraviolet completion. These alternatives and their distinction from genuine $5d$ SCFT completions are discussed in Refs. [23, 26, 27].

For $SU(N)$, the roots are $\alpha_{ij} = e_i - e_j$. The two roots connecting the eigenvalues $+a$ and $-a$ contribute $2(2a)^3 = 16a^3$, while for each of the $N-2$ vanishing eigenvalues there are four further roots of absolute value a.

$$\sum_{\alpha\in\Delta} |\alpha(\phi)|^3 = 4(N+2)a^3, \qquad \sum_{w\in\mathbf{N}} |w(\phi)|^3 = 2a^3. \tag{2.53}$$

The weights of the fundamental representation evaluate directly to the eigenvalues of ϕ, so each massless fundamental contributes $2a^3$. Substituting these results into the exact prepotential gives,

$$\mathcal{F}(a) = \frac{1}{6}\,(2N + 4 - N_f)\,a^3. \tag{2.54}$$

The induced Coulomb-branch metric is therefore,

$$\frac{d^2\mathcal{F}}{da^2} = (2N + 4 - N_f)a, \tag{2.55}$$

and requiring it to remain non-negative for $a > 0$ gives,

$$N_f \leq 2N + 4. \tag{2.56}$$

This bound is necessary rather than sufficient: other Coulomb directions may impose stronger constraints, while positivity of the protected low-energy data alone does not establish the existence of an ultraviolet completion. Independent string-theoretic constructions will provide a stronger test of existence in the following chapter. For the present discussion, the gauge theory determines when the strong-coupling limit is compatible with its low-energy dynamics, but does not itself define the theory reached at that limit. To do so requires an intrinsic description of the ultraviolet fixed point as a superconformal field theory.

2.6 The superconformal fixed point

A Lagrangian formulation speciies a quantum field theory through fields and an action. A conformal field theory may instead be characterised intrinsically by its local operators and their correlation functions. The product of nearby operators admits an operator product expansion,

$$\mathcal{O}_i(x)\mathcal{O}_j(0) \sim \sum_k C_{ij}{}^k(x)\,\mathcal{O}_k(0). \tag{2.57}$$

Once operator normalisations are fixed, the spectrum of conformal primaries and their OPE coefficients constitute the local CFT data from which correlation functions may be reconstructed, subject to conformal symmetry, unitarity and OPE associativity, or crossing [28, 29].

For a five-dimensional $\mathcal{N} = 1$ SCFT, the superconformal symmetry is the exceptional Lie superalgebra $F(4)$ [30, 31], whose bosonic subalgebra is,

$$\mathfrak{so}(5,2) \oplus \mathfrak{su}(2)_R. \tag{2.58}$$

A superconformal primary may be labelled schematically by,

$$\mathcal{O}: \qquad [j_1, j_2]_{\Delta,R}, \tag{2.59}$$

where $[j_1, j_2]$ are Dynkin labels of the rotation algebra $\mathfrak{so}(5) \simeq \mathfrak{sp}(2)$, Δ is the scaling dimension, and $R \in \frac{1}{2}\mathbb{Z}_{\geq 0}$ denotes the $SU(2)_R$ isospin (for example the supercharges Q transform in $[1,0]_{\frac{1}{2},\frac{1}{2}}$). The scaling dimension acquires a spectral interpretation through radial quantisation; writing the radial coordinate of Euclidean $\mathbb{R}^5$ as $r = e^\tau$ gives, up to a conformal factor,

$$\mathbb{R}^5 \setminus \{0\} \simeq \mathbb{R}_\tau \times S^4. \tag{2.60}$$

Dilatations thus become translations in τ, giving the operator-state correspondence,

$$\mathcal{O}(0) \longleftrightarrow |\mathcal{O}\rangle_{S^4}, \qquad E_{S^4} = \Delta_{\mathcal{O}}, \tag{2.61}$$

for a unit-radius sphere [29]. Supersymmetry protects a distinguished subset of this spectrum. Let $\mathcal{O}_R$ denote a Lorentz-scalar superconformal primary transforming with isospin R under $SU(2)_R$, chosen to be the highest-weight state of that representation $[0,0]_{\Delta,R}$. In radial quantisation the Poincaré and conformal supercharges are conjugate, $Q^\dagger = S$, and for an appropriate supercharge one component of the $F(4)$ algebra gives schematically,

$$\left\| Q_{\mathrm{BPS}} |\mathcal{O}_R\rangle \right\|^2 \propto \Delta - 3R. \tag{2.62}$$

Positivity therefore requires $\Delta \geq 3R$. When the bound is saturated, the corresponding descendant is null and the multiplet shortens,

$$\mathcal{O}_R: \qquad [0,0]_{3R,R}, \qquad \Delta = 3R. \tag{2.63}$$

This is the superconformal BPS protection: earlier the shortening fixed a particle mass through its central charge, while here it fixes the scaling dimension of a local operator through its $SU(2)_R$ representation. Since R is a discrete representation label, Δ cannot vary independently while the operator remains in the same short multiplet [31].

2.6.1 The Higgs branch as a protected operator ring

The Higgs branch can now be identified intrinsically within the protected operator spectrum. The relevant operators are the half-BPS scalar primaries $[0,0]_{3R,R}$ introduced above, with $R > 0$; their non-trivial $SU(2)_R$ charge distinguishes them from the $SU(2)_R$-singlet scalar operators associated with Coulomb-branch directions. Choosing a complex structure on the hyperkähler Higgs branch selects a Cartan $U(1)_R \subset SU(2)_R$. For an isospin-R operator,

$$\mathcal{O}_m, \qquad m = -R, \ldots, R, \tag{2.64}$$

the highest-weight component $\mathcal{O}^{(R)}_{\mathrm{HW}} = \mathcal{O}_{m=R}$ is holomorphic in that complex structure. Their $SU(2)_R$ weights, and hence their protected dimensions are additive,

$$R(\mathcal{O}_1\mathcal{O}_2) = R_1 + R_2, \qquad \Delta(\mathcal{O}_1\mathcal{O}_2) = 3(R_1 + R_2). \tag{2.65}$$

Therefore products of these highest-weight operators remain in the protected holomorphic sector and, modulo operator relations, form a graded commutative ring $\mathcal{R}_H$, referred to as the Higgs-branch *chiral ring* [32, 33]. A supersymmetric Higgs-branch vacuum is encoded by the expectation values of its operators, thus the protected algebra is identified with the holomorphic coordinate ring of the vacuum space,

$$\mathcal{R}_H \simeq \mathbb{C}[\mathcal{H}_{\mathrm{UV}}]. \tag{2.66}$$

At the conformal point, its positive grading by protected dimension reflects the action of dilatations, which rescale the Higgs-branch coordinates while leaving the conformal vacuum fixed. The resulting vacuum geometry is therefore a hyperkähler cone. This gives a concrete algebraic object that can be compared with the finite-coupling Higgs-branch ring, allowing the change in the protected vacuum structure at strong coupling to be studied directly.

2.6.2 Instanton operators

The instanton particles discussed above give rise at the fixed point to local *instanton operators*: disorder insertions defined by imposing non-zero instanton number on a small S^4 linking the insertion point [32]. They therefore carry the same conserved $U(1)_I$ charge as the corresponding particle states.

Radial quantisation makes this relation precise. An instanton operator of charge I inserted at the origin creates a state on S^4,

$$\mathcal{I}_I(0) \longleftrightarrow |\mathcal{I}_I\rangle \in \mathscr{H}_I, \tag{2.67}$$

while conservation of instanton number decomposes the Hilbert space into topological sectors,

$$\mathscr{H}_{S^4} = \bigoplus_{I\in\mathbb{Z}} \mathscr{H}_I. \tag{2.68}$$

Instanton number therefore labels a non-perturbative sector of the local operator spectrum.

2.6.3 Global-symmetry enhancement

Continuous global symmetries are encoded by their conserved-current multiplets. For each generator of a flavour symmetry G_F, the corresponding multiplet contains a half-BPS scalar primary,

$$\mu^A : \qquad [0,0]_{3,1}, \qquad \mu^A \in \mathrm{Adj}(G_F),$$

whose descendants include the conserved current J^A_μ. At the ultraviolet fixed point, non-zero instanton sectors may supply additional operators with precisely these quantum numbers, and the symmetry is enlarged,

$$G_F^{\mathrm{finite}} \subseteq G_F^{\mathrm{UV}}.$$

The origin of these additional operators can be seen by decomposing representations of G_F^{UV} under the finite-coupling symmetry, where the resulting components are distinguished by the gauge-theory charges, including $U(1)_I$. Non-zero instanton charge then identifies the contributions that are non-perturbative from the finite-coupling perspective, a distinction that will be central to the later analysis.

2.6.4 Alternative constructions of the UV geometry

The canonical example of the global symmetry enhancement is $SU(2)$ gauge theory with N_f fundamental hypermultiplets,

$$SO(2N_f) \times U(1)_I \longrightarrow E_{N_f+1}, \qquad 1 \leq N_f \leq 7. \tag{2.69}$$

Seiberg originally inferred this enhancement from the spacetime gauge symmetry seen by a Type I′ D4-brane probe and did more than reveal the exceptional symmetry: it also identified the ultraviolet Higgs branch with the reduced moduli space of a single instanton of the enhanced group [1]. For the simple members of the E_n sequence,

$$\mathcal{H}_{E_n} \simeq \widetilde{\mathcal{M}}_1(E_n), \qquad n = 4, \ldots, 8, \tag{2.70}$$

where $\widetilde{\mathcal{M}}_1(G)$ is the reduced moduli space of one G-instanton on $\mathbb{R}^4 \simeq \mathbb{C}^2$. More precisely, the framed one-instanton moduli space and its separation into translational and centred degrees of freedom may be written schematically as,

$$\mathcal{M}_1(G) = \frac{\{A \mid F = *_4 F,\ I[A] = 1\}}{G}, \qquad \mathcal{M}_1(G) \simeq \mathbb{C}^2 \times \widetilde{\mathcal{M}}_1(G), \tag{2.71}$$

the factor $\mathbb{C}^2$ parametrises the position of the instanton on $\mathbb{R}^4$, while $\widetilde{\mathcal{M}}_1(G)$ describes the remaining centred moduli. For simple G, this reduced space has an independent description in Lie theory,

$$\widetilde{\mathcal{M}}_1(G) \simeq \overline{\mathcal{O}_{\min}(\mathfrak{g}_{\mathbb{C}})}, \qquad \overline{\mathcal{O}_{\min}} = \mathcal{O}_{\min} \cup \{0\}, \tag{2.72}$$

as the closure of the minimal non-zero nilpotent orbit of the complexified Lie algebra [34]. These orbit closures are prototypical examples of *conical symplectic singularities*, a class that has become central in modern symplectic and algebraic geometry [35, 36]. Their coordinate rings carry natural graded Poisson structures, and their singularities are organised by symplectic-leaf stratifications. The same space therefore appears simultaneously as an SCFT Higgs branch, an instanton moduli space, and a distinguished singular variety in symplectic geometry.

This multiplicity of descriptions extends well beyond the rank-one example. The ADHM construction realises classical-group instanton spaces as finite-dimensional hyperkähler quotients [37]; brane constructions and small-instanton limits provide complementary string-theoretic realisations [38, 39]; and $3d$ $\mathcal{N} = 4$ mirror symmetry and subsequent monopole-operator constructions realise related symplectic singularities as Coulomb branches of auxiliary gauge theories [40–43]. The latter perspective, realised through magnetic quivers, will be the focal point of this thesis.

2.7 $SU(3)_k$

The rank-one $SU(2)$ theories above provide a useful prototype, but $SU(3)$ introduces two features absent at rank one: a two-dimensional Coulomb branch and a non-vanishing symmetric cubic invariant d_{ijk}. It is therefore the simplest $SU(N)$ theory in which higher-rank Coulomb dynamics and an ordinary five-dimensional Chern–Simons interaction coexist with the instanton sectors discussed above.[10]

The coefficient k already encountered in the cubic prepotential,

$$\mathcal{F}_{\rm CS} = \frac{k}{6} d_{ijk}\phi^i\phi^j\phi^k, \tag{2.73}$$

and is also the coefficient of the five-dimensional Chern–Simons interaction,

$$S_{\rm CS} = \frac{k}{24\pi^2}\int_{\mathbb{R}^{1,4}} \omega_5(A), \qquad d\omega_5(A) = \mathrm{Tr}(F^3). \tag{2.74}$$

The Chern–Simons 5-form is not itself invariant under large gauge transformations. For $SU(N \geq 3)$, these are classified by $\pi_5(SU(N)) \simeq \mathbb{Z}$, and a transformation of winding n shifts the action by,

$$S_{\rm CS} \longrightarrow S_{\rm CS} + 2\pi k n. \tag{2.75}$$

Since the quantum theory depends on the phase e^{iS}, invariance under all such transformations requires $k \in \mathbb{Z}$ for the pure theory in the standard normalisation. The integer k is conventionally called the *Chern–Simons level*. Since the Chern–Simons interaction changes sign under orientation reversal, parity sends $k \to -k$; thus $SU(N)_k$ and $SU(N)_{-k}$ are parity conjugates, with $k = 0$ the parity-invariant case [2, 23].

Fundamental matter modifies this quantisation through the five-dimensional parity anomaly. Each hypermultiplet contains a Dirac fermion, and for a massless fermion one cannot define the quantum determinant while preserving both parity and invariance under large gauge transformations. Choosing a gauge-invariant regularisation leaves a parity-odd Chern–Simons contribution with half-integral fractional part, defined modulo an integer local counterterm [23].

In this sense the fermion determinant and the Chern–Simons term are not separately invariant; rather, their gauge variations must cancel in the complete quantum theory. For N_f fundamental hypermultiplets, the accumulated fractional contribution is therefore $N_f/2$. In the convention used here, consistency requires,

$$k + \frac{N_f}{2} \in \mathbb{Z}, \tag{2.76}$$

Thus,

$$N_f \text{ even} \quad \Longrightarrow \quad k \in \mathbb{Z}, \qquad N_f \text{ odd} \quad \Longrightarrow \quad k \in \mathbb{Z} + \frac{1}{2}. \tag{2.77}$$

Thus for odd N_f, the half-integral value of k should be understood as part of the combined matter–Chern–Simons quantum theory: its gauge variation compensates that of the fermion determinant.

[10] For $SU(2)$, the additional discrete datum is instead a $\mathbb{Z}_2$-valued theta angle, $\theta_{\rm disc} = 0, \pi$, associated with $\pi_4(SU(2)) = \mathbb{Z}_2$. For $SU(N \geq 3)$, $\pi_4(SU(N)) = 0$, so no analogous discrete theta parameter occurs.

The same parity-odd contribution controls real-mass flows. Giving a fundamental hypermultiplet a large positive or negative mass and integrating it out shifts the Chern–Simons coefficient by a half-unit, so in the conventions used here,

$$SU(N)_k + N_f F \xrightarrow{m\to\pm\infty} SU(N)_{k\mp\frac{1}{2}} + (N_f - 1)F. \tag{2.78}$$

The Chern–Simons interaction has a further effect that is important for the instanton sector. The equation of motion for the temporal gauge field A_0 gives the non-Abelian Gauss-law constraint. Because the Chern–Simons action depends on A_0, this constraint acquires a contribution proportional to the spatial instanton density,

$$(D_i E^i)^a + \frac{k}{8\pi^2} *_4 \mathrm{Tr}(T^a F \wedge F) + \cdots = 0, \tag{2.79}$$

where the omitted terms contain the remaining fields of the supersymmetric theory [14]. An instanton background therefore carries an induced gauge-electric charge whose value depends on k and on its embedding in the gauge group. When the instanton sector is quantised, one begins with a state associated with the instanton background and then quantises its fermionic zero modes. The Chern–Simons contribution means that this bare state can carry non-zero charge under the gauge symmetry left unbroken by the instanton embedding. Since a physical state must satisfy Gauss's law, only zero-mode excitations whose gauge charge compensates this contribution survive as gauge-invariant instanton states. Changing k therefore changes which combinations are allowed, and hence can change the protected instanton operators appearing in the ultraviolet theory. The resulting gauge-invariance condition on the instanton zero-mode spectrum will be implemented explicitly for the $SU(3)$ theories in the later chapters, with the one-instanton construction reviewed in App. F.

The remainder of this dissertation focuses on the six theories,

$$\begin{aligned} &SU(3)_{\frac{1}{2}} + 5F, \quad SU(3)_{\frac{3}{2}} + 5F, \quad SU(3)_{\frac{5}{2}} + 5F, \\ &SU(3)_0 + 6F, \quad SU(3)_1 + 6F, \quad SU(3)_2 + 6F. \end{aligned} \tag{2.80}$$

Together they offer a natural setting in which to isolate the effect of k on the non-perturbative instanton contributions to the ultraviolet theory, through the structure of its Higgs-branch geometry.

The discussion of this chapter has accounted for the data that constitute five-dimensional gauge theory: the gauge group, matter content, Coulomb and mass parameters, the gauge coupling, and Chern–Simons level. Field-theoretic considerations alone, however, do not provide a constructive description of the strongly coupled endpoint or its full operator spectrum. In string theory, these same data acquire a direct interpretation in the structure and deformations of brane configurations. The next chapter therefore turns its study, where the strongly coupled endpoint can become accessible.

3 Type IIB Brane-Webs

To motivate the brane perspective developed in this chapter, we return to Seiberg's rank-one construction from the previous chapter. It begins in Type I′ string theory, with a D4-brane probe moving along an interval bounded by orientifold eight-planes and populated by background D8-branes. When the D4 probe lies at an orientifold fixed plane with N_f coincident D8-branes, its worldvolume theory is five-dimensional $SU(2)$ with N_f fundamental hypermultiplets. Crucially, the probe gauge coupling is controlled by the string background, and varying the background moduli can reduce the inverse gauge coupling to zero. Although the weakly coupled gauge-theory description then breaks down, the D4-brane remains part of a consistent string construction, and therefore its long-distance worldvolume theory realises the corresponding interacting fixed point.[1]

To move beyond rank-one examples, we turn to the Type IIB construction using webs of (p, q) five-branes, introduced by Aharony and Hanany and developed further with Kol [17, 44]. Brane webs provide a highly practical and systematic framework for realising a broad range of five-dimensional gauge theories; this chapter will present the construction and its application to the six $SU(3)_k$ theories, producing strong-coupling webs that allow us to uncover their ultraviolet geometry.

3.1 Constructing supersymmetric (p, q) five-brane webs

Type IIB possesses an $SL(2, \mathbb{Z})$ duality which acts on the axiodilaton,

$$\tau = C_0 + ie^{-\Phi},$$

where C_0 is the RR scalar and Φ the dilaton, and simultaneously rotates NS5- and D5-brane charges. We use,

$$(1, 0) = \text{D5}, \qquad (0, 1) = \text{NS5}, \tag{3.1}$$

and label a general five-brane by an integral charge vector (p, q), with $\gcd(p, q) = 1$ for a primitive brane. All five-branes share the directions $x^{0,\dots,4}$, while their remaining worldvolume direction lies in a common (x^5, x^6) plane.

Dimension	0	1	2	3	4	5	6	7	8	9
D5	×	×	×	×	×	×	–	–	–	–
NS5	×	×	×	×	×	–	×	–	–	–
(p, q) 5-brane	×	×	×	×	×	line		–	–	–

Table 3.1: Worldvolume directions of the five-branes used in the Type IIB web construction.

[1] The full Type I/Type I′ duality and probe argument are reviewed in Appendix D.

The common $x^{0,\dots,4}$ directions support the five-dimensional theory, while the remaining directions form a planar web in the (x^5, x^6) plane. Its geometry is constrained by supersymmetry. A D5-brane along x^5 and an NS5-brane along x^6 preserve a common set of eight real supercharges. A general (p, q) five-brane carries the corresponding $SL(2, \mathbb{Z})$-rotated supersymmetry projection, and preserving the same supercharges requires its orientation in the web plane to rotate together with its charge [17, 44].

For constant τ, this relation is encoded by the phase of $p + q\tau$. The value of τ is redundant from the perspective of the low-energy five-dimensional theory: changes in the string coupling may be compensated by rescalings of the web coordinates, while a non-zero axion may be absorbed by a linear shear of the plane [17]. We therefore choose the convenient normalized frame $\tau = i$, for which the BPS slope condition becomes,

$$\Delta x^5 : \Delta x^6 = p : q. \tag{3.2}$$

Thus D5-branes are horizontal, NS5-branes are vertical, and the direction of a general (p, q) segment can be read directly from its charge vector.

3.1.1 Junctions and basic building blocks

Individual five-branes are assembled into a web by allowing several segments to meet at junctions. Orienting all charge vectors outwards from a junction, conservation of five-brane charge requires

$$\sum_i p_i = 0, \qquad \sum_i q_i = 0. \tag{3.3}$$

The same BPS condition that relates charge to orientation also fixes the relative five-brane tensions. In the normalized frame $\tau = i$,

$$T_{(p,q)} \propto \frac{|p + q\tau|}{\sqrt{\operatorname{Im}\tau}} \propto \sqrt{p^2 + q^2}, \qquad \sum_i \vec{T}_i = 0. \tag{3.4}$$

Since each tension vector points along the charge-determined direction of its segment, conservation of (p, q) charge implies force balance and therefore the junction is in mechanical equilibrium. Trivalent junctions form the elementary vertices of a generic web (Figure 3.1); higher-valent intersections may be resolved into networks of trivalent junctions while preserving the external charge data, introducing finite internal segments and, when several such junctions are combined, compact faces.

3.1.2 Seven-branes

A complete web contains such finite internal segments together with external legs extending away from the central configuration. The asymptotic (p, q) charges carried by these external legs form part of the boundary data of the web and distinguish different five-dimensional backgrounds. However external legs need not remain semi-infinite, a (p, q) five-brane may terminate on a compatible $[p, q]$ seven-brane. Seven-branes can be introduced without breaking any additional supersymmetry and allow different presentations of the same five-brane system to be related by their rearrangement [45].

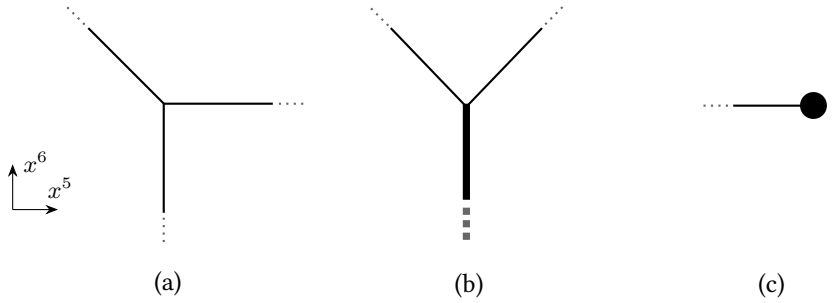


Figure 3.1: Examples of vertices: (a) a common trivalent junction, (b) vertex conservation inducing higher NS5 charge (c) a D5 ending on a D7.

Crucially, the seven-brane sources the axiodilaton and carries a non-trivial $SL(2,\mathbb{Z})$ monodromy, represented by a branch cut across which the charge of a five-brane transforms; moving seven-branes may also create or annihilate five-brane segments through the Hanany–Witten effect.[2] Since τ varies in the presence of seven-branes, physical five-brane trajectories need not remain globally straight. The straight-line diagrams should therefore be understood as a convenient bookkeeping of the charge and junction data, with the normalized constant-τ picture supplying the local construction rules.

3.2 Engineering five-dimensional gauge theories

To begin relating the web geometry to gauge-theory data, we first recover its gauge symmetry from the familiar Chan–Paton construction [46, 47].

A single D5-brane supports massless open-string modes, containing a worldvolume gauge field together with scalars describing its transverse position. For N coincident D5-branes, an oriented open string may begin on brane i and end on brane j, with $i, j = 1, \ldots, N$. These N^2 Chan–Paton sectors may be represented by the elementary matrices E_{ij}, whose products reproduce the joining of oriented strings, $E_{ij}E_{kl} = \delta_{jk}E_{il}$. The corresponding massless vector states therefore assemble into a matrix-valued gauge field with algebra $\mathfrak{u}(N)$.

For a web with fixed asymptotic data, the common translation of all colour branes is not an independent local deformation; charge conservation forces the five-branes supporting the D5 stack to leave with different (p, q) charges, therefore they cannot deform consistently along their initial direction. Removing this centre-of-mass $U(1)$ leaves the traceless $SU(N)$ sector, giving $N - 1$ independent Coulomb moduli.

The associated adjoint scalar measures the displacement of the horizontal D5 segments in the transverse x^6 direction. Writing

$$\phi = \mathrm{diag}(a_1, \ldots, a_N), \qquad \sum_{i=1}^{N} a_i = 0, \qquad a_i = T_{\mathrm{F}} x_i^6, \tag{3.5}$$

an off-diagonal string in the (i, j) sector obeys $[\phi, E_{ij}] = (a_i - a_j)E_{ij}$.

[2] These operations will be further discussed in Subsection 3.4.3.

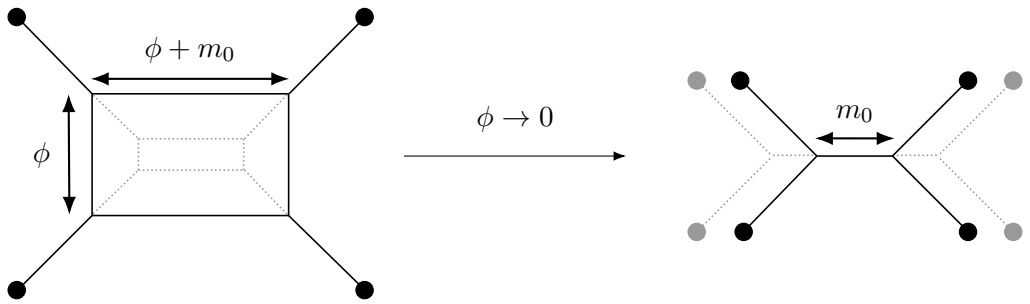


Figure 3.2: Pure $SU(2)$ or E_1 web with separated colour branes (left) and an internal deformation (dotted). At the Coulomb origin (right) the colour branes coincide, restoring $SU(2)$, showing the external deformation (dotted) associated with the gauge coupling.

Thus E_{ij} is the root generator associated with $\alpha_{ij} = e_i - e_j$. A fundamental string stretched between the two D5-branes has length $L_{ij} = |x_i^6 - x_j^6|$ and hence BPS mass,

$$M_{ij} = T_{\mathrm{F}} L_{ij} = |a_i - a_j| = |\alpha_{ij}(\phi)|. \tag{3.6}$$

These states are precisely the off-diagonal W-bosons, and separating the colour branes geometrically realises the Coulomb-branch breaking $SU(N) \to U(1)^{N-1}$.

3.2.1 Magnetic strings and the prepotential

A D3-brane filling the compact face of a web appears in the common five-dimensional spacetime as a magnetic string. Its tension is set by the wrapped D3-brane area, which may be then identified with the first derivative of the Coulomb-branch prepotential,

$$T_i^{\mathrm{mag}} = T_{\mathrm{D3}} A_i = \left| \frac{\partial \mathcal{F}}{\partial \phi_i} \right|, \qquad A_i \longleftrightarrow \frac{\partial \mathcal{F}}{\partial \phi_i}. \tag{3.7}$$

The compact-face geometry therefore directly encodes the magnetic-string tensions and provides a geometric check of the Coulomb-branch dynamics [17]. For the elementary rank-one web this relation is explicit:

$$A(\phi) = \phi(m_0 + \phi) \qquad \Longrightarrow \qquad \mathcal{F}(\phi) = \frac{1}{2} m_0 \phi^2 + \frac{1}{3} \phi^3 + c. \tag{3.8}$$

The quadratic term reproduces the bare inverse-coupling contribution, while the cubic dependence is the one-loop contribution of the massive W-bosons.

3.2.2 Flavours and continuous parameters

Fundamental hypermultiplets are introduced by attaching external D5 flavour legs to the colour web. If the i-th colour D5 lies at $y = a_i$ and the flavour leg at $y_f = -m_f$, an F-string stretched between them has BPS mass,

$$M_i = |a_i - y_f| = |a_i + m_f| = |w_i(\phi) + m_f|,$$

where w_i is a weight of the fundamental representation (see Figure 3.3).

Transverse motion of the flavour leg changes the fixed asymptotic data of the web and therefore implements the real mass deformation. A component becomes massless when $y_f = a_i$, where the string has zero length; coincident flavour legs similarly represent several flavours with equal mass, with their multiplicity retained [17]. The remaining continuous asymptotic parameter is the bare inverse coupling $m_0 \propto g_0^{-2}$, encoded by the relative separation of the NS5-type external legs.

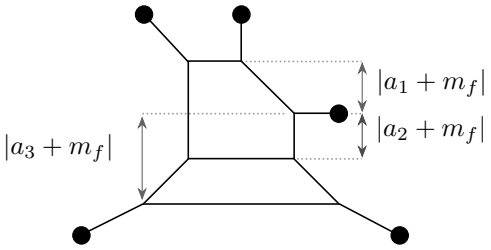


Figure 3.3: $SU(3) + 1F$ web in the convention $y_f = -m_f$. For colour D5 positions a_i, the stretched F-string masses are $M_i = |a_i - y_f| = |a_i + m_f|$, $i = 1, 2, 3$.

3.2.3 Flop transitions and flavour decoupling

The effective prepotential is piecewise cubic on the extended Coulomb branch, with adjacent chambers separated by loci at which a charged BPS state becomes massless. In the web this occurs when the finite segment measuring its mass shrinks to zero. Continuing through this degeneration replaces the local configuration by the alternative resolution, producing a *flop transition*[17]. The two resolutions describe neighbouring chambers of the same theory, with different polynomial expressions for $\mathcal{F}_{\text{eff}}$ corresponding to the change of sign of the relevant BPS mass.

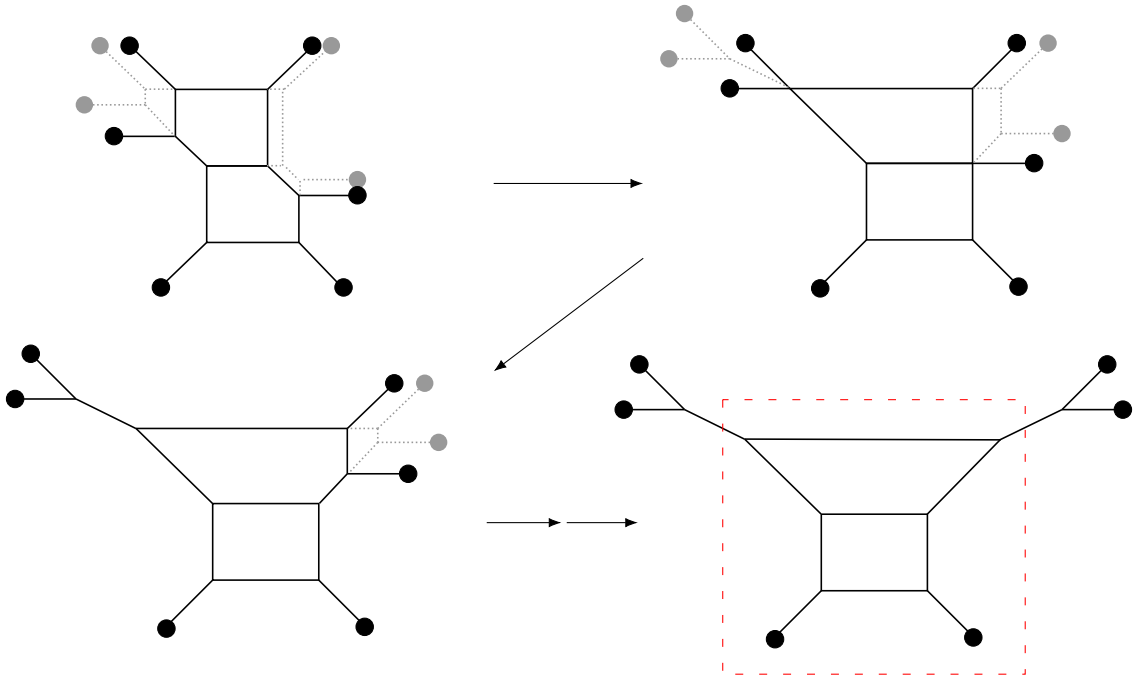


Figure 3.4: Flop transitions and flavour decoupling in an $SU(3) + 2F$ web. Dotted segments indicate the deformation to the next configuration; the red dashed box marks the resulting pure $SU(3)$ web after both flavours are decoupled.

Figure 3.4 illustrates this with a $SU(3) + 2F$ web. Following the left-hand flavour D5 through the sequence, its finite segment first shrinks at the flop wall and re-emerges in the adjacent chamber. Continuing the deformation sends the flavour to infinite distance along its asymptotic leg, corresponding to the large-mass limit $m_f \to \pm\infty$ in which the hypermultiplet is integrated out. The right-hand sequence instead moves the second flavour D5 across the internal junction and onto the neighbouring compact face. At the transition the associated BPS state again becomes massless, while beyond it the web lies in a different Coulomb chamber with the same asymptotic charge data. Repeating the decoupling transition for this second flavour removes both hypermultiplets, leaving the pure $SU(3)$ web indicated in the final configuration.

3.3 Chern–Simons data and the $SU(N)_k + N_f F$ web family

The continuous parameters discussed above are encoded by distances and deformations of a web with fixed asymptotic charge data. The Chern–Simons level k, by contrast, is discrete and is encoded in the (p, q) charges of the external legs. Consequently, webs with the same local colour and flavour content can realise different $SU(N)_k$ theories when their asymptotic charge assignments differ. A convenient way to relate these charges to k is through real-mass flows. Integrating out a fundamental hypermultiplet of large mass shifts the Chern–Simons level by,

$$\Delta k = \frac{1}{2}\,\mathrm{sign}(m_f), \tag{3.9}$$

the parity-odd contribution of the massive fermion discussed in Chapter 2 [2, 23]. In the web, the same limit sends the corresponding flavour D5 to infinity along an asymptotic leg; which asymptotic direction is selected is fixed by the sign of m_f. Charge conservation then changes the external (p, q) data accordingly. For N_f fundamentals,

$$k_{\mathrm{pure}} = k + \frac{1}{2}\sum_{f=1}^{N_f}\mathrm{sign}(m_f). \tag{3.10}$$

Thus the asymptotic charges of a flavoured web may be fixed by the pure $SU(N)_{k_{\mathrm{pure}}}$ theory reached after decoupling the flavours, after which the flavour D5-branes are restored.

3.3.1 The standard web family

The standard $SU(N)_k + N_f F$ web family may be generated from the maximally flavoured configuration,

$$SU(N)_0 + (2N + 4)F,$$

which saturates the upper bound of Eq. 2.56. The remaining members of the family are obtained by giving fundamentals large real masses and integrating them out. Let n_+ flavours be removed with the sign that increases k, and n_- with the opposite sign. The resulting theory has

$$N_f = 2N + 4 - n_+ - n_-, \qquad k = \frac{1}{2}(n_+ - n_-). \tag{3.11}$$

It follows that,

$$\begin{aligned} N_f + 2|k| &= 2N + 4 - (n_+ + n_-) + |n_+ - n_-| \\ &= 2N + 4 - 2\min(n_+, n_-) \leq 2N + 4, \end{aligned}$$

and hence,

$$N_f + 2|k| \leq 2N + 4. \tag{3.12}$$

Thus, within this standard mass-deformation family, the maximal-flavour bound of the field-theory analysis is refined by the Chern–Simons level. This relation characterises the webs generated from the $SU(N)_0 + (2N+4)F$ configuration rather than all possible $SU(N)$ brane-web constructions; more general higher-Chern–Simons rank-two webs are known [48].

For $SU(3)$ the bound becomes $N_f + 2|k| \leq 10$. Combining it with the five-dimensional quantisation condition $k + N_f/2 \in \mathbb{Z}$ gives, up to charge conjugation $k \leftrightarrow -k$,

$$N_f = 5: \qquad k = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \tag{3.13}$$

and

$$N_f = 6: \qquad k = 0, 1, 2. \tag{3.14}$$

Precisely the six theories we have chosen to study.

3.4 Constructing the $SU(3) + N_f$ webs

The preceding sections have assembled the ingredients required to follow the theories of interest towards strong coupling, and these allow for many interesting investigations [48, 49]. For the present use, the web construction can be simplified significantly. It is sufficient to determine the asymptotic charge data without explicit consideration to the internal charge bookkeeping and retain the inverse gauge coupling as the final continuous deformation.

3.4.1 Encoding the Chern-Simons Level

The construction can be organised by first fixing the asymptotic charges of the corresponding pure theory and then restoring the required flavour D5-branes by reversing the mass-decoupling flows described above. For a pure $SU(N)$ web with N horizontal colour D5-branes, charge conservation relates the external charges above and below the colour stack. Writing the upper pair as $(p_L, 1)$ and $(p_R, 1)$, crossing the N colour branes fixes the lower pair to $(-p_L - N, -1)$ and $(N - p_R, -1)$. Hence, if Δ_{top} and Δ_{bottom} denote the differences between the corresponding p-charges, charge conservation imposes,

$$\Delta_{\text{top}} + \Delta_{\text{bottom}} = 2N,$$

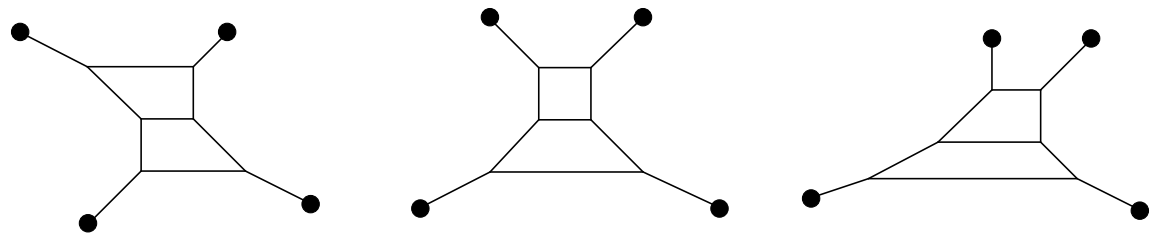
leaving one discrete asymmetry unfixed. The individual charge labels are also not unique. An $SL(2,\mathbb{Z})$ shear $(p, q) \mapsto (p + sq, q)$ preserves the horizontal D5 direction and shifts the absolute p-charges without changing Δ_{top} or Δ_{bottom}. Modulo this freedom, the remaining discrete choice is identified with the Chern–Simons level, and is encoded by the imbalance between the upper and lower charge separations,

$$\Delta_{\text{top}} = N - k, \qquad \Delta_{\text{bottom}} = N + k.$$

Thus $k = (\Delta_{\text{bottom}} - \Delta_{\text{top}})/2$, so the Chern–Simons data are carried by the complete asymptotic charge configuration rather than by any single external leg. For the family shown in Fig. 3.5, we use the shear freedom to keep the upper- and lower-right legs fixed at $(1, 1)$ and $(N - 1, -1)$. The required asymmetry is then carried entirely by the left pair,

$$(k - N + 1, 1), \qquad (-k - 1, -1).$$

Specialising to $N = 3$ gives the pure $SU(3)_k$ webs displayed for $k = 0, 1, 2$.



$SU(3)_0$ $SU(3)_1$ $SU(3)_2$

Figure 3.5: Pure $SU(3)$ webs with varying Chern–Simons level, encoded in the asymptotic legs.

3.4.2 Collapsing to the Coulomb origin

We illustrate the construction with the non-trivial example $SU(3)_{\frac{3}{2}} + 5F$. Choosing three flavours to be decoupled with positive mass and two with negative mass gives $k_{\text{pure}} = 2$, so the finite-coupling web may be reconstructed by starting from a pure $SU(3)_2$ web and reversing these five mass flows.

Among the equivalent charge presentations of the pure theory, we choose the frame in which the upper-right external leg is vertical. This choice is particularly useful for the subsequent seven-brane rearrangements, while leaving the underlying gauge theory unchanged. Restoring the three and two flavour D5-branes on the corresponding sides then gives the required $SU(3)_{\frac{3}{2}} + 5F$ web.

We now remove the continuous deformations that are not part of the ultraviolet point of interest. The real masses are set to zero, $m_f = 0$, so the corresponding flavour legs become coincident, while $\phi_1 = \phi_2 = 0$ places the theory at the origin of its Coulomb branch and collapses the two compact Coulomb faces.

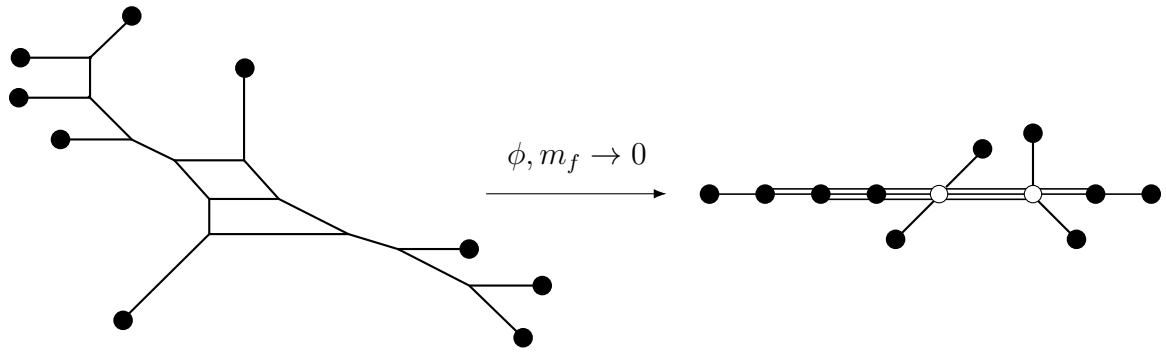


Figure 3.6: Finite-coupling Coulomb origin of $SU(3)_{\frac{3}{2}} + 5F$, the white circles indicate an unresolved junction.

3.4.3 Hanany–Witten transitions

From the Coulomb origin, the ultraviolet limit is reached by sending $m_0 \to 0$. In the present web frame, however, the ordering of the external five-branes and their terminating seven-branes is not adapted to this collapse. Before taking the limit, we therefore pass to an equivalent presentation using Hanany–Witten transitions. A $[P, Q]$ seven-brane carries an $SL(2, \mathbb{Z})$ monodromy. When its branch cut is swept across a five-brane of charge (p, q), the charge label transforms as

$$\begin{pmatrix} p \\ q \end{pmatrix} \longmapsto K_{[P,Q]} \begin{pmatrix} p \\ q \end{pmatrix}, \qquad K_{[P,Q]} = \begin{pmatrix} 1 + PQ & -P^2 \\ Q^2 & 1 - PQ \end{pmatrix}, \tag{3.15}$$

with $K^{-1}_{[P,Q]}$ for the opposite crossing [45]. The branch-cut placement is conventional: changing it alters the charge presentation, not the underlying seven-brane configuration. When the seven-brane itself is moved through the web, the monodromy-induced change of charge must be accompanied by local five-brane charge conservation. This creates or annihilates

$$|pQ - qP|. \tag{3.16}$$

five-brane prongs ending on the $[P, Q]$ seven-brane, with their orientation fixed by the junction charges. This is the generalized Hanany–Witten transition [41, 45]; an example is given in Figure 3.7.

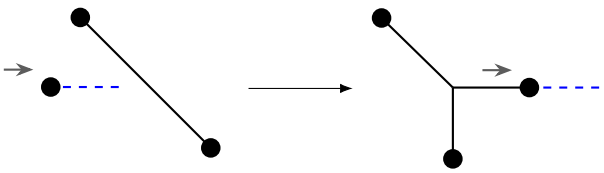


Figure 3.7: Elementary Hanany–Witten transition: a $[1, 0]$ D7 crosses a $(1, -1)$ five-brane. In the orientation shown, the latter transforms to $(0, -1)$ and a single $(1, 0)$ D5 prong is created, since $|pQ - qP| = 1$, giving a charge-conserving trivalent junction.

3.4.4 Rearrangement and the strong-coupling limit

For the $SU(3)_{\frac{3}{2}} + 5F$ example, we start from the finite-coupling origin constructed above. The top seven-brane terminating the diagonal external leg is first displaced outward along that leg, which leaves the five-dimensional theory unchanged but opens the path for the neighbouring vertical seven-brane. Moving the latter through the diagonal branch then produces a Hanany–Witten transition: the monodromy changes the charge of the original diagonal leg so that it becomes a horizontal D5 segment, while the required additional prong is fixed by charge conservation. In this presentation the newly horizontal D5 can coincide with the flavour D5 stack below as $m_0 \to 0$, so the strong-coupling degeneration is obtained directly from the rearranged web.

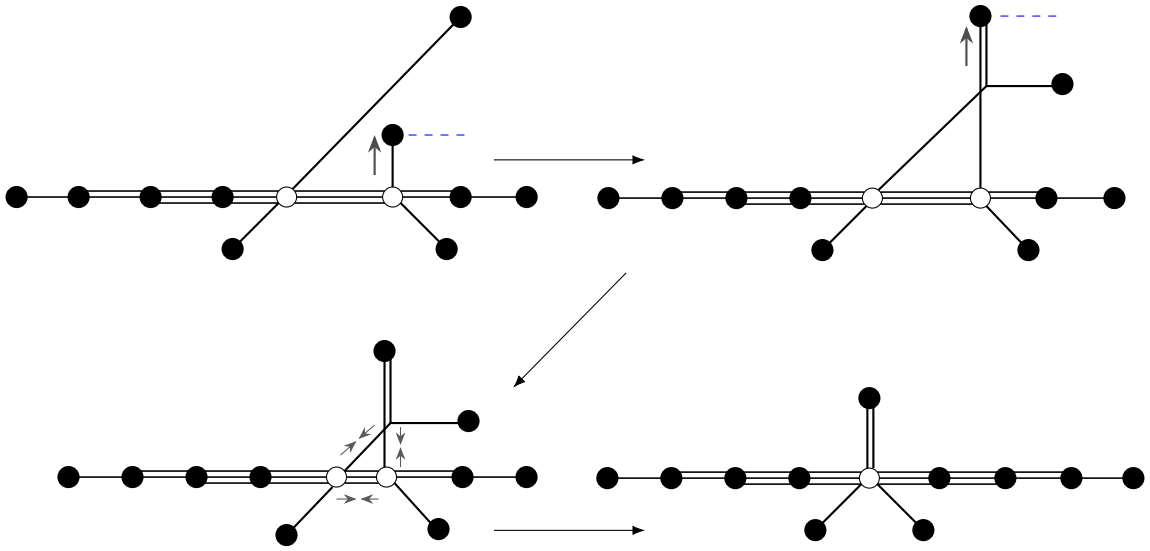


Figure 3.8: Hanany–Witten rearrangement of the $SU(3)_{\frac{3}{2}} + 5F$ web. The diagonal seven-brane is first displaced along its external leg, after which the neighbouring seven-brane is moved through it. The induced monodromy converts the relevant external segment into a D5, which joins the lower flavour stack as $m_0 \to 0$.

The $SU(3)_{\frac{5}{2}} + 5F$ theory provides a second, less trivial example. Here we choose a different asymptotic charge frame for the pure parent before restoring the flavours, arranged so that the two lower external legs of the finite-coupling web are diagonal. This choice is particularly suited to the strong-coupling limit. The seven-branes terminating these diagonal legs are moved through the neighbouring flavour stacks. Their monodromies act on the attached D5s, and the accompanying Hanany–Witten transitions reorganise the flavour legs into vertical five-brane stacks. Once both sides have been transformed in this way, the remaining m_0 separation can be collapsed, and the vertical stacks meet naturally at the common strong-coupling junction.

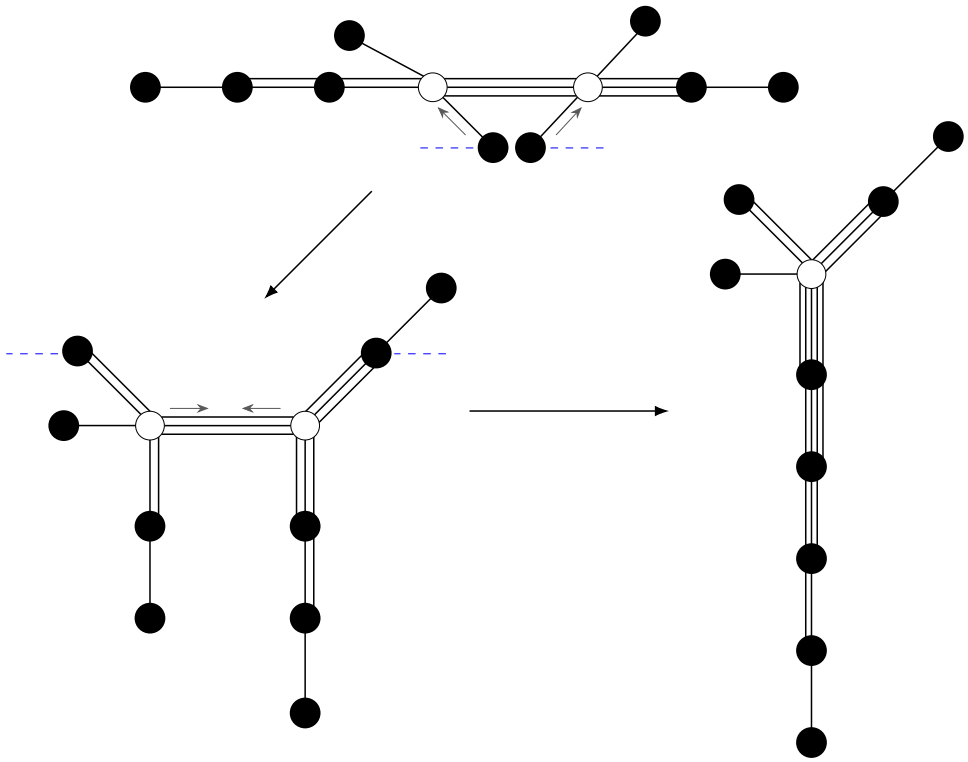


Figure 3.9: Hanany–Witten rearrangement of the $SU(3)_{\frac{5}{2}} + 5F$ web. Moving the two terminating seven-branes through the adjacent flavour stacks reorganises the corresponding five-branes into vertical stacks, which meet at the strong-coupling junction when $m_0 \to 0$.

At $m_f = \phi_i = m_0 = 0$, all continuous deformations of the web have been removed. In particular, the compact-face areas vanish, so the corresponding magnetic-string tensions $T_i = \partial\mathcal{F}/\partial\phi_i$ vanish simultaneously. The resulting configuration therefore lies at the infinite-coupling Coulomb origin, where the perturbative $SU(3)$ gauge description ceases to be valid but the brane web continues to encode the ultraviolet theory [17, 45].

3.4.5 Remaining $SU(3)$ webs

The preceding examples contain the ingredients required for the remaining theories. The resulting finite- and infinite-coupling webs are collected in Figure 3.10. These configurations reproduce the corresponding five-brane constructions studied in the literature [49].

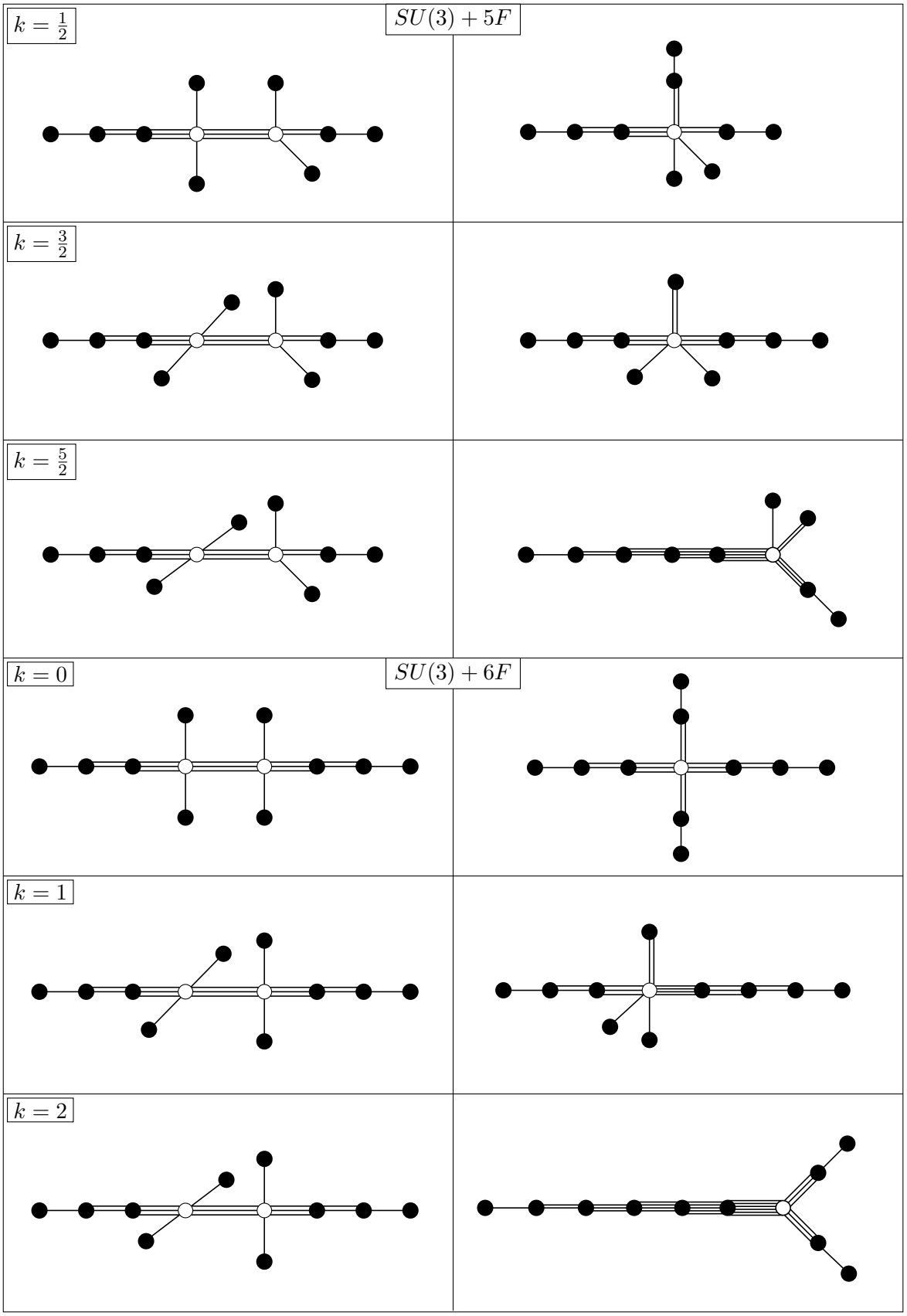


Figure 3.10: Finite- and infinite-coupling five-brane webs for the six $SU(3)_k + N_f F$ theories considered here. Each row pairs the finite-coupling origin (left) with the corresponding $m_0 \to 0$ web (right). The upper three rows show $N_f = 5$ with $k = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}$, while the lower three show $N_f = 6$ with $k = 0, 1, 2$. White circles are unresolved junctions.

4 Magnetic Quivers for HyperKähler Constructions

Brane webs provide a physical realisation of five-dimensional theories at infinite coupling, but identifying the geometry of their Higgs branches required tools developed in other theories with eight supercharges. A central lesson of these developments was that moduli spaces arising from distinct physical constructions can be isomorphic, suggesting a strategy for strongly coupled problems: realise the same geometry in an auxiliary theory where it is more accessible.

Seiberg's rank-one E_n fixed points provided an early example with their infinite-coupling Higgs branches identified with E_n instanton moduli spaces [1]. In three dimensions, Intriligator and Seiberg introduced $3d\ \mathcal{N} = 4$ mirror symmetry, which exchanges Higgs and Coulomb branches, and exhibited examples involving ADE instanton moduli spaces [40]. Chalmers and Hanany related quantum Coulomb branches to BPS monopole moduli spaces [50], while Hanany and Witten embedded these ideas in Type IIB string theory, giving a systematic brane realisation of three-dimensional gauge theories and a string-theoretic explanation of mirror symmetry [41]. These results showed that Higgs branches, Coulomb branches and instanton or monopole moduli spaces can provide distinct constructions of the same hyperkähler geometries.

Quivers had already become established as a convenient language for gauge theories and their moduli [51, 52]. Their usefulness for $3d\ \mathcal{N} = 4$ theories emerged from difficulty with Coulomb branch, where determining its full geometry traditionally required a non-trivial analysis of the metric's quantum corrections or the use of a known mirror description [42]. A major simplification came with the development of monopole-operator methods by Cremonesi, Hanany and collaborators, together with algebraic-geometric formulations of the Coulomb branch [42, 43, 53–55]. Rather than reconstructing the quantum-corrected metric, these methods made the complex geometry of the Coulomb branch accessible directly from the gauge and matter data encoded by the quiver. This framework was then used by Cremonesi, Ferlito, Hanany and Mekareeya to realise instanton moduli spaces for arbitrary simple groups as Coulomb branches of three-dimensional quivers, including the exceptional cases for which the conventional ADHM construction was unavailable [56].

These ideas were then brought to 5d $\mathcal{N} = 1$; Ferlito, Hanany, Mekareeya and Zafrir used known ultraviolet symmetry enhancement and structural properties of $3d\ \mathcal{N} = 4$ quivers to identify the required quivers for the Higgs branches of the exeptional sequence[33]. Cabrera, Hanany and Yagi subsequently reversed this logic and gave a method for the auxiliary quiver to be derived directly from the five-brane configuration [49].

This chapter follows this development. We first review the moduli identifications that naturally motivate the use of $3d\ \mathcal{N} = 4$ Coulomb branches, then turn to their direct extraction from five-brane webs and apply the resulting constructions to the strong-coupling webs of the $SU(3)$.

4.1 $3d\,\mathcal{N}=4$ Theory

In three dimensions, eight real supercharges have an R-symmetry whose simply connected cover is [57],

$$\mathrm{Spin}(4)_R \simeq SU(2)_H \times SU(2)_C.$$

A hypermultiplet contains four real scalars transforming under $SU(2)_H$, while the vector multiplet contains the gauge field and three real adjoint scalars forming a triplet of $SU(2)_C$. An essential additional ingredient in the three-dimensional theory is provided by monopoles: local disorder operators carrying quantised magnetic charge which capture the non-perturbative information on the quantum Coulomb branch.

4.1.1 Monopole charge and gauge fields

A local monopole operator is defined by prescribing a magnetic singularity at its insertion point [58]. On a small sphere S^2 surrounding the insertion, a configuration with non-zero magnetic flux cannot be described by a single smooth gauge potential. Instead one introduces potentials A_N and A_S on northern and southern patches, related on their overlap S^1 by a gauge transformation $g_{NS} : S^1 \to G$ [42, 59].

For a Dirac-type monopole singularity, the magnetic charge specifies how this transition function winds through the gauge group as one moves around the equator. This is described by a one-parameter subgroup. Since the image of $U(1)$ is Abelian, for compact G it is contained in a maximal torus $T \subset G$, up to gauge conjugation. We may therefore write,

$$\lambda_m : U(1) \longrightarrow T, \qquad \lambda_m(e^{i\phi}) = e^{i\phi m}, \tag{4.1}$$

where $m \in \mathfrak{t}$ determines how the circle surrounding the monopole winds through the Cartan directions of the gauge group. For $G = U(N)$, for example,

$$\lambda_m(e^{i\phi}) = \mathrm{diag}(e^{im_1\phi}, \ldots, e^{im_N\phi}). \tag{4.2}$$

Since ϕ and $\phi + 2\pi$ describe the same point on the equator, the map must be single-valued, requiring $e^{2\pi i m} = 1_G$ and hence $m_i \in \mathbb{Z}$ for $U(N)$. This is the non-Abelian form of magnetic charge quantisation.

The same transition function determines how charged fields are glued between the two patches. If Ψ transforms in a representation R of G, then on the equator $\Psi_N = R(\lambda_m)\Psi_S$. Since λ_m takes values only in the Abelian torus T, only the restriction of R to T is relevant. Its commuting action decomposes the representation into simultaneous eigenspaces, $V_R = \bigoplus_\rho V_\rho$, labelled by weights ρ. On a component $v_\rho \in V_\rho$, a torus transformation therefore acts only by a phase,

$$R(t)v_\rho = \chi_\rho(t)v_\rho, \qquad t \in T. \tag{4.3}$$

The map $\chi_\rho : T \to U(1)$ is the *character* associated with the weight ρ: it assigns to each Cartan gauge transformation the phase acquired by that component.

At this point the electric and magnetic constructions take a parallel form. Homomorphisms $\lambda : U(1) \to T$ are called *cocharacters*, and the allowed magnetic charges therefore lie in the cocharacter lattice,

$$X_*(T) = \mathrm{Hom}(U(1), T). \tag{4.4}$$

Similarly, the character homomorphisms define the *character lattice* $X^*(T) = \mathrm{Hom}(T, U(1))$, with the weights ρ labelling the electric charges associated with each such transformation law.

This parallel between the electric and magnetic lattices motivates the GNO dual description. The dual group $G^\vee$ is defined by exchanging the corresponding root data, so that the cocharacters of G become ordinary characters, or weights, of $G^\vee$,

$$X^*(T_{G^\vee}) \simeq X_*(T_G), \tag{4.5}$$

with roots and coroots exchanged. Thus the magnetic charge m of G may equivalently be regarded as a weight of $G^\vee$ [42, 60].

With these actions specified, we can now determine how the monopole winding acts on a charged field component. As one moves around the equator, the monopole specifies the Cartan gauge transformation through λ_m, while the character χ_ρ specifies how that transformation acts on a component of weight ρ. The two therefore compose as,

$$U(1)_{\mathrm{eq}} \xrightarrow{\lambda_m} T \xrightarrow{\chi_\rho} U(1)_{\mathrm{phase}}, \qquad (\chi_\rho \circ \lambda_m)(e^{i\phi}) = e^{i\rho(m)\phi}. \tag{4.6}$$

The composition is itself a homomorphism $U(1) \to U(1)$, so it is characterised by an integer winding number $\rho(m) \in \mathbb{Z}$. This defines the natural character–cocharacter pairing $X^*(T) \times X_*(T) \to \mathbb{Z}$. In a basis with $\rho = (q_1, \ldots, q_r)$ and $m = (m_1, \ldots, m_r)$, it is simply $\rho(m) = \sum_i q_i m_i$.

The same integer has a direct field-theoretic interpretation. On the weight space V_ρ, the non-Abelian covariant derivative reduces to the Abelian coupling $D\Psi_\rho = (d - i\rho(A))\Psi_\rho$. The magnetic flux seen by this component is therefore,

$$\frac{1}{2\pi}\int_{S^2} \rho(F) = \rho(m). \tag{4.7}$$

Thus m specifies the magnetic background, ρ the electric charge of the field component, and $\rho(m)$ the effective magnetic flux experienced by that component. Tthe gauge field itself transforms in the adjoint representation. Its non-zero weights are the *roots* α, defined by $[H, E_\alpha] = \alpha(H)E_\alpha$. The off-diagonal gauge-boson components therefore carry electric charges α under the Cartan and experience the same monopole background through the pairing $\alpha(m)$.

4.1.2 BPS completion of the monopole sector

While the magnetic charge m introduced above specifies an allowed disorder sector, the operators relevant to the supersymmetric Coulomb branch are the BPS insertions within these sectors. To describe a holomorphic Coulomb-branch sector, choose an $\mathcal{N} = 2 \subset \mathcal{N} = 4$ subalgebra; the $\mathcal{N} = 4$ vector multiplet decomposes into an $\mathcal{N} = 2$ vector multiplet V, containing A_μ and one real adjoint scalar σ, and an adjoint chiral multiplet Φ.

The scalar singularity accompanying the magnetic boundary condition can be motivated from the corresponding Bogomolny equation. As in the instanton analysis of Sec. 2.6.1, the corresponding first-order equation can be exposed by completing the square with the bosonic gauge and scalar terms. Writing $B_i = \frac{1}{2}\epsilon_{ijk}F_{jk}$, one has schematically, up to the overall normalisation and the orientation of the inner boundary B_ϵ,

$$\int_{\mathbb{R}^3\setminus B_\epsilon} \mathrm{Tr}(B_iB_i + D_i\sigma D_i\sigma) = \int_{\mathbb{R}^3\setminus B_\epsilon} \mathrm{Tr}(B_i \mp D_i\sigma)^2 \pm 2\int_{S^2_\epsilon} \mathrm{Tr}(\sigma B_i)\, dS_i. \tag{4.8}$$

For fixed monopole boundary data, the square vanishes when,

$$B_i = \pm D_i\sigma, \qquad \Longleftrightarrow \qquad F = \pm \star_3 D\sigma, \tag{4.9}$$

which is the Bogomolny equation [15]. The same condition follows from setting the appropriate gaugino variation to zero, and hence characterises the monopole boundary condition preserving the chosen supercharges [42, 57, 61].

For magnetic charge m, the radial magnetic field behaves as $B_r \sim m/(2r^2)$. Since the leading singular fields lie in the Cartan and therefore commute, the radial Bogomolny equation reduces to $\partial_r\sigma = \pm m/(2r^2)$, giving,

$$\sigma(r) \sim \mp\frac{m}{2r}, \qquad r \to 0. \tag{4.10}$$

Thus the singular behaviour of σ is not independent boundary data: it is fixed by supersymmetry once the magnetic sector m and the preserved supercharges have been chosen. The resulting insertion defines a BPS monopole operator V_m.

4.1.3 Radial quantisation and the protected dimension

At a superconformal fixed point, radial quantisation maps a local monopole operator to a state on the cylinder. Setting $r = e^\tau$, the punctured space is conformally equivalent to $S^2 \times \mathbb{R}_\tau$,

$$\mathbb{R}^3 \setminus \{0\} \simeq_{\mathrm{conf}} S^2 \times \mathbb{R}_\tau, \qquad V_m(0) \longleftrightarrow |m\rangle_{S^2}, \qquad D \longleftrightarrow H_{S^2}. \tag{4.11}$$

The conformal dimension of V_m is therefore the energy of the corresponding state on the unit S^2. The monopole singularity becomes a fixed magnetic flux through the sphere, while under the Weyl map the scalar background rescales as,

$$\sigma_{\mathrm{cyl}} = r\,\sigma_{\mathbb{R}^3} \longrightarrow \mp\frac{m}{2}. \tag{4.12}$$

This gives the radial-quantisation description of the BPS monopole background [57, 61]. The chosen $\mathcal{N} = 2 \subset \mathcal{N} = 4$ subalgebra supplies a $U(1)_R$ grading, which for Coulomb-branch chiral operators is identified with the Cartan of $SU(2)_C$. Since the bare BPS monopole is the bottom component of an $\mathcal{N} = 2$ chiral multiplet, the superconformal shortening condition gives,

$$\Delta = R_{\mathcal{N}=2}. \tag{4.13}$$

Its conformal dimension may therefore be determined from the induced R-charge of the magnetic-sector vacuum [25]. The pairings introduced above determine the charged spectrum in this background. A matter component of gauge weight ρ experiences flux $q = \rho(m)$, while an adjoint

component associated with a root α experiences $q = \alpha(m)$. Quantising the charged fields on S^2 produces a vacuum-charge shift proportional to $|q|$, arising from the spectral asymmetry of the fermionic modes in the monopole background [57, 61]. For an $\mathcal{N} = 2$ chiral multiplet of R-charge r_Φ in a representation R_Φ, the resulting contribution is [62],

$$\delta R_\Phi(m) = \frac{1 - r_\Phi}{2} \sum_{\rho \in R_\Phi} |\rho(m)|. \tag{4.14}$$

For an $\mathcal{N} = 4$ hypermultiplet, the two $\mathcal{N} = 2$ chiral multiplets transform in conjugate representations R and $\bar{R}$ and both have $r_\Phi = \frac{1}{2}$. Since the two representations give the same absolute-value sum, while the adjoint chiral multiplet in the $\mathcal{N} = 4$ vector multiplet has $r_\Phi = 1$, the hypermultiplet and vector-multiplet contributions are respectively,

$$\Delta_{\text{hyp}}(m) = \frac{1}{2} \sum_{\rho \in R} |\rho(m)|, \qquad \Delta_{\text{vec}}(m) = -\frac{1}{2} \sum_{\alpha \in \Delta} |\alpha(m)| = - \sum_{\alpha \in \Delta_+} |\alpha(m)|.$$

Summing over the hypermultiplet representations R_i therefore gives the protected dimension of the bare monopole [42],

$$\Delta(m) = - \sum_{\alpha \in \Delta_+} |\alpha(m)| + \frac{1}{2} \sum_i \sum_{\rho \in R_i} |\rho(m)|. \tag{4.15}$$

Thus the same pairings $\rho(m)$ and $\alpha(m)$ that describe the response of charged fields to the monopole background also determine its protected superconformal grading. Their systematic use in counting Coulomb-branch operators will be developed through the monopole formula in the following chapter.

4.1.4 Quiver gauge theories and auxiliary Coulomb branches

The gauge and matter data entering the monopole construction can be encoded compactly by a *quiver*. A circular node labelled N denotes a $U(N)$ gauge factor, an edge joining two gauge nodes denotes a bifundamental hypermultiplet, and a square node denotes an ungauged flavour symmetry, with an adjoining edge supplying fundamental matter to the neighbouring gauge group. Multiple edges denote the corresponding multiplicity of matter. We denote the Coulomb branch of the theory specified by a quiver Q by $\mathcal{C}(Q)$.

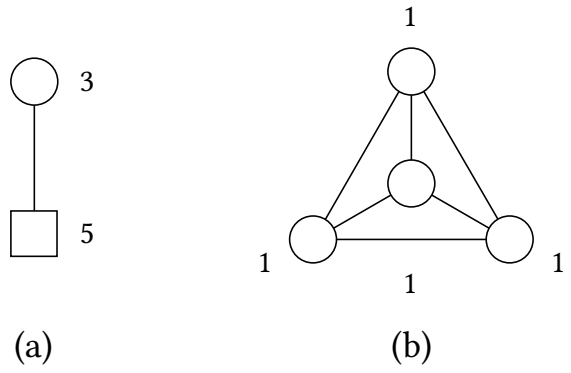


Figure 4.1: Example quivers; (a) shows $U(3) + 5F$ while (b) gives $U(1)^4$ with each group sharing bifundamental matter.

Mirror symmetry supplied early examples in which the Coulomb branch of a $3d\ \mathcal{N}=4$ quiver reproduced a moduli space that was more naturally realised elsewhere [40, 41, 63]. The later monopole-operator description made this correspondence directly computable from the quiver itself, allowing Coulomb branches to be treated as explicit constructions of hyperkähler varieties. A particularly important application is the construction of instanton moduli spaces. Cremonesi, Ferlito, Hanany and Mekareeya showed that the reduced one-instanton moduli space of a simple group G can be realised as the Coulomb branch of an appropriate $3d\ \mathcal{N}=4$ quiver [56]. For simply-laced G, the quiver is obtained from the affine Dynkin diagram of G, with unitary gauge ranks given by the affine comarks and the decoupled diagonal $U(1)$ removed:

$$\mathcal{C}(Q_{\widehat{G}}) \simeq \widetilde{\mathcal{M}}_1(G). \tag{4.16}$$

This immediately connects the three-dimensional construction to the rank-one five-dimensional theories discussed above. Their infinite-coupling Higgs branches satisfy,

$$\mathcal{H}^{5d}_{E_n,\infty} \simeq \widetilde{\mathcal{M}}_1(E_n),$$

and hence admit the equivalent description,

$$\mathcal{H}^{5d}_{E_n,\infty} \simeq \widetilde{\mathcal{M}}_1(E_n) \simeq \mathcal{C}(Q_{\widehat{E_n}}). \tag{4.17}$$

Thus the strongly coupled $5d$ Higgs branch can be encoded by an auxiliary $3d\ \mathcal{N}=4$ quiver whose Coulomb branch realises the same hyperkähler geometry. The remaining question is how such a quiver can be identified when the corresponding moduli space is not already known.

4.2 From inferred magnetic quivers to direct web extraction

4.2.1 Balance and Global Symmetry

The monopole dimension formula provides a direct way to extract global-symmetry information from a unitary quiver. Consider a gauge factor $U(N_i)$ and the elementary monopole $m_i = (1, 0, \ldots, 0)$, with vanishing magnetic charge on the remaining nodes. From the viewpoint of this node, each fundamental hypermultiplet contributes one flavour, while a bifundamental joining $U(N_i)$ to $U(N_j)$ contributes N_j. The effective number of flavours seen by the node is therefore,

$$N_{f,i} = F_i + \sum_j n_{ij} N_j. \tag{4.18}$$

For the elementary monopole, the $N_i - 1$ positive roots which pair non-trivially with m_i contribute $-(N_i - 1)$, while the matter contributes $N_{f,i}/2$. Hence,

$$\Delta_i = \frac{N_{f,i}}{2} - (N_i - 1) = 1 + \frac{e_i}{2}, \qquad e_i \equiv N_{f,i} - 2N_i. \tag{4.19}$$

The *excess* e_i therefore measures how far the elementary monopole lies from dimension one. A node is *balanced* when $e_i = 0$, so that its elementary monopoles have $\Delta = 1$.

The same quantity underlies the good–ugly–bad classification of Gaiotto and Witten [64]. For the unitary quivers considered here, non-negative excess at every node gives the good class; $e = -1$ produces an elementary monopole at the scalar unitarity bound $\Delta = \frac{1}{2}$, characteristic of an ugly theory with a free sector, while more negative excess signals a bad ultraviolet description. The balanced case is distinguished by the appearance of dimension-one monopoles.

Each $U(N_i)$ gauge factor possesses a conserved *topological* current,

$$j_{J_i} = \frac{1}{2\pi} * \operatorname{Tr} F_i, \qquad d{*}j_{J_i} = \frac{1}{2\pi} \operatorname{Tr}(DF_i) = 0. \tag{4.20}$$

With conservation following from the Bianchi identity $DF_i = 0$, defining a global $U(1)_{J_i}$ symmetry. Perturbative fields are neutral under this symmetry, whereas a monopole operator of charge $m_i = \operatorname{diag}(m_{i,1}, \ldots, m_{i,N_i})$ carries,

$$J_i(m) = \frac{1}{2\pi} \int_{S^2} \operatorname{Tr} F_i = \sum_a m_{i,a}. \tag{4.21}$$

The unitary factors of the quiver consequently provide a manifest Abelian topological symmetry acting on the Coulomb branch [56].

At the infrared fixed point these Abelian symmetries may enhance. The importance of the balanced condition is that it places the elementary monopole at $\Delta = 1$. For a $3d$ $\mathcal{N} = 4$ Coulomb-branch chiral primary, this is precisely the quantum number of the moment-map operator in a flavour-current multiplet, whose superconformal descendants include a conserved current [64]. Each such monopole therefore supplies a generator of the infrared flavour symmetry E_m, with the independent topological $U(1)_{J_i}$ symmetries furnishing the commuting Cartan generators H_i, satisfying $[H_i, E_m] = J_i(m)E_m$. Its topological-charge vector may then be identified with a root of the enhanced symmetry, with the elementary monopoles of balanced nodes provide the simple roots, and the dimension-one monopoles supported on connected combinations of nodes supply the remaining roots.

The edges determine which sums of simple roots occur. If two balanced nodes i and j are joined by a bifundamental, a monopole carrying one unit of flux on both would naively receive the sum of the two elementary dimensions, but the shared bifundamental now sees one pair of equal magnetic charges and reduces the total contribution by one. Thus,

$$\Delta(m_i + m_j) = 1 + 1 - 1 = 1, \tag{4.22}$$

producing the root $\alpha_i + \alpha_j$. If the nodes are disconnected, this cancellation does not occur and the combined monopole instead has $\Delta = 2$, so no corresponding current appears. Iterating this construction reproduces the root system, and for a simply-laced quiver the connected balanced subquiver can therefore be read directly as the Dynkin diagram of the non-Abelian Coulomb-branch symmetry [33, 64].

Conversely, the balance condition can be used as an inference rule. This was the logic of the early magnetic-quiver constructions: the expected non-Abelian symmetry fixes the balanced Dynkin subquiver, while additional flavour or unbalanced-node data fix the remaining ranks [33]. As a simple example, suppose the desired symmetry is $A_2 \simeq SU(3)$. Its Dynkin diagram requires two connected balanced gauge nodes of ranks N_1 and N_2. Without additional matter, the balance equations are homogeneous and admit only the trivial solution, so a non-trivial finite-rank quiver must be framed. Attaching F fundamental hypermultiplets to one end gives,

$$2N_1 = N_2, \qquad 2N_2 = N_1 + F.$$

Hence $F = 3N_1$, and the primitive positive-integer solution is $N_1 = 1$, $N_2 = 2$, $F = 3$, giving the quiver $U(1) - U(2) - [3]$ shown in Fig. 4.2. The three flavours are therefore the minimal framing compatible with a non-trivial balanced A_2 quiver. More generally, once the balanced graph and its framing or unbalanced attachments are specified, the balance equations fix the rank vector. For connected unframed quivers built only from bifundamental matter, the common diagonal gauge $U(1)$ acts trivially and is removed, so u unbalanced nodes leave $u-1$ independent Abelian symmetry factors [33].

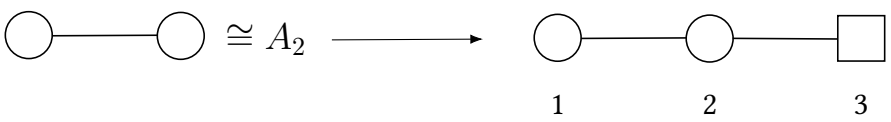


Figure 4.2: Inferring a balanced A_2 quiver from its symmetry. The A_2 Dynkin diagram fixes two connected balanced gauge nodes; requiring a non-trivial positive-rank solution framed from one side introduces the minimal flavour $[3]$, and the balance equations then uniquely determine the ranks 1 and 2, giving $U(1) - U(2) - [3]$.

4.2.2 Early inference and its limitation

This reverse use of balance was employed by Ferlito, Hanany, Mekareeya and Zafrir to construct magnetic quivers for families of $5d$ $\mathcal{N} = 1$ theories [33]. Starting from the known rank-one E_n theories, they selected appropriate maximal A- and D-type subalgebras and generalised the corresponding symmetry patterns to higher-rank $SU(n)$ sequences. The expected ultraviolet symmetry determined the balanced Dynkin subquiver and the number of residual $U(1)$ factors fixed the number of unbalanced nodes. Their attachment points were inherited from the rank-one exceptional diagram, after which the balance equations fixed the gauge ranks. For the families considered, these inputs determined the candidate quiver uniquely. The symmetry-based construction therefore remained indirect: the ultraviolet symmetry and its relation to the rank-one seed had to be known before the quiver could be inferred. The subsequent development was to extract this information directly from the strong-coupling five-brane web.

4.3 Direct extraction of a magnetic quiver from the five-brane web

The preceding chapter gave the Coulomb moduli a direct geometric interpretation as deformations of the finite internal structure of the web with its asymptotic data held fixed. What remained to be identified was how the Higgs moduli are encoded in the same construction. The relevant geometric operation is different. At special loci, and in particular at the strong-coupling degeneration, the web can separate into distinct supersymmetric subwebs whose relative motions preserve the common supersymmetry. These relative motions furnish Higgs-branch directions. The problem is therefore to determine how the combinatorial data of such decompositions specifies the auxiliary quiver.

Cabrera, Hanany and Yagi developed this correspondence by comparing strong-coupling five-brane decompositions with magnetic quivers whose Coulomb branches were already known [49]. Their strategy was to use the exceptional sequences previously inferred from ultraviolet symmetry and balance, and test how those quivers were encoded directly in the corresponding brane webs [33]. In the rank-one cases, the E_n enhancement is realised at infinite coupling by $SU(2) + (n-1)F$. Proceeding through the E_3, E_4, E_5 and E_6 examples, together with pure $SU(N_c)$ Yang–Mills, each case isolates a further ingredient of the web-to-quiver dictionary before the observations are assembled into a general prescription.

4.3.1 The E_3 example: subwebs and stable intersection

The principle rules of the web-to-quiver map start with the E_3 enhancement, realised by $SU(2)+2F$. At infinite coupling its Higgs branch is the union of two cones,

$$\mathcal{H}_\infty = \overline{\mathcal{O}}_{\min}(A_2) \cup \overline{\mathcal{O}}_{\min}(A_1), \tag{4.23}$$

and the strong-coupling web admits two inequivalent maximal decompositions into independently movable subwebs [49]. Let the k distinct subwebs be assigned a rank-one $U(1)$ gauge node in the magnetic quiver; only their relative positions correspond to Higgs-branch moduli, therefore a common translation of all subwebs is a decoupled centre-of-mass motion and a common diagonal $U(1) \subset U(1)^k$ is removed from the magnetic-quiver gauge group. The resulting Coulomb branch therefore has quaternionic dimension $k-1$. In Figure 4.3, the three-subweb decomposition gives a two-dimensional component, while the two-subweb decomposition gives a one-dimensional component, in agreement with the two cones of the E_3 Higgs branch.

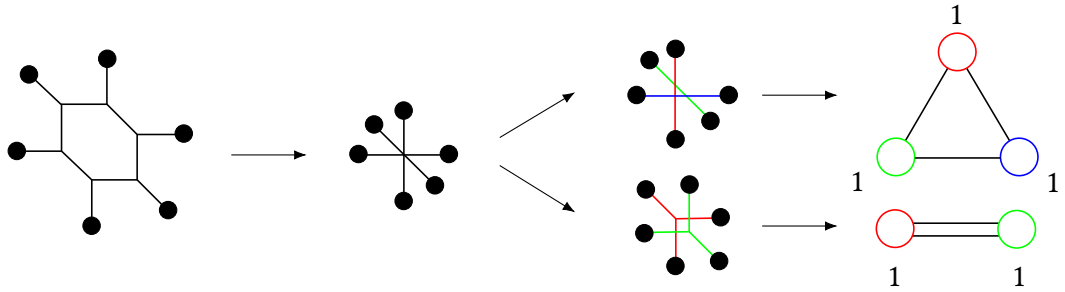


Figure 4.3: The E_3 example isolates the basic node and edge rules: independently movable subwebs produce magnetic-quiver nodes, while their stable intersections determine bifundamental multiplicities.

The remaining information is the adjacency between these nodes, determined by the intersections of the corresponding subwebs. For two transverse five-brane segments carrying charges $v_1 = (p_1, q_1)$ and $v_2 = (p_2, q_2)$, the local intersection multiplicity is,

$$I(v_1, v_2) = |p_1 q_2 - p_2 q_1| \,. \tag{4.24}$$

In the three-subweb decomposition, the constituents carry charges $(1, 0)$, $(0, 1)$ and $(1, -1)$, so every pair has intersection multiplicity one. The corresponding three $U(1)$ nodes are therefore pairwise connected by a single bifundamental hypermultiplet, reproducing the affine-A_2 quiver whose Coulomb branch is $\overline{\mathcal{O}}_{\min}(A_2)$ [49]. The second decomposition requires a more general notion of intersection. Complete subwebs may overlap or meet non-transversely, so the number of crossings visible in a particular presentation of the web is not invariant. Instead the two subwebs are displaced slightly until their intersections are transverse, after which the local intersection multiplicities are summed. This defines the *stable intersection*,

$$SI(A, B) = \sum_{x \in A \cap B} I_x(A, B), \tag{4.25}$$

which is independent of the generic displacement used to resolve the intersection. For the two subwebs in the second E_3 decomposition one finds,

$$SI(A, B) = 2.$$

The corresponding two $U(1)$ nodes are therefore joined by two bifundamental hypermultiplets, giving the affine-A_1 quiver and hence the second component $\overline{\mathcal{O}}_{\min}(A_1)$. The link multiplicity may be understood physically in terms of D3-branes stretched between the subwebs, with each independent intersection contributing a bifundamental degree of freedom [49].

However, not every subdivision of the web corresponds to an allowed Higgs motion. Each constituent must itself define a supersymmetric brane configuration, requiring both charge conservation at its junctions and the s-rule. In the original Hanany–Witten setting, more than one D3-brane suspended between the same NS5- and D5-brane is incompatible with supersymmetry [41, 65]. For general (p, q) webs this condition has an $SL(2, \mathbb{Z})$-covariant extension: between a $[P, Q]$ seven-brane and a (p, q) five-brane, the allowed number of suspended (P, Q) five-branes is controlled by the intersection number $|pQ - Pq|$, as follows from the associated Hanany–Witten brane-creation effect [65, 66]. A subdivision violating these conditions does not define independent supersymmetric subwebs; the offending pieces must instead remain part of a larger constituent. A decomposition is maximal when none of its constituents can be further split into admissible independently movable subwebs.

4.3.2 The E_4 example: seven-brane endpoint contributions

The E_4 theory, $SU(2)$ with $N_f = 3$, shows that stable intersection alone is not sufficient. Its strong-coupling web has a unique maximal decomposition, but distinct subwebs can terminate on the same seven-brane. In this situation the planar stable intersection does not by itself reproduce the known affine-A_4 quiver.

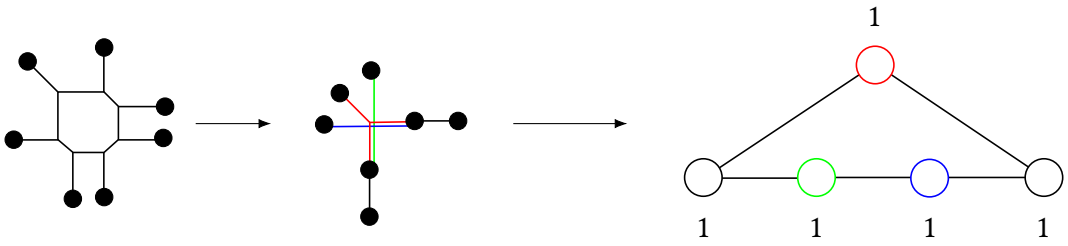


Figure 4.4: Shared seven-brane endpoints modify the stable-intersection count. Opposite-side terminations contribute positively, while same-side terminations contribute negatively to the quiver adjacency.

For every seven-brane A_i shared by two subwebs A and B, let $X_i(A, B)$ count pairs of five-branes, one from each subweb, which terminate on A_i from opposite sides, and let $Y_i(A, B)$ count the corresponding pairs which terminate from the same side. Opposite-side endpoints increase the adjacency, whereas same-side endpoints reduce that inferred from the planar stable intersection. The edge multiplicity is therefore corrected according to,

$$E(A, B) = SI(A, B) + \sum_i X_i(A, B) - \sum_i Y_i(A, B). \tag{4.26}$$

In the E_4 example, the same-side contributions precisely remove links which would otherwise have been predicted by stable intersection alone [49].

4.3.3 The E_5 example: node multiplicity

For $SU(2)$ with $N_f = 4$, the maximal decomposition introduces a different feature: several movable subwebs can be identical and coincident. Rather than producing separate rank-one nodes, n identical copies are encoded by a single $U(n)$ gauge node. Thus the multiplicity of a subweb determines the rank of the corresponding magnetic-quiver node.

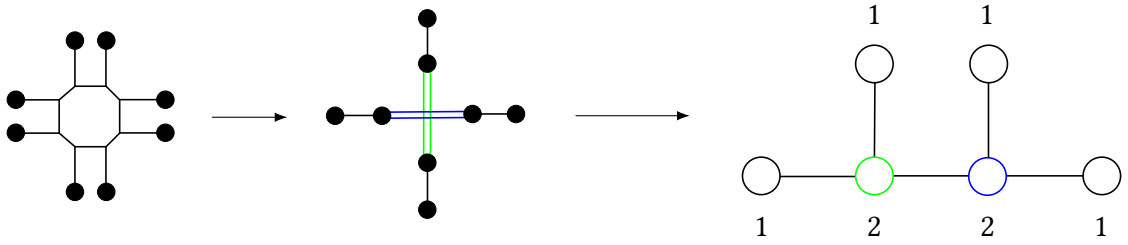


Figure 4.5: Identical coincident subwebs are encoded by the rank of a single magnetic-quiver node.

The E_5 decomposition therefore yields higher-rank nodes directly from coincident constituents. When computing the adjacency of such a node to another, one uses a single representative of the repeated subweb, since its multiplicity has already been incorporated into the gauge rank [49]. In this way the web reproduces the affine-D_5 magnetic quiver associated with the E_5 Higgs branch.

4.3.4 The E_6 example: adjacency without stable intersection

The E_6 theory, realised by $SU(2)$ with $N_f = 5$, shows that the seven-brane contributions introduced above are not merely corrections to the planar stable intersection. Its infinite-coupling Higgs branch is the reduced one-E_6-instanton moduli space, whose magnetic quiver is the affine-E_6 Dynkin diagram. In the unique maximal decomposition of the strong-coupling web, some subwebs have vanishing stable intersection but terminate on a common seven-brane from opposite sides. Their corresponding quiver nodes are therefore connected entirely through the X_i endpoint contribution. Applying the full edge-counting rule to the decomposition reproduces the known affine-E_6 quiver [49].

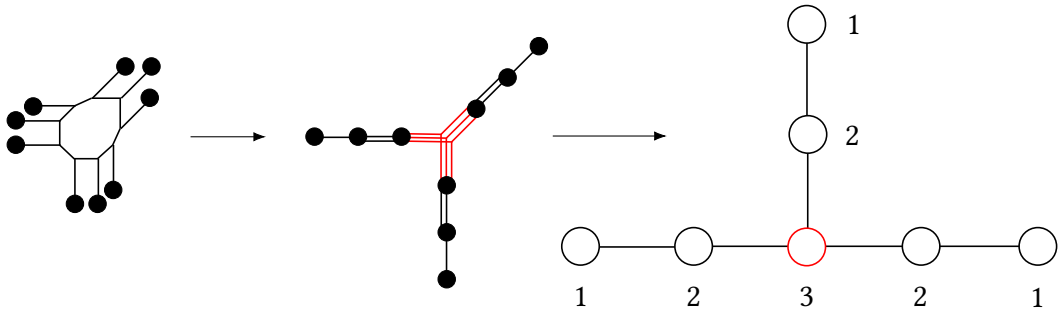


Figure 4.6: In the E_6 example, shared seven-brane endpoints can generate quiver adjacency even when the stable intersection vanishes.

4.3.5 Pure $SU(N_c)$: edge multiplicity

The edge multiplicity in Eq. 4.26 also exhibits invariance under Hanany–Witten rearrangements. Moving a seven-brane across a five-brane may change the planar stable intersection, but the accompanying creation or annihilation of five-brane prongs changes the endpoint contribution by the compensating amount. Pure $SU(N_c)_0$ Yang–Mills provides a simple example. In Fig. 4.7, the stable intersection changes from N_c to zero while the opposite-side endpoint contribution changes from zero to N_c, leaving the total edge multiplicity unchanged. The same mechanism accounts for the same-side contribution. The full edge-counting rule therefore gives the appropriate generalised intersection for subwebs ending on seven-branes, independent of the particular Hanany–Witten presentation of the web. The resulting magnetic quiver reproduces the expected Coulomb branch $\mathbb{C}^2/\mathbb{Z}_{N_c}$ [49].

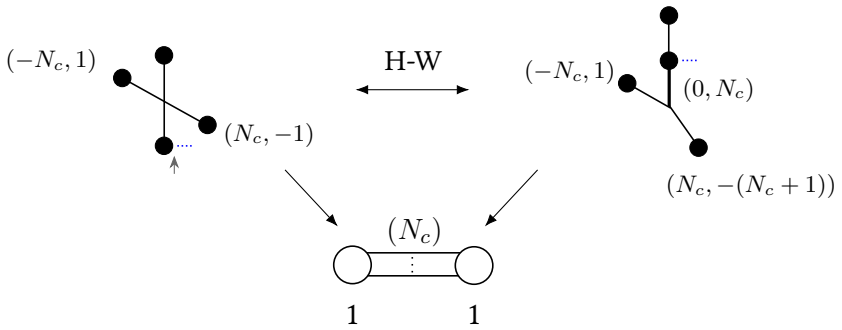


Figure 4.7: The generalised intersection is invariant under Hanany–Witten rearrangements even though its stable-intersection and seven-brane contributions may change separately.

4.4 Web-to-quiver prescription and the $SU(3)$ theories

To recap, a maximal supersymmetric decomposition of the five-brane web determines the corresponding magnetic quiver as follows. Each inequivalent subweb type W_a defines a unitary gauge node, with n_a identical copies giving $U(n_a)$. The adjacency between any two nodes is fixed by the edge multiplicity in Eq. 4.26, evaluated using one representative of each subweb type. The Coulomb branch of the resulting quiver is then identified with the Higgs-branch geometry generated by the relative motions of the subwebs. A maximal decomposition determines one hyperkähler cone. If the same strong-coupling web admits several inequivalent maximal decompositions, each gives a distinct magnetic quiver and hence a distinct component,

$$\mathcal{H}_{\mathrm{UV}} = \bigcup_{\lambda} \mathcal{C}(Q_\lambda). \tag{4.27}$$

The intersections of these components are encoded by coarser decompositions. Given two maximal decompositions, a common coarsening is obtained by recombining subwebs so that both maximal decompositions may be recovered by further splitting it. The relative motions retained by this coarser configuration are precisely those common to the two components, and the magnetic quiver extracted from it therefore describes their intersection [49].

These rules can now be applied to the $SU(3)$ webs constructed in Chapter 3. It is useful to begin with the finite-coupling configurations, which provide a baseline for the strong-coupling enhancement. For fixed matter content, the three Chern–Simons levels considered here give the same finite-coupling magnetic quiver, reproducing the expected absence of any change in the Higgs branch: the instanton states remain massive and the available Higgs directions are determined entirely by the perturbative hypermultiplet sector. Thus the three $N_f = 5$ theories share one finite-coupling quiver, and likewise the three $N_f = 6$ theories share another.

At strong coupling this degeneracy is lifted. Instanton states can become massless, opening additional Higgs-branch directions whose structure depends on the Chern–Simons level. The six strong-coupling webs therefore lead to distinct magnetic quivers, whose balance properties expose the corresponding enhanced non-Abelian global symmetries. These web–quiver pairs are presented below. The following chapter then develops the monopole methods introduced above to analyse the chiral rings of these Coulomb branches.

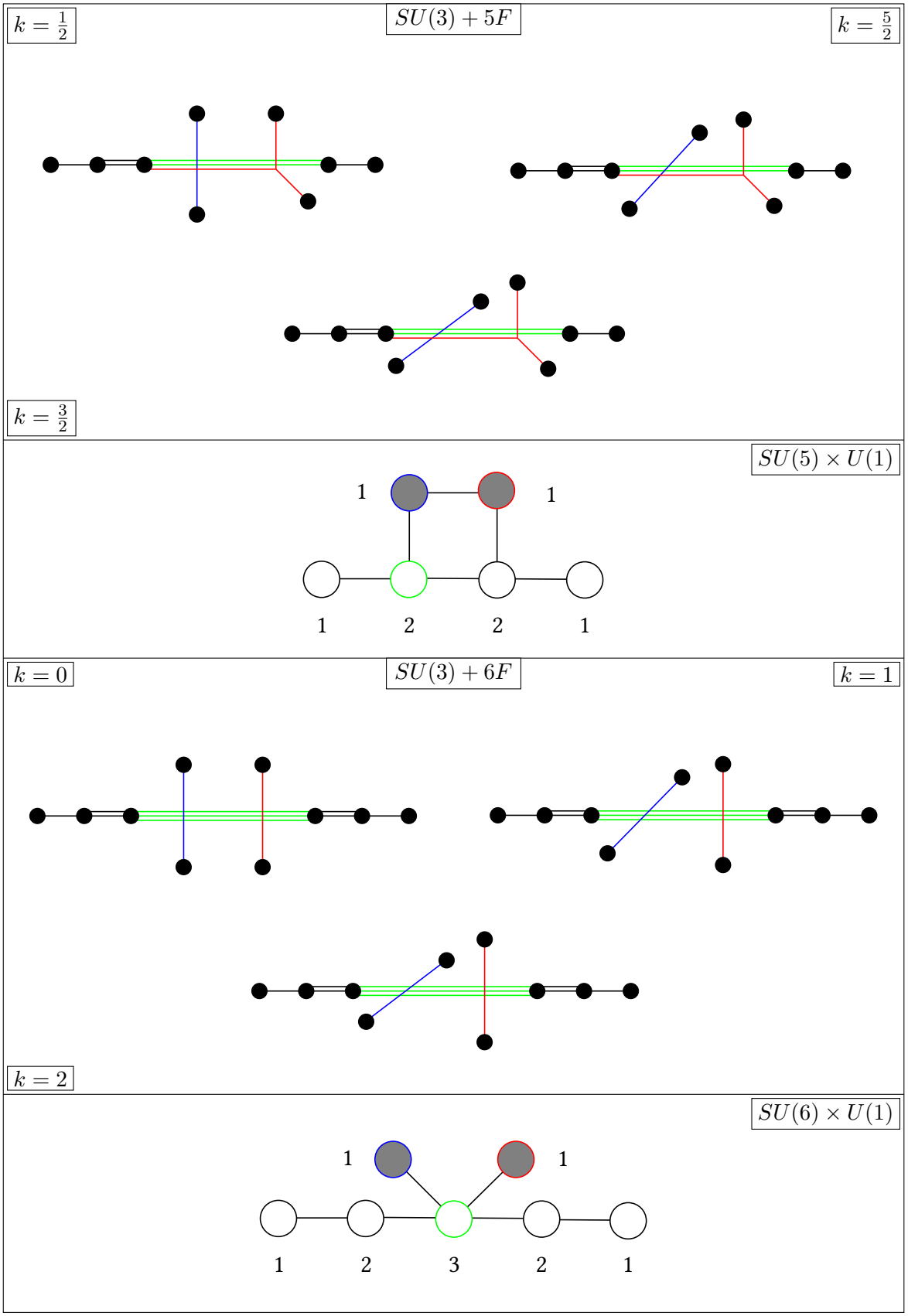


Figure 4.8: Finite-coupling webs and Higgs-branch quivers for $SU(3)_k + 5F$, $SU(3)_k + 6F$. Shaded circles are unbalanced nodes, while coloured boundaries indicate the associated web contribution.

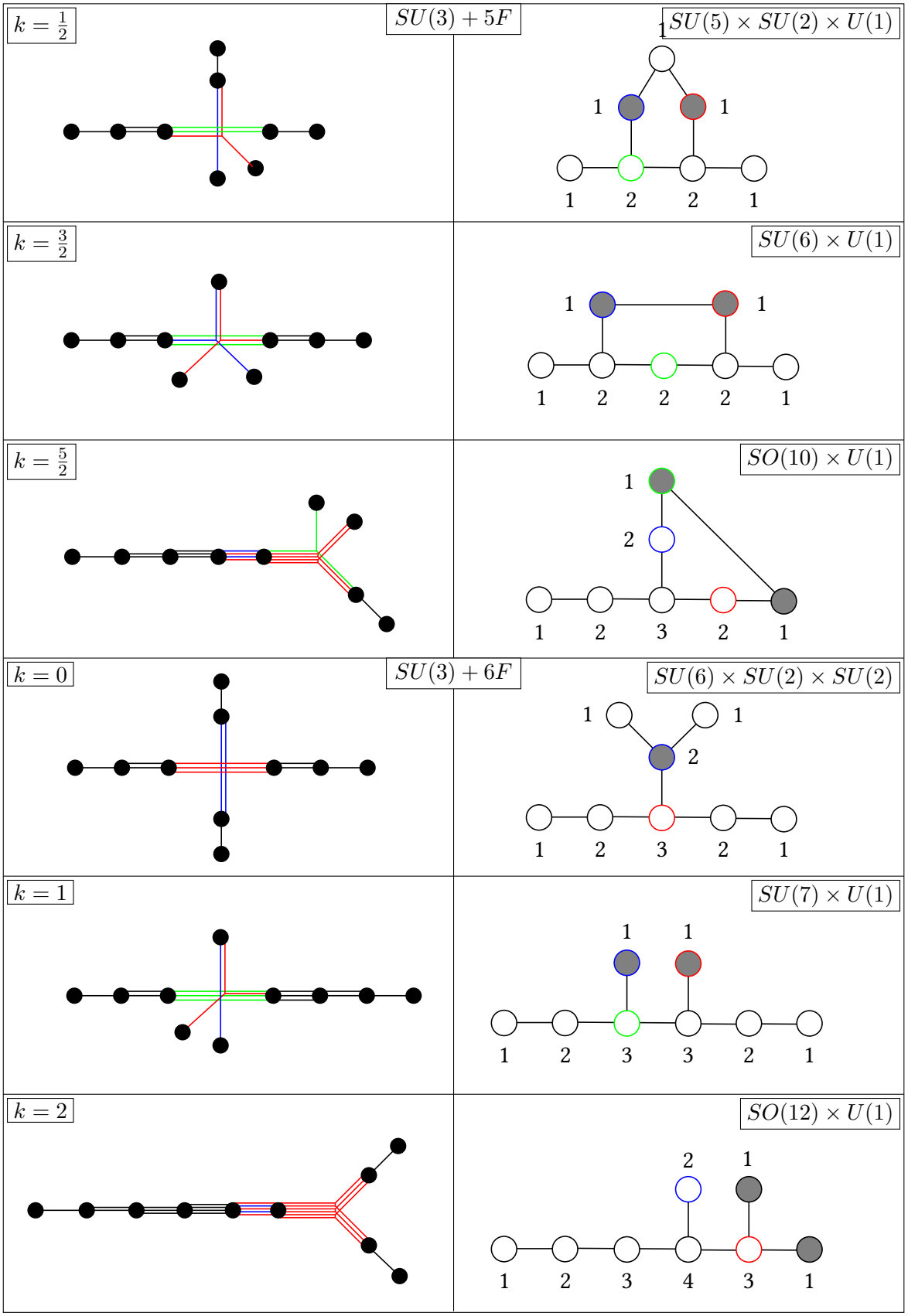


Figure 4.9: Strong-coupling five-brane webs and magnetic quivers for $SU(3)_k + 5F$, $SU(3)_k + 6F$. The enhanced global symmetry is indicated in each quiver panel, shaded circles are unbalanced nodes, while coloured boundaries indicate the non-trivial web contributions.

5 Chiral Rings and the Hilbert Series

We now turn to the machinery required to extract the algebraic structure of the Higgs branch from its magnetic quiver. As discussed in Chapter 2, the protected holomorphic operators on the Higgs branch form a graded commutative ring under multiplication and may be identified with its regular holomorphic functions. Since the Higgs branch parametrises gauge-inequivalent vacua, these functions must be gauge invariant. Its holomorphic coordinate ring is therefore,

$$\mathcal{R}_H \equiv \mathbb{C}[\mathcal{H}], \tag{5.1}$$

with elements of $\mathcal{R}_H$ corresponding to gauge-invariant protected operators. Ideally, one would determine a finite collection of such operators,

$$X_1, \ldots, X_s \tag{5.2}$$

from which every element of $\mathcal{R}_H$ can be constructed. These generators need not be algebraically independent, and may therefore satisfy polynomial relations,

$$P_a(X_1, \ldots, X_s) = 0. \tag{5.3}$$

If I denotes the ideal generated by these relations, the Higgs-branch coordinate ring may be presented as,

$$\mathbb{C}[\mathcal{H}_{\mathrm{UV}}] \simeq \frac{\mathbb{C}[X_1, \ldots, X_s]}{I}. \tag{5.4}$$

In the affine setting relevant here, the generators X_i provide coordinates on an ambient complex space, while the equations in I cut out the subvariety corresponding to $\mathcal{H}_{\mathrm{UV}}$. Determining the generators and their relations therefore amounts to determining the affine complex-algebraic structure of the Higgs branch.

For the strongly coupled Higgs branches of interest, however, this presentation is not known in advance: an operator may be composite rather than an independent generator, while a relation at a given degree may itself follow from lower-degree constraints. One must therefore first consider the protected ring as a whole. Since the ring is naturally graded, its operator content can be organised by degree and encoded in the *Hilbert series*, which counts the independent holomorphic operators at each degree after the relations have been imposed. Its structure then provides the information from which the generators and relations can be identified.

5.1 The Hilbert Series

The protected R-charge equips the ring with a natural grading. As in the preceding chapter, we choose a complex structure on the $3d\ \mathcal{N}=4$ Coulomb branch, equivalently an $\mathcal{N}=2$ subalgebra, so that the holomorphic operators are represented by highest-weight components under $SU(2)_C$. If j_C denotes the $SU(2)_C$ isospin of the corresponding representation, these chiral operators obey the protected relation $\Delta = j_C$. We use the integer degree $n \equiv 2j_C = 2\Delta$. Since this charge is additive under multiplication, the coordinate ring decomposes as,

$$\mathcal{R} = \bigoplus_{n\geq 0} \mathcal{R}_n, \qquad \mathcal{R}_n\,\mathcal{R}_m \subseteq \mathcal{R}_{n+m}, \tag{5.5}$$

where $\mathcal{R}_n$ is the vector space of independent holomorphic operators of degree n. The Hilbert series is the generating function for the dimensions of these graded components,

$$H_{\mathcal{R}}(t) = \sum_{n=0}^{\infty} \dim_{\mathbb{C}} \mathcal{R}_n\, t^n. \tag{5.6}$$

The coefficient of t^n therefore records the number of independent holomorphic operators of degree n, after the algebraic relations of the ring have been imposed. Its combinatorial nature is analogous to that of a partition function in statistical mechanics, in which a large set of states is organised according to a conserved or grading quantity. Consider a freely generated bosonic operator X of degree d. Its powers,

$$1, \qquad X, \qquad X^2, \qquad X^3, \ldots \tag{5.7}$$

occur at degrees $0, d, 2d, 3d, \ldots$, and hence contribute,

$$1 + t^d + t^{2d} + \cdots = \frac{1}{1-t^d} \tag{5.8}$$

to the Hilbert series. If the ring is freely generated by several operators X_i of degrees d_i, arbitrary monomials in these generators are counted by,

$$H_{\mathcal{R}}(t) = \prod_i \frac{1}{1-t^{d_i}}. \tag{5.9}$$

This construction is compactly encoded by the *plethystic exponential*,

$$\mathrm{PE}[f(t)] \equiv \exp\left[\sum_{k=1}^{\infty} \frac{1}{k} f(t^k)\right]. \tag{5.10}$$

For a single generator,

$$\mathrm{PE}[t^d] = \exp\left(\sum_{k=1}^{\infty} \frac{t^{dk}}{k}\right) = \exp\Big[-\log(1-t^d)\Big] = \frac{1}{1-t^d}, \tag{5.11}$$

It therefore follows that,

$$\mathrm{PE}\left[\sum_i t^{d_i}\right] = \prod_i \frac{1}{1-t^{d_i}}. \tag{5.12}$$

The plethystic exponential thus converts the degrees of the fundamental generators into the generating function that counts all of their symmetric products. Relations modify this free counting by removing combinations of generators that are not independent. In the simple case of a complete intersection[1], with generators of degrees d_i and independent relations of degrees r_a, the Hilbert series is,

$$H_{\mathcal{R}}(t) = \frac{\prod_a (1 - t^{r_a})}{\prod_i (1 - t^{d_i})} = \mathrm{PE}\left[\sum_i t^{d_i} - \sum_a t^{r_a}\right]. \tag{5.13}$$

More general rings may contain *syzygies*, namely relations among the relations, together with further higher-order structure, and therefore need not admit such a finite plethystic expression. However, for the strongly coupled Coulomb branches considered here, these generators and relations are precisely the unknown data. The problem is therefore reversed: the Hilbert series must first be computed from the magnetic quiver, after which its generator and relation content can be reconstructed.

5.2 The Monopole Formula: Hilbert Series from Quivers

The magnetic quiver provides a concrete way to organise the Coulomb-branch chiral ring according to the magnetic charges carried by its operators. Each magnetic sector contributes a set of protected operators, together with the gauge-invariant scalar dressings that remain available in that background. The Coulomb-branch Hilbert series is obtained by summing these contributions over all allowed sectors. For a unitary quiver with gauge group,

$$G = \prod_{i=1}^{L} U(N_i), \tag{5.14}$$

a magnetic sector is specified by a magnetic charge for every gauge node,

$$m = \left(m^{(1)}, \ldots, m^{(L)}\right), \qquad m^{(i)} = (m_{i,1}, \ldots, m_{i,N_i}), \tag{5.15}$$

with the entries at each node taken in a Weyl chamber, $m_{i,1} \geq \cdots \geq m_{i,N_i}$. Thus m denotes a complete magnetic assignment across the quiver. The corresponding BPS bare monopole V_m has protected dimension $\Delta(m)$, determined by the vector- and hypermultiplet contributions described in Chapter 4. The bare monopole does not, however, exhaust the operators associated with a fixed magnetic sector. A non-zero magnetic background generally breaks the gauge group to the subgroup,

$$H_m = Z_G(m), \tag{5.16}$$

consistent with the gauge transformations commuting with the full magnetic background. After choosing an $\mathcal{N} = 2 \subset \mathcal{N} = 4$ subalgebra the vector multiplet contains a real scalar σ and a complex adjoint scalar ϕ. The singular profile of σ is fixed by the BPS monopole condition and is therefore included in the definition of V_m. By contrast, a constant value of ϕ may be added while preserving the same supersymmetric background provided that $[\phi, m] = 0$, with its surviving components taking values in the complexified Lie algebra of H_m.

[1]More precisely, a complete intersection is obtained when the defining ideal can be generated by a regular sequence, so that the number of independent defining equations equals the codimension of the variety. For g generators and r such relations, $\dim_{\mathbb{C}} \mathcal{M} = g - r$, and the Hilbert series takes the product form given; see [42].

These surviving scalar components may be used to dress the monopole,

$$V_m \longrightarrow V_m P(\phi). \tag{5.17}$$

Since H_m remains a gauge redundancy within the sector, the polynomial $P(\phi)$ must be invariant under H_m. Such operators are the *dressed monopole operators*. The invariant polynomial algebra of an adjoint scalar is freely generated by the Casimir invariants of the residual group. If their polynomial degrees are $d_a(m)$, where a labels the independent Casimir generators of H_m, their contribution to the Hilbert series is,

$$P_G(t;m) = \prod_a \frac{1}{1-t^{2d_a(m)}}. \tag{5.18}$$

This is the freely generated counting function of the previous section, applied to the scalar invariants that survive in a fixed magnetic sector. For the unitary factors appearing in the magnetic quivers, the residual group is read directly from the degeneracies of the magnetic charges. At a node $U(N_i)$, if the distinct entries of $m^{(i)}$ occur with multiplicities $\lambda_{i,1},\dots,\lambda_{i,s_i}$, then,

$$H_{m^{(i)}} \simeq \prod_{p=1}^{s_i} U(\lambda_{i,p}). \tag{5.19}$$

Equal magnetic charges may still be mixed by gauge transformations, whereas directions carrying different fluxes cannot; for example, a charge $(2,2,0)$ at a $U(3)$ node leaves $U(2)\times U(1)$ unbroken. The invariant polynomials of a $U(k)$ factor have degrees $1,\dots,k$, e.g. $\mathrm{Tr}\,\phi, \mathrm{Tr}\,\phi^2,\dots,\mathrm{Tr}\,\phi^k$.

Computationally, this residual-group analysis is performed independently at each gauge node. For a quiver-wide magnetic assignment $m=(m^{(1)},\dots,m^{(L)})$, one determines the multiplicities $\lambda_{i,p}$ of equal charges at every node and hence,

$$H_m = \prod_{i,p} U(\lambda_{i,p}). \tag{5.20}$$

$$P_G(t;m) = \prod_{i,p}\prod_{j=1}^{\lambda_{i,p}} \frac{1}{1-t^{2j}}. \tag{5.21}$$

For example, a $U(2)$ node contributes $\left[(1-t^2)(1-t^4)\right]^{-1}$ when its two magnetic charges coincide, but $(1-t^2)^{-2}$ when they are distinct. A magnetic sector therefore contributes $t^{2\Delta(m)}P_G(t;m)$, where the first factor counts the bare monopole and the second its allowed dressings. Summing over all inequivalent magnetic assignments gives,

$$H_{\mathcal{C}}(t) = \sum_{\substack{m^{(1)}\in\mathbb{Z}^{N_1}/S_{N_1}\\ \vdots \\ m^{(L)}\in\mathbb{Z}^{N_L}/S_{N_L}}} t^{2\Delta(m^{(1)},\dots,m^{(L)})} P_G\left(t;m^{(1)},\dots,m^{(L)}\right), \tag{5.22}$$

with the ungauging convention for any decoupled diagonal $U(1)$ understood as in the preceding chapter.

More compactly,

$$H_{\mathcal{C}}(t) = \sum_{[m]\in X_*(T)/W} t^{2\Delta(m)} P_G(t;m). \tag{5.23}$$

The zero-flux sector contains the ordinary scalar Casimirs, while non-zero sectors supply the bare and dressed monopole operators.

This is the *monopole formula* of Cremonesi, Hanany and Zaffaroni [42]. It provides a direct route from the combinatorial data of a magnetic quiver to the Hilbert series of its Coulomb branch, which may then be evaluated to the desired order using modern computational methods. The resulting series organises the protected operator content by degree, but the same counting can be refined to retain the additional quantum numbers carried by those operators. We now turn to this symmetry refinement.

5.3 Symmetry Refinement and Highest-Weight Generating Functions

The Higgs branches of interest carry a global symmetry G_F, whose action on the geometry induces a corresponding action on the coordinate ring. If $p \in \mathcal{H}$ and $f \in \mathbb{C}[\mathcal{H}]$, the corresponding action on functions is given by $(g \cdot f)(p) = f(g^{-1} \cdot p)$. Since the global symmetry preserves the protected grading, each graded component $\mathcal{R}_n$ therefore carries a representation of G_F. The necessary charge consideration arises from the monopole description; Chapter 4 showed that monopole operators carry topological charges $J_i(m)$ associated with the unitary gauge nodes, and that for the good quivers considered here these symmetries enhance according to the balanced Dynkin subquiver to the non-Abelian Coulomb-branch symmetry.

The magnetic charges entering the monopole formula therefore also distinguish operators by their charges under the Cartan of this global symmetry. Introducing fugacities z_i for these charges refines the monopole formula to,

$$H(t,\mathbf{z}) = \sum_{[m]} t^{2\Delta(m)} P_G(t;m) \prod_i z_i^{J_i(m)}. \tag{5.24}$$

The variable t records the protected degree as before, while the z_i retain the topological charges of each monopole sector and hence its position in the weight lattice of the global symmetry. The variables $\mathbf{z}$ may consequently be regarded as maximal-torus fugacities for for G_F. At fixed degree, the monomials in $\mathbf{z}$ record the individual weights carried by the operators, which assemble into characters of irreducible representations. Writing,

$$\mathcal{R}_n \simeq \bigoplus_{\lambda\in P_+} a_{n,\lambda} V_\lambda, \qquad H(t,\mathbf{z}) = \sum_{n\geq 0}\sum_{\lambda\in P_+} a_{n,\lambda}\chi_\lambda(\mathbf{z}) t^n, \tag{5.25}$$

the coefficient $a_{n,\lambda}$ gives the multiplicity of the irreducible representation V_λ at degree n. Evaluating the character at the identity, $\chi_\lambda(\mathbf{1}) = \dim V_\lambda$, removes the symmetry refinement and reproduces the ordinary Hilbert series.

Although the character expansion retains the full representation content, the explicit character expressions can become cumbersome. Since each irreducible representation is uniquely determined by its highest weight, the same information can be encoded more compactly in terms of highest-weight labels. For a rank-r symmetry, let $\lambda = \sum_{i=1}^{r} \lambda_i \omega_i$ have Dynkin labels $[\lambda_1, \ldots, \lambda_r]$, and introduce formal highest-weight fugacities μ_i associated with the fundamental weights ω_i. The representation V_λ is then represented by the monomial $\boldsymbol{\mu}^\lambda = \prod_i \mu_i^{\lambda_i}$. Replacing each character by its corresponding highest-weight monomial defines the *highest-weight generating function*,

$$\mathrm{HWG}(t, \boldsymbol{\mu}) = \sum_{n \geq 0} \sum_{\lambda \in P_+} a_{n,\lambda}\, \boldsymbol{\mu}^\lambda t^n. \tag{5.26}$$

It should not be confused with an unrefined Hilbert series; in general $\mathrm{HWG}(t, \boldsymbol{\mu} = \mathbf{1}) \neq H(t, \mathbf{1})$, since the former counts irreducible representations with multiplicity whereas the latter counts all states within them. The advantage of the highest-weight form is that regularities in the representation content become considerably more transparent. Knowledge of the global symmetry, together with the degrees and representations of the low-lying monopole operators, can therefore strongly constrain and in favourable cases motivate the form of the HWG itself [33].

5.4 Highest-Weight Generating Functions for SU(3)

For the six $SU(3)$ theories studied in this dissertation, Ferlito, Hanany, Mekareeya and Zafrir proposed HWGs for the corresponding families of $SU(n)$ theories and tested them using lower-rank limits, dimension checks and direct Coulomb-branch computations in several cases [33]. Although these HWGs admit compact plethystic expressions, the terms in the exponent should not yet be identified with the fundamental generators and defining relations of the coordinate ring. Determining which operators are algebraically fundamental, and which relations among them are independent, is the inverse problem addressed in the following chapter.

The fugacities μ_i label highest weights of the indicated higher-rank symmetry factor, while ν, or ν_1, ν_2, label highest weights of additional $SU(2)$ factors. The fugacity q records an additional $U(1)$ charge.

For the five-flavour theories,

$$\mathrm{HWG}_{\frac{1}{2},5} = \mathrm{PE}\Big[\mu_1\mu_4 t^2 + \mu_2\mu_3 t^4 + (\nu^2+1)t^2 + \nu\Big(\mu_2 q + \mu_3 q^{-1}\Big)\, t^3 - \nu^2\mu_2\mu_3 t^6\Big], \tag{5.27}$$

$$\mathrm{HWG}_{\frac{3}{2},5} = \mathrm{PE}\Big[\mu_1\mu_5 t^2 + \mu_2\mu_4 t^4 + t^2 + \Big(\mu_2 q + \mu_4 q^{-1}\Big)\, t^3 - \mu_2\mu_4 t^6\Big], \tag{5.28}$$

$$\mathrm{HWG}_{\frac{5}{2},5} = \mathrm{PE}\Big[\mu_2 t^2 + t^2 + \Big(q\mu_4 + q^{-1}\mu_5\Big)\, t^3 + \mu_4\mu_5\left(t^4 - t^6\right)\Big]. \tag{5.29}$$

These correspond respectively to

$$SU(3)_{\frac{1}{2}} + 5F, \qquad SU(3)_{\frac{3}{2}} + 5F, \qquad SU(3)_{\frac{5}{2}} + 5F,$$

with non-Abelian symmetry factors $SU(5) \times SU(2)$, $SU(6)$ and $SO(10)$, respectively, together with the additional $U(1)$ recorded by q.

For the six-flavour theories,

$$\mathrm{HWG}_{0,6} = \mathrm{PE}\Big[\mu_1\mu_5 t^2 + \mu_2\mu_4 t^4 + \mu_3^2 t^6 + (\nu_1^2+\nu_2^2)t^2 + t^4 + \nu_1\nu_2\mu_3\left(t^3+t^5\right) - \nu_1^2\nu_2^2\mu_3^2 t^{10}\Big], \tag{5.30}$$

$$\mathrm{HWG}_{1,6} = \mathrm{PE}\Big[\mu_1\mu_6 t^2 + \mu_2\mu_5 t^4 + t^2 + \left(\mu_3 q + \mu_4 q^{-1}\right) t^3\Big], \tag{5.31}$$

$$\mathrm{HWG}_{2,6} = \mathrm{PE}\Big[\mu_2 t^2 + \mu_4 t^4 + t^2 + \left(q + q^{-1}\right)\mu_6 t^3\Big]. \tag{5.32}$$

These correspond respectively to

$$SU(3)_0 + 6F, \qquad SU(3)_1 + 6F, \qquad SU(3)_2 + 6F,$$

with non-Abelian symmetry factors $SU(6)\times SU(2)\times SU(2)$, $SU(7)$ and $SO(12)$, respectively; the latter two also carry the additional $U(1)$ recorded by q.

6 Plethystic Reconstruction of Generators and Relations

6.1 The Plethystic Logarithm

We now work backwards from the graded counting data of the ultraviolet chiral rings to identify the intrinsic algebraic structure that generates it: the fundamental operators and the constraints among them. Chapter 5 showed how the plethystic exponential packages a set of generators and their symmetric products into the Hilbert series. This naturally motivates applying its inverse, the *plethystic logarithm*, to recover the elementary algebraic data underlying the counting function.

For a refined Hilbert series $H(t, \mathbf{x})$, the plethystic logarithm is defined by,

$$\mathrm{PL}[H](t, \mathbf{x}) = \sum_{r=1}^{\infty} \frac{\mu(r)}{r} \log H(t^r, \mathbf{x}^r), \tag{6.1}$$

where $\mu(r)$ is the Möbius function, characterised by $\sum_{r|n} \mu(r) = \delta_{n1}$. Its role as the inverse of the plethystic exponential follows immediately from,

$$\mathrm{PL}[\mathrm{PE}[f]] = \sum_{r,k\geq 1} \frac{\mu(r)}{rk} f(t^{rk}, \mathbf{x}^{rk}) = \sum_{n\geq 1} \frac{1}{n} \left(\sum_{r|n} \mu(r) \right) f(t^n, \mathbf{x}^n) = f(t, \mathbf{x}). \tag{6.2}$$

Thus the ordinary logarithm removes the exponential structure, while Möbius inversion removes the repeated-power contributions generated by the plethystic exponential.

Because the representation-valued expressions become cumbersome beyond the first few orders, the plethystic analysis is performed computationally. For each theory, the program takes the HWG obtained in Chapter 5 and expands it to a chosen order in t. Each highest-weight monomial is then replaced by the full character of the corresponding irreducible representation, reconstructing the refined Hilbert series in a form suitable for evaluating (6.1). The plethystic logarithm is computed symbolically order by order, and each coefficient is decomposed into irreducible characters and recorded by its Dynkin labels. The calculation may be truncated consistently at any degree D, since the coefficient of t^d depends only on Hilbert-series coefficients through degree d.

For example, consider $\mathrm{SU}(3)_{\frac{1}{2}} + 5F$, whose ultraviolet symmetry is $\mathrm{SU}(5) \times \mathrm{SU}(2) \times \mathrm{U}(1)_q$. Using Dynkin-label notation $([a_1, a_2, a_3, a_4]; [b])_q$, the procedure gives,

$$\begin{aligned}\mathrm{PL}[H] = &\Big(([1,0,0,1];[0])_0 + ([0,0,0,0];[2])_0 + ([0,0,0,0];[0])_0\Big)t^2 \\ &+ \Big(([0,1,0,0];[1])_{+1} + ([0,0,1,0];[1])_{-1}\Big)t^3 \\ &- \Big(([1,0,0,1];[0])_0 + 2([0,0,0,0];[0])_0\Big)t^4 \\ &- \Big(([2,0,0,0];[1])_{+1} + ([0,0,1,1];[1])_{+1} + 2([0,1,0,0];[1])_{+1} \\ &\qquad + ([0,0,0,2];[1])_{-1} + ([1,1,0,0];[1])_{-1} + 2([0,0,1,0];[1])_{-1}\Big)t^5 + \mathcal{O}(t^6).\end{aligned} \tag{6.3}$$

At these first orders the interpretation is direct. The positive contributions at degrees two and three identify the candidate fundamental generator multiplets, while the negative contributions at degrees four and five identify the representations in which the first primitive relations occur. At higher degree, descendants of lower relations and syzygies can contribute to the same plethystic coefficient, so this direct reading must eventually be refined.

6.2 Perturbative Gauge-Invarient Operators

To give physical intepretations to the ultraviolet generators, we must first understand the gauge-invariant operators already present in the finite-coupling theory. After choosing a $\mathcal{N} = 2$ subalgebra used to define the chiral ring, each fundamental hypermultiplet is represented holomorphically by chiral fields $Q^i{}_a$ and $\widetilde{Q}^a{}_i$ in conjugate bifundamental representations of the gauge and flavour groups, with a and i denoting colour and flavour indices. The hypermultiplet scalars transform as $\mathrm{SU}(2)_R$ doublets, so each elementary holomorphic field has $R = \frac{1}{2}$, or degree $2R = 1$ in the grading used throughout this work. Gauge invariance therefore first permits mesonic bilinears at degree two and, for $\mathrm{SU}(3)$, baryonic products of three squarks at degree three [6]. Explicitly,

$$X^i{}_j = Q^i{}_a \widetilde{Q}^a{}_j, \qquad B^{ijk} = \epsilon^{abc} Q^i{}_a Q^j{}_b Q^k{}_c, \qquad \widetilde{B}_{ijk} = \epsilon_{abc} \widetilde{Q}^a{}_i \widetilde{Q}^b{}_j \widetilde{Q}^c{}_k. \tag{6.4}$$

The traceful meson may be separated into its flavour-adjoint and singlet components by defining,

$$\eta' \equiv \frac{1}{3} \widetilde{Q}^a{}_i Q^i{}_a = \frac{1}{3} \operatorname{tr} X, \qquad M^i{}_j = X^i{}_j - \frac{3\eta'}{N_f} \delta^i{}_j, \tag{6.5}$$

so that

$$X = M + \frac{3\eta'}{N_f} \mathbf{1}_{N_f}. \tag{6.6}$$

Thus M and η' are the degree-two classical mesonic operators, while B and $\widetilde{B}$ are the degree-three baryonic operators. The complete classical generator set and its relations are reviewed in Appendix E.

The finite-coupling chiral ring also contains chiral operators constructed from the vector multiplet. In the four-supercharge description, the field-strength superfield W_α admits the Lorentz-scalar composite,

$$S = -\frac{1}{32\pi^2}\,\mathrm{Tr}\, W^\alpha W_\alpha, \tag{6.7}$$

whose lowest component is the gaugino bilinear $\mathrm{Tr}\,\lambda^\alpha\lambda_\alpha$. This is the glueball operator. It has $\mathrm{SU}(2)_R$ spin one and therefore occurs at degree two, alongside the mesonic operators. Unlike η', however, S has no representative on the classical bosonic scalar ring: it vanishes when the fermionic fields are set to zero, while remaining an independent operator in the finite-coupling chiral ring [32].

6.3 Identifying the Generators

To identify how these perturbative operators are completed at infinite coupling, we embed the finite-coupling symmetry as,

$$\mathrm{SU}(5)\times\mathrm{U}(1)_B\times\mathrm{U}(1)_I \subset \mathrm{SU}(5)\times\mathrm{SU}(2)\times\mathrm{U}(1)_q. \tag{6.8}$$

The degree-two generators have a particularly direct interpretation. For each continuous flavour symmetry, the conserved-current multiplet contains an $\mathrm{SU}(2)_R$ triplet of moment-map operators; choosing the holomorphic highest-weight component gives a degree-two chiral operator transforming in the adjoint of the flavour symmetry. The three degree-two representations found in the plethystic logarithm are therefore precisely the moment maps for the three factors of the enhanced symmetry.

The $\mathrm{SU}(5)$ adjoint branches trivially and is identified with the traceless meson $M^i{}_j$, while the singlet is denoted by Σ and corresponds to the $\mathrm{U}(1)_q$ moment map. The remaining triplet,

$$\Phi \in (\mathbf{1},\mathbf{3})_0, \tag{6.9}$$

is the moment map of the enhanced $\mathrm{SU}(2)$ and contains the non-perturbative information.

Its raw branching under the $\mathrm{SU}(2)$ Cartan is,

$$([0,0,0,0];[2])_0 \longrightarrow [0,0,0,0]_{(2,0)} \oplus [0,0,0,0]_{(0,0)} \oplus [0,0,0,0]_{(-2,0)}, \tag{6.10}$$

where the subscripts temporarily denote (x,q). The charged components can be identified from the one-instanton spectrum. Quantising the fermionic zero modes around the canonical $\mathrm{SU}(2)\subset\mathrm{SU}(N)$ charge-one instanton and imposing gauge neutrality gives the current-type states $Q_B(E_\pm)=\kappa\mp N$ in the baryon-number convention used here [67, 68]; the derivation is given in Appendix F. For $\mathrm{SU}(3)_{\frac{1}{2}}+5F$, the surviving one-instanton current therefore carries $(Q_B,Q_I)=(-\frac{5}{2},1)$. It supplies one charged component of the enhanced $\mathrm{SU}(2)$ moment map, while the anti-instanton supplies the conjugate component. Thus,

$$\Phi = \begin{pmatrix} H & I_+ \\ I_- & -H \end{pmatrix}, \tag{6.11}$$

with H the neutral Cartan component and $I_\pm$ the instanton and anti-instanton operators.

At degree three the two ultraviolet $SU(2)$ doublets branch as,

$$\begin{aligned}([0,1,0,0];[1])_{+1} &\longrightarrow [0,1,0,0]_{(1,1)} \oplus [0,1,0,0]_{(-1,1)},\\ ([0,0,1,0];[1])_{-1} &\longrightarrow [0,0,1,0]_{(1,-1)} \oplus [0,0,1,0]_{(-1,-1)}.\end{aligned} \tag{6.12}$$

The classical operators B and $\widetilde{B}$ must occupy one component of each doublet. With the baryon-number convention used here, $B \sim Q^3$ has $(Q_B, Q_I) = (-3, 0)$, while $\widetilde{B}$ has $(3, 0)$. The remaining components are supplied by the one-instanton sectors in the same flavour representations, so the ultraviolet degree-three generators combine the perturbative baryons with their non-perturbative partners. This joining of baryonic and instanton states into single enhanced multiplets is precisely the structure found in the one-instanton analysis of these SQCD sequences [33].

The Abelian embedding is now fixed by the instanton current, for which $(x, q) = (2, 0)$ has $(Q_B, Q_I) = (-\frac{5}{2}, 1)$, together with the classical baryon, for which $(x, q) = (1, 1)$ has $(Q_B, Q_I) = (-3, 0)$. These two assignments determine,

$$Q_B = -\frac{5x + 7q}{4}, \qquad Q_I = \frac{x - q}{2}. \tag{6.13}$$

The neutral degree-two generators admit a parallel interpretation in terms of moment maps. In the ultraviolet basis, H is the neutral component of the $\mathrm{SU}(2)$ moment map associated with its Cartan generator x, while Σ is the moment map for $\mathrm{U}(1)_q$ [66, 69]. At finite coupling, the corresponding Abelian moment maps are instead supplied by the flavour-singlet meson $3\eta' = \operatorname{tr} X$, associated with $\mathrm{U}(1)_B$, and the glueball S, associated with the topological $\mathrm{U}(1)_I$. Thus (H, Σ) and $(3\eta', S)$ are the same moment maps written in the ultraviolet and finite-coupling Cartan bases, respectively. Since moment maps are linear in the symmetry generators, the charge map derived above induces the corresponding change of basis between these neutral operators. With the normalisations used here,

$$H = 3\eta' - \frac{7}{2}S, \qquad \Sigma = 3\eta' + \frac{5}{2}S. \tag{6.14}$$

The distinction between the two ultraviolet neutral scalars is therefore precisely the vector-multiplet contribution: on the classical scalar locus $S = 0$, both reduce to $3\eta' = \operatorname{tr} X$.

The physical identification of the degree-two generators may therefore be written as,

$$M^i{}_j, \qquad \Sigma = 3\eta' + \frac{5}{2}S, \qquad \Phi = \begin{pmatrix} 3\eta' - \frac{7}{2}S & I_+^{(-\frac{5}{2},1)} \\ I_-^{(\frac{5}{2},-1)} & -3\eta' + \frac{7}{2}S \end{pmatrix}, \tag{6.15}$$

while the degree-three multiplets are,

$$B^{ij\alpha} = \left(B^{ij}_{(-3,0)},\, B^{ij}_{(-\frac{1}{2},-1)}\right), \qquad \widetilde{B}_{ij}{}^{\alpha} = \left(\widetilde{B}_{ij}^{(\frac{1}{2},1)},\, \widetilde{B}_{ij}^{(3,0)}\right). \tag{6.16}$$

6.4 From Relation Representations to Covariant Equations

Having identified the generators, the higher order negative terms can now be used constructively: at each degree we form the allowed polynomial combinations of the generators and decompose them into the representations required by the plethystic logarithm.

Let $\mathcal{G}_2$ and $\mathcal{G}_3$ denote the complete degree-two and degree-three generator spaces identified above. The free polynomial ring is $P = \mathrm{Sym}(\mathcal{G}_2 \oplus \mathcal{G}_3)$, with for example,

$$P_5 = \mathcal{G}_2 \otimes \mathcal{G}_3, \qquad P_6 = \mathrm{Sym}^3 \mathcal{G}_2 \oplus \mathrm{Sym}^2 \mathcal{G}_3. \tag{6.17}$$

Consider the charge-$+1$ part of the degree-five plethystic logarithm. It requires relations transforming as,

$$([2,0,0,0];[1])_{+1} \oplus ([0,0,1,1];[1])_{+1} \oplus 2([0,1,0,0];[1])_{+1}, \tag{6.18}$$

or, in dimension notation,

$$(\mathbf{15},\mathbf{2})_{+1} \oplus (\overline{\mathbf{40}},\mathbf{2})_{+1} \oplus 2(\mathbf{10},\mathbf{2})_{+1}. \tag{6.19}$$

Since the charge-$+1$ degree-three generator is $B \in (\mathbf{10},\mathbf{2})_{+1}$, these representations must arise from multiplying B by the degree-two generators M, Φ and Σ. For the mesonic product,

$$\mathbf{24} \otimes \mathbf{10} = \mathbf{10} \oplus \mathbf{15} \oplus \overline{\mathbf{40}} \oplus \mathbf{175}. \tag{6.20}$$

The $\mathbf{15}$ and $\overline{\mathbf{40}}$ each occur once and match the corresponding negative plethystic terms. Therefore the relations are

$$P_{\overline{\mathbf{40}}}(M \otimes B) = 0, \qquad P_{\mathbf{15}}(M \otimes B) = 0. \tag{6.21}$$

In these multiplicity-one channels the representation content fixes the tensor structure of the relation up to an overall normalization.

The repeated $(\mathbf{10},\mathbf{2})_{+1}$ channel requires more care. One copy is obtained from the $\mathbf{10} \subset \mathbf{24} \otimes \mathbf{10}$ component of MB, while the products ΦB and ΣB supply two further copies. A convenient basis is,

$$C_M^{ij\alpha} = M^i{}_k B^{kj\alpha} - M^j{}_k B^{ki\alpha}, \qquad C_\Phi^{ij\alpha} = \Phi^\alpha{}_\beta B^{ij\beta}, \qquad C_\Sigma^{ij\alpha} = \Sigma B^{ij\alpha}. \tag{6.22}$$

Thus the available degree-five operator space contains three copies of $(\mathbf{10},\mathbf{2})_{+1}$, whereas the plethystic logarithm requires two relations in this representation. The relation space is therefore a two-dimensional subspace of,

$$\mathrm{span}\{C_M, C_\Phi, C_\Sigma\}. \tag{6.23}$$

At each negative term appearing in the plethystic logarithm, we repeat this procedure: its degree determines the allowed products of generators, while its representation selects the corresponding covariant components. Multiplicity-one channels are then fixed directly, whereas repeated representations leave undetermined linear combinations. To resolve these multiplicities, we may use the known classical relations by restricting the ultraviolet generators to the classical chiral ring.

6.5 Classical Restriction

The remaining coefficients are constrained by requiring that their restriction reproduce the classical Higgs-branch chiral ring of the weakly coupled gauge theory. More precisely, the comparison is made with the *zero-coupling scalar ring*, obtained in the limit $g \to 0$. In this limit the instanton mass scales as $1/g^2$, so the non-perturbative instanton sectors decouple, while the fermionic gaugino bilinear is absent from the classical bosonic ring [32].

For $\mathrm{SU}(3)_{\frac{1}{2}} + 5F$, the ultraviolet generators therefore restrict[1] as,

$$S \longrightarrow 0, \qquad I_{\pm} \longrightarrow 0, \qquad M^{i}{}_{j} \longrightarrow M^{i}{}_{j}. \tag{6.24}$$

Using the moment-map identifications obtained above, the neutral ultraviolet scalars consequently obey,

$$H = 3\eta' - \frac{7}{2}S \longrightarrow 3\eta', \qquad \Sigma = 3\eta' + \frac{5}{2}S \longrightarrow 3\eta'. \tag{6.25}$$

Thus the two distinct neutral directions of the ultraviolet ring collapse onto the same flavour-singlet mesonic coordinate on the classical scalar locus; their difference is entirely carried by the vector-multiplet contribution,

$$\Sigma - H = 6S \longrightarrow 0. \tag{6.26}$$

The degree-three multiplets similarly retain only their perturbative baryonic components,

$$B^{ij\alpha} = \left(B^{ij}_{(-3,0)},\, B^{ij}_{(-\frac{1}{2},-1)} \right) \longrightarrow \left(B^{ij}, 0 \right), \tag{6.27}$$

and

$$\widetilde{B}_{ij}{}^{\alpha} = \left(\widetilde{B}^{(\frac{1}{2},1)}_{ij},\, \widetilde{B}^{(3,0)}_{ij} \right) \longrightarrow \left(0, \widetilde{B}_{ij} \right). \tag{6.28}$$

The traceful meson on this locus is therefore,

$$X = M + \frac{3\eta'}{5}\mathbf{1}_5, \tag{6.29}$$

and satisfies the classical relations

$$X^2 = \eta' X, \qquad XB = 0, \qquad \widetilde{B}X = 0. \tag{6.30}$$

For the three degree-five covariants introduced above, the zero-coupling restriction gives,

$$C_M^{ij\alpha} \longrightarrow -\frac{6}{5}\eta' B^{ij}, \qquad C_\Phi^{ij\alpha} \longrightarrow 3\eta' B^{ij}, \qquad C_\Sigma^{ij\alpha} \longrightarrow 3\eta' B^{ij}. \tag{6.31}$$

Hence a general covariant,

$$aC_M + bC_\Phi + cC_\Sigma \tag{6.32}$$

[1] These restrictions should initially be written with undetermined coefficients: the Hilbert-series data determines which operators are allowed, but not whether their coefficients vanish. Requiring the ultraviolet relations to reproduce the classical chiral ring fixes which coefficients are non-zero, after which such factors may be absorbed into the generator definitions.

vanishes on the classical locus when,

$$-\frac{2}{5}a + b + c = 0. \tag{6.33}$$

The kernel is therefore two-dimensional, in agreement with the two copies of $(\mathbf{10}, \mathbf{2})_{+1}$ appearing in the plethystic logarithm. A convenient basis is,

$$C_M + \frac{2}{5}C_\Sigma = 0, \qquad C_\Phi - C_\Sigma = 0, \tag{6.34}$$

or explicitly,

$$M^i{}_k B^{kj\alpha} - M^j{}_k B^{ki\alpha} + \frac{2}{5}\Sigma B^{ij\alpha} = 0, \tag{6.35}$$

$$\Phi^\alpha{}_\beta B^{ij\beta} - \Sigma B^{ij\alpha} = 0. \tag{6.36}$$

The same procedure is applied at each relation degree: representation theory determines the available covariant relation space, while restriction to the zero-coupling classical scalar ring fixes the combinations that reduce to the known mesonic and baryonic relations. At higher degree, products of lower-degree relations and their syzygies must additionally be separated from genuinely new constraints.

6.6 Syzygies and Ideal Verification

At the first degrees at which relations appear, negative terms in the plethystic logarithm may be identified directly with generators of the defining ideal. This interpretation is no longer sufficient at higher degree. Relations already imposed at lower order generate further equations when multiplied by the ring generators, and these descendants need not be linearly independent. Relations among them are *syzygies*; at still higher degree, further dependencies among the syzygies may also occur. Consequently, higher-order plethystic coefficients encode an alternating contribution from primitive relations and their dependencies, rather than a direct count of new constraints.

Let P denote the polynomial ring generated by the ultraviolet operators, and let $\mathcal{R}_d \subset P_d$ be the space of primitive relations introduced at degree d. Once these relations have been fixed, their contributions to degree n are generated by multiplication,

$$\Psi_n : \bigoplus_{d \le n} P_{n-d} \otimes \mathcal{R}_d \longrightarrow P_n, \qquad f \otimes r \longmapsto fr. \tag{6.37}$$

The degree-n component of the ideal generated by the known relations and the corresponding quotient dimension are therefore,

$$I_n = \operatorname{im} \Psi_n, \qquad \dim(P/I)_n = \dim P_n - \operatorname{rank} \Psi_n. \tag{6.38}$$

The kernel of Ψ_n records linear dependencies among the degree-n descendants and hence contains the first syzygies relevant at that degree. In this way, the rank of the multiplication map, rather than the number of naively generated descendants, determines how many independent constraints are actually imposed.

A simple example occurs for $\mathrm{SU}(3)_0 + 6F$. Its degree-two generators include the $\mathrm{SU}(6)$ adjoint moment map M, while the degree-four relations include

$$Q = M^2 - \frac{\operatorname{tr} M^2}{6}\mathbf{1}_6 = 0. \tag{6.39}$$

Multiplication by M produces degree-six descendants. However, the two possible matrix products are not independent, since

$$MQ - QM = \left[M, M^2 - \frac{\operatorname{tr} M^2}{6}\mathbf{1}_6\right] = 0 \tag{6.40}$$

holds identically in the free polynomial ring. This is therefore not a new constraint on the quotient ring, but a syzygy among descendants of the degree-four relation. Because the commutator is traceless and transforms by conjugation, this syzygy lies in the $\mathrm{SU}(6)$ adjoint $\mathbf{35}$.

At degree six the reconstructed ideal also contains an independent primitive relation in the $(\mathbf{35}, \mathbf{1}, \mathbf{1})$. The primitive relation and the adjoint syzygy contribute with opposite signs to the plethystic logarithm,

$$-(\mathbf{35}, \mathbf{1}, \mathbf{1}) + (\mathbf{35}, \mathbf{1}, \mathbf{1}) = 0. \tag{6.41}$$

The absence of an adjoint contribution from the net degree-six plethystic coefficient therefore does not imply that no adjoint relation is present: its contribution is cancelled by the syzygy generated by the lower-degree ideal.

This motivates the final check on the reconstructed ring. After choosing the primitive relations, the ideal is generated degree by degree through the multiplication maps Ψ_n, and the resulting quotient is compared with the target Hilbert series,

$$\dim(P/I)_n \xrightarrow{?} [t^n]H(t). \tag{6.42}$$

A stronger test retains the symmetry refinement,

$$\dim(P/I)_{n,\omega} \xrightarrow{?} [t^n \mathbf{x}^\omega]H(t, \mathbf{x}), \tag{6.43}$$

where ω denotes a weight of the global symmetry. Agreement weight by weight verifies not only the total number of operators at each degree, but also their complete representation content. The ideal calculation therefore provides the appropriate consistency check once descendants and syzygies begin to obscure the direct interpretation of the plethystic logarithm.

7 Infinite Coupling Relations for $\mathrm{SU}(3)_k$

Using the procedure outlined in the previous chapter, we now turn to the explicit chiral-ring relations for the six theories. As a consistency check, the ideals they generate were tested degree by degree as described in Section 6.6; in every case, after accounting for descendants and syzygies, the resulting quotient P/I reproduces the corresponding Hilbert series through order t^{10}.

Each theory is presented by degree, beginning with the corresponding plethystic-logarithm coefficient, written in highest-weight notation for brevity, followed by the explicit covariant equation written in the ultraviolet operators. The four non-maximal Chern–Simons levels are treated first, where the relations admit a tensorial description, before turning to the two maximal chambers and their pure-spinor structure.

7.1 Non-maximal Levels

7.1.1 Five flavours, $|k| = \frac{1}{2}$

For $SU(3)_{\frac{1}{2}} + 5F$, the ultraviolet symmetry is $SU(5) \times SU(2) \times U(1)_q$. The degrees two and three terms of the plethystic logarithm are,

$$\mathrm{PL}[t^2] = \mu_1\mu_4 + \nu^2 + 1, \qquad \mathrm{PL}[t^3] = \nu\Big(q\mu_2 + q^{-1}\mu_3\Big)\,. \tag{7.1}$$

Using the charge embedding fixed in the previous chapter, these decompose into the physical operators,

d	Ultraviolet multiplet	Branched components
2	$(24,1)_0$	M
2	$(1,1)_0$	Σ
2	$(1,3)_0$	$H_{(0,0)},\ I_{(-5/2,\,1)},\ \widetilde{I}_{(5/2,\,-1)}$
3	$(10,2)_{+1}$	$B_{(-3,\,0)},\ (B_{-1})_{(-1/2,\,-1)}$
3	$(\overline{10},2)_{-1}$	$(\widetilde{B}_{+1})_{(1/2,\,1)},\ \widetilde{B}_{(3,\,0)}$

(7.2)

where the parenthesised subscripts record (Q_B, Q_I), while the labels on B_{-1} and $\widetilde{B}_{+1}$ record their instanton charge.

The infinte coupling generators are written,

$$\Phi = \begin{pmatrix} H & I \\ \widetilde{I} & -H \end{pmatrix}, \qquad \mathcal{B} = \begin{pmatrix} B \\ B_{-1} \end{pmatrix}, \qquad \widetilde{\mathcal{B}} = \begin{pmatrix} \widetilde{B}_{+1} \\ \widetilde{B} \end{pmatrix}. \tag{7.3}$$

The neutral ultraviolet scalars and shifted meson are,

$$H = 3\eta' - \frac{7}{2}S, \qquad \Sigma = 3\eta' + \frac{5}{2}S, \qquad \widehat{X} = M + \frac{\Sigma}{5}\mathbf{1}_5. \tag{7.4}$$

Degree four

The first relations occur at degree four,

$$\mathrm{PL}[t^4] = \mu_1\mu_4 + 2. \tag{7.5}$$

The adjoint and two singlet channels combine into,

$$\widehat{X}\left(\widehat{X} - \frac{\Sigma}{3}\mathbf{1}_5\right) = 0, \qquad \Sigma^2 - H^2 = I\widetilde{I}. \tag{7.6}$$

The first equation constrains the degree-two flavour current, while the second directly couples the neutral ultraviolet moment maps to the instanton–anti-instanton bilinear.

Degree five

At degree five,

$$\mathrm{PL}[t^5] = q\nu(\mu_1^2 + 2\mu_2 + \mu_3\mu_4) + q^{-1}\nu(\mu_4^2 + 2\mu_3 + \mu_1\mu_2). \tag{7.7}$$

These modules impose incidence relations between the degree-two currents and the baryonic doublets. For $\mathcal{B}$,

$$\widehat{X}^i{}_k\mathcal{B}^{kj}_\alpha = 0, \qquad P_{\overline{40}}(M \otimes \mathcal{B}_\alpha) = 0, \qquad \Sigma\mathcal{B} = \Phi\mathcal{B}, \tag{7.8}$$

together with the conjugate equations,

$$\widetilde{\mathcal{B}}_{\alpha,ik}\widehat{X}^k{}_j = 0, \qquad P_{40}\Big(M \otimes \widetilde{\mathcal{B}}_\alpha\Big) = 0, \qquad \Sigma\widetilde{\mathcal{B}} = -\Phi\widetilde{\mathcal{B}}. \tag{7.9}$$

Where P_R denotes the equivariant projection onto the indicated irreducible $SU(5)$ representation. Branching the enhanced $SU(2)$ relation gives,

$$(\Sigma - H)B = IB_{-1}, \quad (\Sigma + H)B_{-1} = \widetilde{I}\,B \quad \longrightarrow \quad 6SB = IB_{-1}, \quad (6\eta' - S)B_{-1} = \widetilde{I}\,B. \tag{7.10}$$

Degree six

The degree-six coefficient is

$$\begin{aligned} \mathrm{PL}[t^6] = {} & q^2(\mu_4\nu^2 + \mu_1\mu_3) + q^{-2}(\mu_1\nu^2 + \mu_2\mu_4) \\ & + (1 + \mu_2\mu_3) + \nu^2(1 + \mu_1\mu_4 + \mu_2\mu_3). \end{aligned} \tag{7.11}$$

The $U(1)_q$ charged products first impose the intrinsic constraints on the degree-three two-forms,

$$P_{(\bar{5},3)}\Big(\mathrm{Sym}^2\,\mathcal{B}\Big)=0, \qquad P_{(45,1)}\Big(\mathrm{Sym}^2\,\mathcal{B}\Big)=0, \tag{7.12}$$

with conjugate equations for $\widetilde{\mathcal{B}}$. Their components include,

$$\epsilon_{ijklm}B^{jk}B^{lm}=0, \qquad B^{ij}B^{kl}_{-1}-B^{ij}_{-1}B^{kl}=0, \tag{7.13}$$

so the ordinary Plücker condition on the classical baryon extends across the full enhanced doublet. The remaining relations constrain products of the two conjugate doublets. Define the enhanced-$SU(2)$ singlet and triplet combinations,

$$K_{\mathrm{s}}=B\widetilde{B}-B_{-1}\widetilde{B}_{+1}, \tag{7.14}$$

$$(K_+,K_0,K_-)=\left(-B\widetilde{B}_{+1},\,\frac{1}{2}(B\widetilde{B}+B_{-1}\widetilde{B}_{+1}),\,B_{-1}\widetilde{B}\right), \quad (h_+,h_0,h_-)=(I,H,\widetilde{I}). \tag{7.15}$$

Here products such as $B\widetilde{B}$ are uncontracted pair tensors,

$$(B\widetilde{B})^{ij}{}_{kl}=B^{ij}\widetilde{B}_{kl},$$

and hence transform in,

$$10\otimes\overline{10}=1\oplus24\oplus75.$$

The 75 component of the mesonic quadratic is most naturally written using the compound matrix. For $T:F\to F$, where F is the fundamental flavour space,

$$\mathrm{Comp}_r(T)^{\,i_1\ldots i_r}{}_{j_1\ldots j_r}=\det\Big(T^{i_a}{}_{j_b}\Big)^{r}_{a,b=1}, \tag{7.16}$$

so the entries of $\mathrm{Comp}_r(T)$ are precisely the $r\times r$ minors of T. Equivalently, it represents the induced map on $\Lambda^r F$, but the compound notation will be used to keep the minor structure explicit. For M,

$$\mathrm{Comp}_2(M)^{ij}{}_{kl}=M^i{}_kM^j{}_l-M^i{}_lM^j{}_k. \tag{7.17}$$

The mixed relations then take the form,

$$\begin{aligned}
&P_1K_{\mathrm{s}}=\frac{\Sigma^3}{27}, \qquad P_{24}K_{\mathrm{s}}=-\frac{\Sigma^2}{9}M, \qquad P_{75}K_{\mathrm{s}}=\frac{\Sigma}{3}P_{75}\,\mathrm{Comp}_2(M),\\
&P_1K_a=\frac{\Sigma^2h_a}{54}, \quad P_{24}K_a=-\frac{\Sigma h_a}{18}M, \quad P_{75}K_a=\frac{h_a}{6}P_{75}\,\mathrm{Comp}_2(M). \qquad a=+,0,-.
\end{aligned}$$

7.1.2 Five flavours, $|k|=\frac{3}{2}$

For $SU(3)_{\frac{3}{2}}+5F$, the ultraviolet symmetry is $SU(6)\times U(1)_q$. Under $SU(6)\supset SU(5)\times U(1)_x$, the required branchings are,

$$\mathbf{35}\longrightarrow\mathbf{24}_0\oplus\mathbf{5}_{+6}\oplus\overline{\mathbf{5}}_{-6}\oplus\mathbf{1}_0, \qquad \mathbf{15}\longrightarrow\mathbf{10}_{+2}\oplus\mathbf{5}_{-4}. \tag{7.18}$$

The degrees two and three terms of the plethystic logarithm are,

$$\mathrm{PL}[t^2] = \mu_1\mu_5 + 1, \qquad \mathrm{PL}[t^3] = q\mu_2 + q^{-1}\mu_4. \tag{7.19}$$

The corresponding generators therefore decompose into the physical operators

d	Ultraviolet multiplet	Branched components
2	$\mathbf{35}_0$	$M,\ H_{(0,0)},\ I_{(-3/2,1)},\ \widetilde{I}_{(3/2,-1)}$
2	$\mathbf{1}_0$	Σ
3	$\mathbf{15}_{+1}$	$B_{(-3,0)},\ (B_{-1})_{(-3/2,-1)}$
3	$\overline{\mathbf{15}}_{-1}$	$(\widetilde{B}_{+1})_{(3/2,1)},\ \widetilde{B}_{(3,0)}$

(7.20)

$$Q_B = -\frac{x+10q}{4}, \qquad Q_I = \frac{x-2q}{6}. \tag{7.21}$$

With the corresponding moment-map normalisation, the neutral ultraviolet scalars and traceful meson are,

$$H = -\frac{\eta'}{15} + \frac{1}{6}S, \qquad \Sigma = 3\eta' + \frac{3}{2}S, \qquad X = M + \frac{3\eta'}{5}\mathbf{1}_5. \tag{7.22}$$

The enhanced $SU(6)$ current and degree-three generators are,

$$\Phi = \begin{pmatrix} M + H\mathbf{1}_5 & I \\ \widetilde{I} & -5H \end{pmatrix}, \qquad \mathcal{B} = \begin{pmatrix} B & B_{-1} \\ -B_{-1}^{\mathsf{T}} & 0 \end{pmatrix}, \qquad \widetilde{\mathcal{B}} = \begin{pmatrix} \widetilde{B} & \widetilde{B}_{+1} \\ -\widetilde{B}_{+1}^{\mathsf{T}} & 0 \end{pmatrix}. \tag{7.23}$$

Degree four

The first relations occur at degree four,

$$\mathrm{PL}[t^4] = 1 + \mu_1\mu_5. \tag{7.24}$$

The singlet and adjoint channels combine into the factorised $SU(6)$-covariant equation,

$$\left(\Phi - \frac{\Sigma}{9}\mathbf{1}_6\right)\left(\Phi + \frac{2\Sigma}{9}\mathbf{1}_6\right) = 0. \tag{7.25}$$

Thus, for $\Sigma \neq 0$, Φ has eigenvalues $\Sigma/9$ and $-2\Sigma/9$, with multiplicities four and two fixed by tracelessness. Defining,

$$Z \equiv \Phi - \frac{\Sigma}{9}\mathbf{1}_6, \qquad Y \equiv \Phi + \frac{2\Sigma}{9}\mathbf{1}_6, \tag{7.26}$$

the fundamental decomposes as $\mathbb{C}^6 = \ker Z \oplus \ker Y$, with dimensions four and two respectively. In branched components, $ZY = 0$ gives

$$(X - \eta'\mathbf{1}_5)\left(X + \frac{S}{2}\mathbf{1}_5\right) = -I\widetilde{I}, \qquad \left(X - \frac{S}{2}\mathbf{1}_5\right) I = 0, \tag{7.27}$$

$$\widetilde{I}\left(X - \frac{S}{2}\mathbf{1}_5\right) = 0, \qquad S\left(\eta' - \frac{S}{2}\right) = \widetilde{I}I. \tag{7.28}$$

The final equation directly couples the glueball to the instanton–anti-instanton bilinear, while the remaining components complete the mesonic quadratic relation into the enhanced current multiplet.

Degree five

At degree five,

$$\mathrm{PL}[t^5] = q(\mu_1^2 + \mu_2 + \mu_3\mu_5) + q^{-1}(\mu_5^2 + \mu_4 + \mu_1\mu_3). \tag{7.29}$$

The incidence channels are compactly expressed as,

$$Y^a{}_c\mathcal{B}^{cb} = 0, \qquad \widetilde{\mathcal{B}}_{ac}Y^c{}_b = 0, \tag{7.30}$$

together with the hook relations

$$P_{105}(Z \otimes \mathcal{B}) = 0, \qquad P_{105}\left(Z^{\mathsf{T}} \otimes \widetilde{\mathcal{B}}\right) = 0. \tag{7.31}$$

Since $\ker Y$ is the two-dimensional eigenspace selected by the quadratic relation, the first incidence equation restricts the baryonic tensor to $\mathcal{B} \in \Lambda^2 \ker Y$, with the conjugate equation imposing the corresponding condition on $\widetilde{\mathcal{B}}$. In terms of branched operators the first relation contains,

$$\left(X + \frac{S}{2}\mathbf{1}_5\right)^i{}_k B^{kj} = I^i B^j_{-1}, \qquad \left(\eta' - \frac{S}{2}\right) B^j_{-1} = \widetilde{I}_k B^{kj}. \tag{7.32}$$

Degree six

The degree-six relation modules appearing in the net plethystic logarithm are,

$$\mathrm{PL}[t^6] = q^2\mu_4 + q^{-2}\mu_2 + 1 + \mu_2\mu_4 + \mu_3^2. \tag{7.33}$$

As in the preceding theory, the primitive relation space also contains an additional neutral adjoint $\mu_1\mu_5$, whose contribution is cancelled in the net plethystic logarithm by a syzygy generated by the degree-four equation $ZY = 0$. The same-charge products obey the Plücker relations,

$$\mathcal{B}^{ab}\mathcal{B}^{cd} - \mathcal{B}^{ac}\mathcal{B}^{bd} + \mathcal{B}^{ad}\mathcal{B}^{bc} = 0, \tag{7.34}$$

and,

$$\widetilde{\mathcal{B}}_{ab}\widetilde{\mathcal{B}}_{cd} - \widetilde{\mathcal{B}}_{ac}\widetilde{\mathcal{B}}_{bd} + \widetilde{\mathcal{B}}_{ad}\widetilde{\mathcal{B}}_{bc} = 0. \tag{7.35}$$

Together with the degree-five incidence equations, these identify the baryonic generators as decomposable two-forms supported on $\ker Y$. The remaining neutral relations are compactly,

$$\mathcal{B}\widetilde{\mathcal{B}} = \frac{\Sigma}{3}\,\mathrm{Comp}_2(Z), \qquad \mathrm{Comp}_3(Z) = 0. \tag{7.36}$$

The first supplies the singlet, adjoint and **189** mixed-product channels, including the primitive adjoint hidden by the degree-six syzygy cancellation. The second compactly represents the remaining primitive **175** relation, since its complementary $\mathbf{1} \oplus \mathbf{35} \oplus \mathbf{189}$ components already lie in the ideal generated by $ZY = 0$.

7.1.3 Six flavours, $k = 0$

For $SU(3)_0 + 6F$, the ultraviolet symmetry is $SU(6) \times SU(2)_x \times SU(2)_q$. The degrees two and three terms of the plethystic logarithm are,

$$\mathrm{PL}[t^2] = \mu_1\mu_5 + \nu_x^2 + \nu_q^2, \qquad \mathrm{PL}[t^3] = \nu_x\nu_q\mu_3. \tag{7.37}$$

The corresponding generators decompose into the physical operators,

d	Ultraviolet multiplet	Branched components
2	$(35,1,1)$	M
2	$(1,3,1)$	$I_{x\,(3,1)},\ H_{(0,0)},\ \widetilde{I}_{x\,(-3,-1)}$
2	$(1,1,3)$	$I_{q\,(3,-1)},\ \Sigma_{(0,0)},\ \widetilde{I}_{q\,(-3,1)}$
3	$(20,2,2)$	$\widetilde{B}_{(3,0)},\ (\widetilde{B}_{+1})_{(0,1)},\ (B_{-1})_{(0,-1)},\ B_{(-3,0)}$

(7.38)

$$Q_B = \frac{3}{2}(x+q), \qquad Q_I = \frac{1}{2}(x-q). \tag{7.39}$$

With the corresponding moment-map normalisation, the neutral ultraviolet scalars and traceful meson are,

$$H = -3\eta' - 3S, \qquad \Sigma = -3\eta' + 3S, \qquad X = M + \frac{3\eta'}{6}\mathbf{1}_6. \tag{7.40}$$

The two enhanced $SU(2)$ currents and the degree-three multiplet are,

$$\Phi_x = \begin{pmatrix} H & -I_x \\ \widetilde{I}_x & -H \end{pmatrix}, \qquad \Phi_q = \begin{pmatrix} \Sigma & -I_q \\ \widetilde{I}_q & -\Sigma \end{pmatrix}, \qquad \mathcal{B} = \begin{pmatrix} \widetilde{B} & \widetilde{B}_{+1} \\ B_{-1} & B \end{pmatrix}. \tag{7.41}$$

Degree four

The first relations occur at degree four,

$$\mathrm{PL}[t^4] = \mu_1\mu_5 + 2. \tag{7.42}$$

The adjoint and two singlet channels give,

$$M^2 - \frac{\operatorname{tr} M^2}{6}\mathbf{1}_6 = 0, \qquad H^2 - 6\operatorname{tr} M^2 = I_x\widetilde{I}_x, \qquad \Sigma^2 - 6\operatorname{tr} M^2 = I_q\widetilde{I}_q. \tag{7.43}$$

Thus the two enhanced $SU(2)$ moment maps are tied to the same quadratic mesonic invariant.

Degree five

At degree five,

$$\mathrm{PL}[t^5] = \nu_x\nu_q(2\mu_3 + \mu_1\mu_2 + \mu_4\mu_5). \tag{7.44}$$

For the **20** incidence channels, let M_{20} denote the action of the mesonic current on each three-form component of $\mathcal{B}$,

$$(M_{20}\mathcal{B})^{ijk} = M^i{}_\ell\mathcal{B}^{\ell jk} + M^j{}_\ell\mathcal{B}^{i\ell k} + M^k{}_\ell\mathcal{B}^{ij\ell}. \tag{7.45}$$

The two incidence relations are then,

$$2M_{20}\mathcal{B} = \Phi_x\mathcal{B} = \mathcal{B}\Phi_q^{\mathsf{T}}, \tag{7.46}$$

while the remaining irreducible channels are,

$$P_{70}(M \otimes \mathcal{B}) = 0, \qquad P_{\overline{70}}(M \otimes \mathcal{B}) = 0. \tag{7.47}$$

The chained equality identifies the mesonic action on the baryonic **20** with the actions of the two enhanced $SU(2)$ moment maps.

Degree six

The degree-six relation modules appearing in the net plethystic logarithm are,

$$\mathrm{PL}[t^6] = \mu_3^2 + \mu_1\mu_5\nu_x^2\nu_q^2 + (1+\mu_2\mu_4)(\nu_x^2+\nu_q^2). \tag{7.48}$$

The primitive relation space also contains an additional $(35,1,1)$ channel, whose contribution is cancelled in the net plethystic logarithm by an adjoint syzygy. The $SU(2)_x \times SU(2)_q$ singlet combination is,

$$K_{\mathrm{s}} = \widetilde{B} \odot B - \widetilde{B}_{+1} \odot B_{-1}, \tag{7.49}$$

where $\odot$ denotes the symmetric tensor product. Its primitive relations are,

$$P_{175}(K_{\mathrm{s}}) = \frac{1}{6}P_{175}\Big(M_{20}^3\Big), \qquad P_{35}(K_{\mathrm{s}}) = \frac{2}{15}P_{35}\Big(M_{20}^3\Big). \tag{7.50}$$

The second equation is the primitive adjoint relation hidden by the degree-six syzygy cancellation. The two triplet bilinears are,

$$T_x = \left(\widetilde{B}\wedge\widetilde{B}_{+1},\ \frac{1}{2}(\widetilde{B}\wedge B - \widetilde{B}_{+1}\wedge B_{-1}),\ -B_{-1}\wedge B\right), \tag{7.51}$$

$$T_q = \left(\widetilde{B}\wedge B_{-1},\ \frac{1}{2}(\widetilde{B}\wedge B + \widetilde{B}_{+1}\wedge B_{-1}),\ -\widetilde{B}_{+1}\wedge B\right), \tag{7.52}$$

where $\wedge$ denotes the antisymmetric tensor product and the entries are ordered by the weights $+,0,-$. Writing,

$$(\Phi_x)_m = (I_x, H, \widetilde{I}_x), \qquad (\Phi_q)_n = (I_q, \Sigma, \widetilde{I}_q), \qquad m,n \in \{+,0,-\}, \tag{7.53}$$

the $SU(2)_x$ triplet relations are,

$$P_1((T_x)_m) = \frac{1}{27}(\Phi_x)_m P_1\Big(M_{20}^2\Big), \qquad P_{189}((T_x)_m) = \frac{1}{12}(\Phi_x)_m P_{189}\Big(M_{20}^2\Big), \tag{7.54}$$

$$P_1((T_q)_n) = \frac{1}{27}(\Phi_q)_n P_1\Big(M_{20}^2\Big), \qquad P_{189}((T_q)_n) = \frac{1}{12}(\Phi_q)_n P_{189}\Big(M_{20}^2\Big). \tag{7.55}$$

Finally, the $(3,3)$ sector is formed from the nine symmetric bilinears,

Y_{mn}	$q:+$	$q:0$	$q:-$
$x:+$	$\widetilde{B}\odot\widetilde{B}$	$\widetilde{B}\odot\widetilde{B}_{+1}$	$\widetilde{B}_{+1}\odot\widetilde{B}_{+1}$
$x:0$	$\widetilde{B}\odot B_{-1}$	$\frac{1}{2}\big(\widetilde{B}\odot B+\widetilde{B}_{+1}\odot B_{-1}\big)$	$\widetilde{B}_{+1}\odot B$
$x:-$	$B_{-1}\odot B_{-1}$	$B_{-1}\odot B$	$B\odot B$

(7.56)

where the rows give the $SU(2)_x$ weight m and the columns the $SU(2)_q$ weight n. The nine relations are captured uniformly by,

$$P_{35}(Y_{mn})=\frac{1}{108}(\Phi_x)_m(\Phi_q)_n\,M_{20},\qquad m,n\in\{+,0,-\}. \tag{7.57}$$

7.1.4 Six flavours, $|k|=1$

For $SU(3)_1+6F$, the ultraviolet symmetry is $SU(7)\times U(1)_q$. Under $SU(7)\supset SU(6)\times U(1)_x$, with

$$\mathbf{7}\longrightarrow\mathbf{6}_{+1}\oplus\mathbf{1}_{-6}, \tag{7.58}$$

the relevant branchings are,

$$\mathbf{48}\longrightarrow\mathbf{35}_0\oplus\mathbf{6}_{+7}\oplus\overline{\mathbf{6}}_{-7}\oplus\mathbf{1}_0,\qquad \mathbf{35}\longrightarrow\mathbf{20}_{+3}\oplus\mathbf{15}_{-4}. \tag{7.59}$$

The degrees two and three terms of the plethystic logarithm are,

$$\mathrm{PL}[t^2]=\mu_1\mu_6+1,\qquad \mathrm{PL}[t^3]=q\mu_3+q^{-1}\mu_4. \tag{7.60}$$

The corresponding generators therefore decompose into the physical operators,

d	Ultraviolet multiplet	Branched components
2	$\mathbf{48}_0$	$M,\ H_{(0,0)},\ I_{(-2,1)},\ \widetilde{I}_{(2,-1)}$
2	$\mathbf{1}_0$	Σ
3	$\mathbf{35}_{+1}$	$B_{(-3,0)},\ (B_{-1})_{(-1,-1)}$
3	$\overline{\mathbf{35}}_{-1}$	$(\widetilde{B}_{+1})_{(1,1)},\ \widetilde{B}_{(3,0)}$

(7.61)

$$Q_B=-\frac{2x+15q}{7},\qquad Q_I=\frac{x-3q}{7}. \tag{7.62}$$

With the corresponding moment-map normalisation, the neutral ultraviolet scalars and traceful meson are,

$$H=-\frac{3\eta'+4S}{42},\qquad \Sigma=-3\eta'+3S,\qquad X=M+\frac{3\eta'}{6}\mathbf{1}_6. \tag{7.63}$$

The enhanced $SU(7)$ current is,

$$\Phi=\begin{pmatrix}M+H\mathbf{1}_6 & I\\ \widetilde{I} & -6H\end{pmatrix}. \tag{7.64}$$

Since,

$$\Lambda^3\mathbf{7} \longrightarrow \Lambda^3\mathbf{6}_{+3} \oplus \Lambda^2\mathbf{6}_{-4}, \tag{7.65}$$

the degree-three generators assemble into conjugate $SU(7)$ three-forms,

$$\mathcal{B}^{abc} = B^{abc}, \qquad \mathcal{B}^{ab7} = B^{ab}_{-1}, \qquad \widetilde{\mathcal{B}}_{abc} = \widetilde{B}_{abc}, \qquad \widetilde{\mathcal{B}}_{ab7} = \widetilde{B}_{+1\,ab}, \tag{7.66}$$

with $a, b, c = 1, \ldots, 6$.

Degree four

The first relations occur at degree four,

$$\mathrm{PL}[t^4] = 1 + \mu_1\mu_6. \tag{7.67}$$

The singlet and adjoint channels combine into the factorised $SU(7)$-covariant equation,

$$\left(\Phi + \frac{\Sigma}{7}\mathbf{1}_7\right)\left(\Phi - \frac{4\Sigma}{21}\mathbf{1}_7\right) = 0. \tag{7.68}$$

Thus, for $\Sigma \neq 0$, Φ has two eigenvalues with multiplicities four and three fixed by tracelessness. Defining,

$$Z \equiv \Phi + \frac{\Sigma}{7}\mathbf{1}_7, \qquad Y \equiv \Phi - \frac{4\Sigma}{21}\mathbf{1}_7, \tag{7.69}$$

the fundamental decomposes as $\mathbb{C}^7 = \ker Z \oplus \ker Y$, with dimensions four and three respectively. In terms of branched operators $ZY = 0$ gives,

$$\left(X + \left(-\eta' + \frac{S}{3}\right)\mathbf{1}_6\right)\left(X - \frac{2S}{3}\mathbf{1}_6\right) = -I\widetilde{I}, \qquad \left(X + \frac{S}{3}\mathbf{1}_6\right) I = 0, \tag{7.70}$$

$$\widetilde{I}\left(X + \frac{S}{3}\mathbf{1}_6\right) = 0, \qquad -\eta' S = \widetilde{I}I. \tag{7.71}$$

The final equation directly couples the glueball to the instanton–anti-instanton bilinear, while the remaining components complete the mesonic quadratic relation into the enhanced $SU(7)$ current.

Degree five

At degree five,

$$\mathrm{PL}[t^5] = q(\mu_3 + \mu_1\mu_2 + \mu_4\mu_6) + q^{-1}(\mu_4 + \mu_5\mu_6 + \mu_1\mu_3). \tag{7.72}$$

Using the $SU(7)$ volume form, define the dual four-forms,

$$\widehat{\mathcal{B}}_{ABCD} = \frac{1}{3!}\varepsilon_{ABCDEFG}\mathcal{B}^{EFG}, \qquad \widehat{\widetilde{\mathcal{B}}}^{ABCD} = \frac{1}{3!}\varepsilon^{ABCDEFG}\widetilde{\mathcal{B}}_{EFG}. \tag{7.73}$$

The incidence relations are,

$$Y^A{}_D\mathcal{B}^{DBC} = 0, \qquad Z^E{}_A\widehat{\mathcal{B}}_{EBCD} = 0, \tag{7.74}$$

together with their conjugates. Since $\dim\ker Y = 3$ and $\dim\ker Z = 4$, these restrict $\mathcal{B}$ to $\Lambda^3\ker Y$ and its dual to $\Lambda^4\ker Z$. The degree-three generator therefore selects the three-dimensional eigenspace of the enhanced current. For example, the first incidence equation contains,

$$\left(X - \frac{2S}{3}\mathbf{1}_6\right)^a{}_d B^{dbc} = -I^a B^{bc}_{-1}, \qquad \eta' B^{bc}_{-1} = -\widetilde{I}_d B^{dbc}. \tag{7.75}$$

Thus the perturbative baryon and its instanton-charged partner are different components of the same enhanced three-form.

Degree six

The degree-six relation modules appearing in the net plethystic logarithm are,

$$\mathrm{PL}[t^6] = q^2\mu_1\mu_5 + q^{-2}\mu_2\mu_6 + 1 + \mu_2\mu_5 + \mu_3\mu_4. \tag{7.76}$$

As in the preceding theories, the primitive relation space also contains an additional neutral adjoint $\mathbf{48}$, whose contribution is cancelled in the net plethystic logarithm by a syzygy generated by the degree-four relations. The same-charge products obey the Grassmannian Plücker relations,

$$P_{140}\Big(\mathrm{Sym}^2\,\mathcal{B}\Big) = 0, \qquad P_{\overline{140}}\Big(\mathrm{Sym}^2\,\widetilde{\mathcal{B}}\Big) = 0. \tag{7.77}$$

Together with the degree-five incidence equations, these make $\mathcal{B}$ a decomposable three-form supported on $\ker Y$, and similarly for its conjugate. The remaining neutral relations are captured by the single mixed equation,

$$\mathcal{B}^{ABC}\widetilde{\mathcal{B}}_{DEF} = -\,\mathrm{Comp}_3(Z)^{ABC}{}_{DEF}. \tag{7.78}$$

Since,

$$\mathbf{35}\otimes\overline{\mathbf{35}} = \mathbf{1}\oplus\mathbf{48}\oplus\mathbf{392}\oplus\mathbf{784}, \tag{7.79}$$

this relation supplies all four neutral channels at once, including the primitive adjoint relation hidden by the degree-six syzygy cancellation.

7.2 Maximal CS Levels and Pure Spinor Geometry

The two maximal Chern–Simons chambers differ from the preceding examples in the representation carried by their degree-three generators. For $(N_f, |k|) = (5, \frac{5}{2})$ and $(6, 2)$, the enhanced semisimple symmetries are respectively $SO(10)$ and $SO(12)$, and the baryonic generators are completed by instanton-charged operators into chiral spinors. Their quadratic relations are therefore naturally organised using the geometry of spinor representations, a perspective which has also recently proved useful in describing instanton sectors of five-dimensional Higgs branches [70].

Let V be a $2n$-dimensional complex vector space with a non-degenerate symmetric form, and let $S^\pm$ denote its two chiral spinor representations. Clifford multiplication associates to a non-zero spinor $\lambda \in S^\pm$ the annihilator,

$$\mathrm{Ann}(\lambda) = \{v \in V \mid \gamma(v)\lambda = 0\}. \tag{7.80}$$

The Clifford relation implies that $\mathrm{Ann}(\lambda)$ is isotropic. The spinor λ is called *pure* when this subspace has the maximal possible dimension n, so that it is maximal isotropic, or maximal totally null. Projectivising either chiral pure-spinor cone gives one connected component of $\mathrm{OGr}(n, 2n)$ in its spinor embedding, of dimension $n(n-1)/2$. For either chirality, the projecctive pure-spinor locus is a homogeneous $\mathrm{Spin}(2n)$ space [70].

Purity also admits a representation-theoretic description. The symmetric square of a pure spinor lies entirely in its highest-weight component; equivalently, the lower-form bilinears permitted by chirality vanish, leaving the middle-degree component. In the notation used below, these conditions are expressed through bilinears of the form $\lambda \cdot \gamma_{a_1 \ldots a_p} \lambda$. The representation containing the defining quadrics therefore depends on the dimension, which accounts for the different pure-spinor equations appearing in the D_5 and D_6 cases.

The relation to the finite-coupling operators becomes transparent after choosing a maximal isotropic splitting $V = F \oplus F^*$, with $F \simeq \mathbb{C}^n$. The spin module is then realised as the exterior algebra of F,

$$S^+ \simeq \Lambda^{\mathrm{even}} F, \qquad S^- \simeq \Lambda^{\mathrm{odd}} F. \tag{7.81}$$

Under $SU(n) \subset SO(2n)$, the separate tensor components of a chiral spinor are therefore identified with its different exterior degrees. In the theories below, the classical baryons and their instanton-charged partners assemble into the enhanced spinor in precisely this way.

For both chambers, take $V = F \oplus F^*$ with ordered basis $(e_1, \ldots, e_n, e^1, \ldots, e^n)$ and split metric,

$$\eta = \begin{pmatrix} 0 & \mathbf{1}_n \\ \mathbf{1}_n & 0 \end{pmatrix}. \tag{7.82}$$

Clifford multiplication on $\Lambda^\bullet F$ is realised by,

$$\gamma_i = e_i \wedge, \qquad \gamma_{\bar{i}} = \iota_i, \qquad \{\gamma_a, \gamma_b\} = \eta_{ab}, \tag{7.83}$$

where $\bar{i} = n+i$, and even and odd exterior degree give the two chiral modules. Thus $\gamma(v) = \sum_a v^a \gamma_a$; the convention with anticommutator 2η is obtained from $\Gamma_a = \sqrt{2}\,\gamma_a$. Antisymmetrised products are taken with unit weight,

$$\gamma_{a_1 \ldots a_p} = \frac{1}{p!} \sum_{\sigma \in S_p} \mathrm{sgn}(\sigma)\, \gamma_{a_{\sigma(1)}} \cdots \gamma_{a_{\sigma(p)}}.$$

Spinor bilinears are formed using the Chevalley pairing,

$$\chi \cdot \psi = \big[\chi \wedge \lambda_0(\psi)\big]_{e_{1\cdots n}}, \qquad \lambda_0(\psi_r) = (-1)^{r(r-1)/2} \psi_r, \tag{7.84}$$

and we write,

$$\beta^{(p)}_{a_1 \ldots a_p}(\chi, \psi) = \chi \cdot \gamma_{a_1 \ldots a_p} \psi. \tag{7.85}$$

Collectively, the tensors $\beta^{(p)}(\chi, \psi)$ are the differential-form components of the rank-one bispinor determined by χ and ψ. The corresponding Clifford, or Fierz, decomposition therefore identifies a spinor pair with a polyform. This will provide the natural language for the mixed spinor relations for the two cases below.

7.2.1 Five flavours, $|k| = \frac{5}{2}$:

For $SU(3)_{\frac{5}{2}} + 5F$, the ultraviolet symmetry is $SO(10) \times U(1)_q$. Under $SO(10) \supset SU(5) \times U(1)_x$, the required branchings are,

$$\mathbf{45} \longrightarrow \mathbf{24}_0 \oplus \mathbf{10}_{+4} \oplus \overline{\mathbf{10}}_{-4} \oplus \mathbf{1}_0, \tag{7.86}$$

$$\mathbf{16}_+ \longrightarrow \mathbf{1}_{-5} \oplus \mathbf{10}_{-1} \oplus \overline{\mathbf{5}}_{+3}, \qquad \mathbf{16}_- \longrightarrow \mathbf{5}_{-3} \oplus \overline{\mathbf{10}}_{+1} \oplus \mathbf{1}_{+5}. \tag{7.87}$$

The degrees two and three terms of the plethystic logarithm are,

$$\mathrm{PL}[t^2] = \mu_2 + 1, \qquad \mathrm{PL}[t^3] = q\mu_4 + q^{-1}\mu_5. \tag{7.88}$$

The corresponding generators therefore decompose into the physical operators,

d	Ultraviolet multiplet	Branched components
2	$\mathbf{45}_0$	$M,\ H_{(0,0)},\ I_{(-1/2,1)},\ \widetilde{I}_{(1/2,-1)}$
2	$\mathbf{1}_0$	Σ
3	$\mathbf{16}_{+,+1}$	$(B_{-1})_{(-5/2,-1)},\ B_{(-3,0)},\ (B_{+1})_{(-7/2,1)}$
3	$\mathbf{16}_{-,-1}$	$(\widetilde{B}_{-1})_{(7/2,-1)},\ \widetilde{B}_{(3,0)},\ (\widetilde{B}_{+1})_{(5/2,1)}$

(7.89)

$$Q_B = -\frac{x + 25q}{8}, \qquad Q_I = \frac{x+q}{4}. \tag{7.90}$$

With the corresponding moment-map normalisation, the neutral ultraviolet scalars are,

$$H = \frac{3\eta' - S}{30}, \qquad \Sigma = 3\eta' + S. \tag{7.91}$$

The enhanced $\mathrm{Spin}(10)$ current is,

$$\Phi = \begin{pmatrix} M + H\mathbf{1}_5 & I \\ \widetilde{I} & -M^{\mathsf{T}} - H\mathbf{1}_5 \end{pmatrix}, \qquad I^{\mathsf{T}} = -I, \qquad \widetilde{I}^{\mathsf{T}} = -\widetilde{I}. \tag{7.92}$$

The degree-three generators transform in the two chiral spinor representations, whose finite-symmetry components may be written as differential forms,

$$\mathcal{B} = (B_{-1})_{(0)} + (B)_{(2)} + (B_{+1})_{(4)} \in \mathbf{16}_+, \qquad \widetilde{\mathcal{B}} = (\widetilde{B}_{-1})_{(1)} + (\widetilde{B})_{(3)} + (\widetilde{B}_{+1})_{(5)} \in \mathbf{16}_-. \tag{7.93}$$

The parenthesised labels denote exterior degree. The classical baryon and antibaryon occupy the two- and three-form components, while the remaining degrees are their instanton-charged partners.

Degree four

The first relations occur at degree four,

$$\mathrm{HW}\Big(-[t^4]\,\mathrm{PL}\Big) = 1 + \mu_1^2. \tag{7.94}$$

The singlet and symmetric-traceless channels combine into the $\mathrm{Spin}(10)$-covariant equation,

$$\Phi^2 = \frac{\Sigma^2}{36}\mathbf{1}_{10}. \tag{7.95}$$

In $SU(5)$ components this gives,

$$(M + H\mathbf{1}_5)^2 - \frac{\Sigma^2}{36}\mathbf{1}_5 = -I\widetilde{I}, \tag{7.96}$$

together with,

$$(M + H\mathbf{1}_5)\, I - I\left(M^{\mathsf{T}} + H\mathbf{1}_5\right) = 0, \qquad \widetilde{I}\,(M + H\mathbf{1}_5) - \left(M^{\mathsf{T}} + H\mathbf{1}_5\right)\widetilde{I} = 0. \tag{7.97}$$

The mesonic and instanton-current quadratic relations are therefore the components of a single orthogonal-current equation.

Degree five

At degree five,

$$\mathrm{HW}\Big(-[t^5]\,\mathrm{PL}\Big) = q(\mu_4 + \mu_1\mu_5) + q^{-1}(\mu_5 + \mu_1\mu_4). \tag{7.98}$$

These modules impose the incidence of the orthogonal current with the two spinor generators,

$$\gamma\left[\left(\Phi - \frac{\Sigma}{6}\mathbf{1}_{10}\right)v\right]\mathcal{B} = 0, \qquad \gamma\left[\left(\Phi + \frac{\Sigma}{6}\mathbf{1}_{10}\right)v\right]\widetilde{\mathcal{B}} = 0, \qquad v \in F \oplus F^*. \tag{7.99}$$

Expanding these equations in exterior degree reproduces the baryonic incidence relations together with their instanton-charged completion.

Degree six

The primitive degree-six relation modules are,

$$(q^2 + q^{-2})\mu_1 + 1 + \mu_2 + 2\mu_4\mu_5. \tag{7.100}$$

A μ_1^2 syzygy among descendants of the degree-four relations also contributes at this order and is therefore not an additional primitive equation. The q-charged products give the $\mathrm{Spin}(10)$ pure-spinor quadrics,

$$\mathcal{B}\cdot\gamma_a\mathcal{B} = 0, \qquad \widetilde{\mathcal{B}}\cdot\gamma_a\widetilde{\mathcal{B}} = 0. \tag{7.101}$$

Each degree-three generator is therefore constrained to the affine cone over the corresponding D_5 spinor variety. Writing the higher exterior components in their Hodge-dual $SU(5)$ form, these equations become,

$$(B_{-1})(B_{+1})_i - \frac{1}{8}\epsilon_{ijklm}B^{jk}B^{lm} = 0, \qquad B^{ij}(B_{+1})_j = 0, \tag{7.102}$$

and

$$(\widetilde{B}_{+1})(\widetilde{B}_{-1})^i - \frac{1}{8}\epsilon^{ijklm}\widetilde{B}_{jk}\widetilde{B}_{lm} = 0, \qquad \widetilde{B}_{ij}(\widetilde{B}_{-1})^j = 0. \tag{7.103}$$

The remaining neutral relations couple the two chiralities. Under $\mathfrak{so}(10) \simeq \Lambda^2 V$, the orthogonal current defines the two-form,

$$\Phi_{(2)} = \sum_{a<b} (\eta\Phi)_{ab}\, a^a \wedge a^b. \tag{7.104}$$

The mixed equations are,

$$\beta^{(0)}(\widetilde{\mathcal{B}}, \mathcal{B}) = -\frac{\Sigma^3}{27}, \qquad \beta^{(2)}(\widetilde{\mathcal{B}}, \mathcal{B}) = -\frac{\Sigma^2}{9}\Phi_{(2)}, \tag{7.105}$$

$$\beta^{(4)}(\widetilde{\mathcal{B}}, \mathcal{B}) = -\frac{\Sigma}{6}\Phi_{(2)}^{\wedge 2}, \qquad \star\Phi_{(2)}^{\wedge 3} = \frac{\Sigma}{2}\Phi_{(2)}^{\wedge 2}. \tag{7.106}$$

The scalar, two-form and four-form components of the mixed bispinor are thus fixed by the orthogonal current, while the final equation supplies the second four-form relation required at this degree.

7.2.2 Six flavours, $|k| = 2$:

For $SU(3)_2 + 6F$, the ultraviolet symmetry is $SO(12) \times U(1)_q$. Under $SO(12) \supset SU(6) \times U(1)_x$, the required branchings are,

$$\mathbf{66} \longrightarrow \mathbf{35}_0 \oplus \mathbf{15}_{+2} \oplus \overline{\mathbf{15}}_{-2} \oplus \mathbf{1}_0, \tag{7.107}$$

$$\mathbf{32}_+ \longrightarrow \mathbf{6}_{-2} \oplus \mathbf{20}_0 \oplus \overline{\mathbf{6}}_{+2}. \tag{7.108}$$

The degrees two and three terms of the plethystic logarithm are,

$$\mathrm{PL}[t^2] = \mu_2 + 1, \qquad \mathrm{PL}[t^3] = (q + q^{-1})\mu_6. \tag{7.109}$$

The corresponding generators therefore decompose into the physical operators

d	Ultraviolet multiplet	Branched components
2	$\mathbf{66}_0$	$M,\ H_{(0,0)},\ I_{(-1,1)},\ \widetilde{I}_{(1,-1)}$
2	$\mathbf{1}_0$	Σ
3	$\mathbf{32}_{+,+1}$	$(\widetilde{B}_{-1})_{(4,-1)},\ \widetilde{B}_{(3,0)},\ (\widetilde{B}_{+1})_{(2,1)}$
3	$\mathbf{32}_{+,-1}$	$(B_{-1})_{(-2,-1)},\ B_{(-3,0)},\ (B_{+1})_{(-4,1)}$

(7.110)

$$Q_B = 3q - \frac{x}{2}, \qquad Q_I = \frac{x}{2}. \tag{7.111}$$

The two neutral degree-two directions admit a direct physical interpretation in this chamber,

$$H = S, \qquad \Sigma = 3\eta'. \tag{7.112}$$

Thus the singlet component contained in the $\mathrm{Spin}(12)$ current is the glueball itself, while the separate ultraviolet singlet is the classical mesonic trace direction. No further rotation of the neutral scalars is required, and the relations below will therefore be written directly in terms of S and η'.

The enhanced $\mathrm{Spin}(12)$ current is,

$$\Phi = \begin{pmatrix} M + S\mathbf{1}_6 & I \\ \widetilde{I} & -M^{\mathsf{T}} - S\mathbf{1}_6 \end{pmatrix}, \qquad I^{\mathsf{T}} = -I, \qquad \widetilde{I}^{\mathsf{T}} = -\widetilde{I}. \tag{7.113}$$

The degree-three generators transform in the same chiral spinor representation but carry opposite $U(1)_q$ charge. Using the exterior-form realisation introduced above, their branched components are,

$$\widetilde{\mathcal{B}} = (\widetilde{B}_{-1})_{(1)} + (\widetilde{B})_{(3)} + (\widetilde{B}_{+1})_{(5)} \in \mathbf{32}_{+,+1}, \qquad \mathcal{B} = (B_{-1})_{(1)} + (B)_{(3)} + (B_{+1})_{(5)} \in \mathbf{32}_{+,-1}. \tag{7.114}$$

The parenthesised labels denote exterior degree. The classical antibaryon and baryon occupy the three-form components, while the one- and five-form components are their instanton-charged partners.

Degree four

The first relations occur at degree four,

$$\mathrm{HW}\Big(-[t^4]\,\mathrm{PL}\Big) = 1 + \mu_1^2. \tag{7.115}$$

The singlet and symmetric-traceless channels combine into the $\mathrm{Spin}(12)$-covariant equation,

$$\Phi^2 = \frac{\eta'^2}{4}\mathbf{1}_{12}. \tag{7.116}$$

In $SU(6)$ components this gives,

$$(M + S\mathbf{1}_6)^2 - \frac{\eta'^2}{4}\mathbf{1}_6 = -I\widetilde{I}, \tag{7.117}$$

together with,

$$(M + S\mathbf{1}_6)\, I - I\left(M^{\mathsf{T}} + S\mathbf{1}_6\right) = 0, \tag{7.118}$$

$$\widetilde{I}\,(M + S\mathbf{1}_6) - \left(M^{\mathsf{T}} + S\mathbf{1}_6\right)\widetilde{I} = 0. \tag{7.119}$$

Thus the mesonic, glueball and instanton-current quadratic constraints are components of a single orthogonal-current relation.

Degree five

At degree five,

$$\mathrm{HW}\Big(-[t^5]\,\mathrm{PL}\Big) = (q + q^{-1})\,(\mu_6 + \mu_1\mu_5)\,. \tag{7.120}$$

The corresponding incidence relations are,

$$\gamma\left[\left(\Phi - \frac{\eta'}{2}\mathbf{1}_{12}\right)v\right]\widetilde{\mathcal{B}} = 0, \qquad \gamma\left[\left(\Phi + \frac{\eta'}{2}\mathbf{1}_{12}\right)v\right]\mathcal{B} = 0, \qquad v \in F \oplus F^*. \tag{7.121}$$

Clifford multiplication by the vector v changes spinor chirality, accounting for the $\mu_6 + \mu_1\mu_5$ relation content. Expanding these equations in exterior degree reproduces the baryonic incidence relations

together with their instanton-charged completion.

Degree six

The primitive degree-six relation modules are

$$(q^2+q^{-2})\mu_2+1+\mu_2+\mu_4+\mu_6^2+\mu_5^2. \tag{7.122}$$

A μ_1^2 syzygy among descendants of the degree-four relations also appears at this order and is not an additional primitive equation. The q-charge products give the $\mathrm{Spin}(12)$ pure-spinor quadrics,

$$\mathcal{B}\cdot\gamma_{ab}\mathcal{B}=0, \qquad \widetilde{\mathcal{B}}\cdot\gamma_{ab}\widetilde{\mathcal{B}}=0. \tag{7.123}$$

Each degree-three generator is therefore constrained to the affine pure-spinor cone. In this dimension,

$$\mathrm{Sym}^2(\mathbf{32})=\mathbf{66}\oplus\mathbf{462}, \tag{7.124}$$

so the pure-spinor equations remove the two-form $\mathbf{66}$ component, leaving the Cartan-square $\mathbf{462}$. The remaining neutral relations couple the two oppositely charged spinors. Under $\mathfrak{so}(12)\simeq\Lambda^2 V$, the enhanced current defines the two-form,

$$\Phi_{(2)}=\sum_{a<b}(\eta\Phi)_{ab}\,a^a\wedge a^b. \tag{7.125}$$

Using the negative ordered twelve-dimensional volume, define the six-form projectors,

$$\mathcal{P}_\pm=\frac{1}{2}(1\pm\star). \tag{7.126}$$

The mixed-bispinor equations are,

$$\beta^{(0)}(\widetilde{\mathcal{B}},\mathcal{B})=-\eta'^3, \qquad \beta^{(2)}(\widetilde{\mathcal{B}},\mathcal{B})=\eta'^2\Phi_{(2)}, \tag{7.127}$$

$$\beta^{(4)}(\widetilde{\mathcal{B}},\mathcal{B})=-\frac{\eta'}{2}\Phi_{(2)}^{\wedge 2}, \qquad \mathcal{P}_+\beta^{(6)}(\widetilde{\mathcal{B}},\mathcal{B})=\frac{1}{6}\mathcal{P}_+\Phi_{(2)}^{\wedge 3}, \tag{7.128}$$

together with,

$$\mathcal{P}_-\Phi_{(2)}^{\wedge 3}=0. \tag{7.129}$$

These occupy the $\mathbf{1}$, $\mathbf{66}$, $\mathbf{495}$, $\mathbf{462}_+$ and $\mathbf{462}_-$ channels, respectively. The final projected current equation supplies the second six-form chirality channel required at this degree.

8 Analysis and Outlook

Having constructed the explicit relations carving out the infinite coupling Higgs branch geometry, we may now begin to extract physical information from their structure, beyond what was possible from counting data alone. Despite being a tremendous exercise in representation theory, these relations allow us to understand how the non-perturbative instantons complete the finite coupling operator content in the enhanced global symmetry. A first consequence concerns the glueball operator.

In four-dimensional $\mathcal{N} = 1$ pure $SU(N)$ Yang–Mills, the glueball superfield is nilpotent in the classical chiral ring. At the classical level, repeated products of the gaugino bilinear cannot remain independent indefinitely: the anticommuting nature of the gauginos, together with the finite gauge-index structure, produces nilpotent relations in the chiral ring. Cachazo, Douglas, Seiberg and Witten showed that for $SU(N)$ the resulting classical relation is $S^N = 0$ [71]. Witten subsequently extended this result to the other classical gauge groups, for which the nilpotency degree is governed by the dual Coxeter number,

$$S^{h^\vee_G} = 0, \tag{8.1}$$

with $h^\vee_G = N$ for $SU(N)$ [72]. In five dimensions, Cremonesi, Ferlito, Hanany and Mekareeya showed that this finite-coupling nilpotency is promoted at infinite coupling to a non-perturbatively corrected relation. With an appropriate relative normalisation of the glueball and instanton operators, the pure Yang–Mills constraint becomes,

$$S^{h^\vee_G} = I\widetilde{I}. \tag{8.2}$$

The glueball is therefore no longer nilpotent at infinite coupling: its powers are continued into the non-perturbative instanton sector rather than terminating. Moving away from the fixed point makes the instanton states massive, and completed relation then reduces once more to $S^{h^\vee_G} = 0$, recovering the finite-coupling nilpotency [32].

For the matter-coupled theories studied here, the finite-coupling constraint is less immediate, since the glueball appears together with the mesonic trace and its nilpotency can require several ultraviolet relations to be combined. This question is particularly relevant for $SU(N)$ theories, Hanany and Van den Driessche raised the possibility that the degree of glueball nilpotency could depend on the the Chern–Simons level [69]. The relations derived here address this question for the six $SU(3)$ theories considered; the branched finite coupling relations show that every Chern–Simons level gives $S^3 = 0$ with $S^2 \neq 0$. Thus, despite the substantial k-dependence on the mechanism with which the non-preturbative operators complete the relations, the nilpotency degree remains fixed at $h^\vee_{SU(3)} = 3$ in all six cases. The explicit derivations are collected in Appendix G.

The other neutral scalar, η', may also a priori be nilpotent. However, for $N_f \geq N_c$ there are enough independent quark flavours to form non-vanishing baryonic operators while the traceful meson remains non-zero.[1] This behaviour is reproduced explicitly by the finite-coupling relations derived here: they couple η' to the mesonic and baryonic operators without generating any pure-power constraint $(\eta')^p = 0$. This is the expected behaviour and is consistent with the general SQCD result in the regime $N_f \geq 2N_c - 1$ [66].

The glueball nilpotency marks an important limitation of the geometric methods used throughout this work. Brane webs and magnetic quivers determine the Higgs branch as an algebraic variety, so they capture only the *reduced coordinate ring*. If the full gauge-invariant Higgs ring is

$$R = \frac{\mathbb{C}[x_i]}{I}, \qquad R_{\text{red}} = \frac{\mathbb{C}[x_i]}{\sqrt{I}} = \frac{R}{\sqrt{(0)}}, \tag{8.3}$$

then passing to R_{red} removes every nilpotent element. Thus the Hilbert series or HWG computed from the magnetic-quiver geometry counts functions on the reduced variety only: any nilpotent structure has already been discarded before the counting problem is formulated. This is the distinction made explicit by Bourget *et al.* [66], a finite-coupling magnetic quiver cannot by itself retain a nilpotent glueball direction.

The infinite-coupling ring avoids this problem as the instanton completions lift the glueball nilpotency. Although the ultraviolet coordinate ring is still reduced, it retains S as a non-vanishing coordinate. Once the explicit ultraviolet relations have been reconstructed, one can recover the full finite coupling ring by branching to the finite symmetry and setting the insanton contributions to zero. Algebraically, this is equivalent to taking a quotient of the reduced ultraviolet ring by the ideal generated by those components, and such a quotient need not remain reduced.

Recovering this information makes it possible to ask more refined questions about the geometric structure carried by the nilpotent glueball sector. For the $SU(3) + 5F$ theories, the finite-coupling relations contain

$$SB = S\widetilde{B} = 0, \tag{8.4}$$

together with

$$S\,M^2 = 0, \qquad S\,\mathrm{Comp}_2(M) = 0, \tag{8.5}$$

where $\mathrm{Comp}_2(M)$ is the second compound matrix, whose entries are the 2×2 minors of M. Thus the S-dependent part of the ring is annihilated by the baryonic operators and by the equations

$$M^2 = 0, \qquad \operatorname{rank} M \leq 1. \tag{8.6}$$

These equations imply a constrained locus over which the glueball structure may be supported, they are precisely the defining mesonic conditions of the minimal height-two nilpotent orbit closure,

$$\overline{\mathcal{O}}_{(2,1^3)} \subset \mathfrak{sl}_5, \tag{8.7}$$

suggesting that the mesonic support is organised by the corresponding nilpotent-orbit geometry.

[1] See Appendix E

A higher-rank example was calculated in search for a similar structure. For $SU(4)_{\frac{1}{2}} + 7F$, [2] the relations include

$$S\ \mathrm{Comp}_3(M) = 0, \qquad S^2\ \mathrm{Comp}_2(M) = 0, \qquad S^3 M = 0, \qquad S^4 = 0. \tag{8.8}$$

Together with the square-zero mesonic constraint, the successive powers of S are shown to be supported on increasingly restrictive rank loci,

$$S:\ \mathrm{rank}\, M \leq 2, \qquad S^2:\ \mathrm{rank}\, M \leq 1, \qquad S^3:\ M = 0. \tag{8.9}$$

The first two correspond naturally to the height-two orbit closures

$$\overline{\mathcal{O}}_{(2,1^5)} \subset \overline{\mathcal{O}}_{(2^2,1^3)} \subset \mathfrak{sl}_7. \tag{8.10}$$

The two examples suggest a possible general pattern for the glueball-supported strata. For $1 \leq p \leq N_c - 1$, a natural conjecture is

$$\mathrm{Supp}\left(\frac{S^p R_{\mathrm{finite}}}{S^{p+1} R_{\mathrm{finite}}}\right)_{\mathrm{red}} = \overline{\mathcal{O}}_{(2^{N_c-p-1},\, 1^{N_f-2N_c+2p+2})}, \tag{8.11}$$

so that each successive power of S is supported on the next lower height-two nilpotent-orbit closure. Correspondingly, one is led to the candidate glueball contribution,

$$\Delta\,\mathrm{HWG}(t;\boldsymbol{\mu}) = \sum_{d=1}^{N_c-1} t^{2d}\,\mathrm{PE}\left[\sum_{j=1}^{N_c-d-1} \mu_j \mu_{N_f-j} t^{2j}\right]. \tag{8.12}$$

As a preliminary check, the difference between the Hilbert series of R_{finite} and of the quotient $R_{\mathrm{finite}}/(S)$ was computed for $SU(3)_{\frac{1}{2}} + 5F$ and $SU(4)_{\frac{1}{2}} + 7F$, and in both cases agrees with this pattern. At present this should only be regarded as a hint at a possible general structure. Further checks require computing the corresponding Hilbert series for the remaining theories and in particular, determining whether the same stratification survives at the maximal Chern–Simons levels, or whether it is instead a special feature of the particular relation between colour and flavour in the family considered here.

The explicit relations also open several directions which have not been pursued here. Most immediately, the Higgs-branch coordinate ring carries a Poisson structure inherited from the holomorphic symplectic form, and determining the Poisson brackets among the mesonic, baryonic, glueball and instanton generators would supply information absent from the commutative relations alone. The relations can likewise be used to investigate singular and symplectic-leaf stratifications and the behaviour of the operator algebra under partial Higgsing and mass deformations. Extending these calculations to higher colour rank would test whether the apparent relation between glueball power and available mesonic rank is a genuine structural property of five-dimensional SQCD or a feature of the low-rank families considered here.

[2] The relations are given in Appendix G.

A The Five-Dimensional Prepotential

The low-energy dynamics on the Coulomb branch of a five-dimensional $\mathcal{N}=1$ gauge theory is encoded by a real prepotential $\mathcal{F}(\phi)$. This appendix derives this structure by dimensional reduction from four-dimensional $\mathcal{N}=2$ supersymmetry, explains the restriction to a locally cubic function, and determines the classical and perturbative contributions to $\mathcal{F}$.

A.1 Four-dimensional origin

Consider a generic point on the Coulomb branch of a four-dimensional $\mathcal{N}=2$ theory, where the gauge symmetry is broken to $U(1)^r$. In $\mathcal{N}=1$ language, each Abelian vector multiplet decomposes as,

$$\mathcal{V}^i_{\mathcal{N}=2} \longrightarrow (\Phi^i, V^i), \qquad a^i \equiv \Phi^i|_{\theta=0}, \qquad i=1,\dots,r. \tag{A.1}$$

The most general local two-derivative action compatible with the manifest $\mathcal{N}=1$ supersymmetry contains,

$$S = \int d^4x\, d^4\theta\, K(\Phi,\bar\Phi) + \frac{1}{16\pi i}\int d^4x\, d^2\theta\, \tau_{ij}(\Phi) W^{i\alpha} W^j_\alpha + \text{h.c.}, \tag{A.2}$$

where K is real and τ_{ij} is holomorphic. Since $W^{i\alpha}W^j_\alpha$ is symmetric,

$$\tau_{ij} = \tau_{ji}. \tag{A.3}$$

For $\mathcal{N}=1$ supersymmetry alone, K and τ_{ij} are independent. The second supersymmetry relates them. Writing the two Weyl fermions in each $\mathcal{N}=2$ vector multiplet as ψ^i and λ^i, its action schematically includes,

$$\delta_\eta a^i \sim \eta\lambda^i, \qquad \delta_\eta A^i_\mu \sim \eta\sigma_\mu\bar\psi^i - \psi^i\sigma_\mu\bar\eta, \tag{A.4}$$

with the remaining transformations fixed by closure of the $\mathcal{N}=2$ algebra. The fermionic kinetic terms extracted from Eq. (A.2) are,

$$\begin{aligned}\mathcal{L}_{\text{kin}} \supset\ & -K_{i\bar j}\,\partial_\mu a^i \partial^\mu \bar a^{\bar j} - iK_{i\bar j}\,\bar\psi^{\bar j}\bar\sigma^\mu\partial_\mu\psi^i \\ & + \frac{\tau_{ij}-\bar\tau_{ij}}{8\pi i}\left(-i\bar\lambda^j\bar\sigma^\mu\partial_\mu\lambda^i + \cdots\right),\end{aligned} \tag{A.5}$$

so invariance under the second supersymmetry requires,

$$K_{i\bar j} = \frac{\tau_{ij}-\bar\tau_{ij}}{8\pi i} = \frac{1}{4\pi}\,\mathrm{Im}\,\tau_{ij}. \tag{A.6}$$

The precise numerical factor depends on convention; the important point is that the scalar metric and gauge-coupling matrix are governed by the same function. The same action also contains interactions of the form,

$$\mathcal{L} \supset \frac{1}{16\pi i}\frac{\partial \tau_{ij}}{\partial a^k}\,\lambda^i \sigma^{\mu\nu}\psi^k F^j_{\mu\nu} + \text{h.c.}, \tag{A.7}$$

and the second supersymmetry therefore imposes,

$$\partial_k \tau_{ij} = \partial_i \tau_{kj}. \tag{A.8}$$

Equivalently, this follows directly from, Eq. (A.6). The Kähler condition and holomorphy give

$$\partial_k K_{j\bar{i}} = \partial_j K_{k\bar{i}} \quad \Longrightarrow \quad \partial_k \tau_{ji} = \partial_j \tau_{ki}, \tag{A.9}$$

and, using Eq. (A.3),

$$\partial_k \tau_{ij} = \partial_j \tau_{ik}. \tag{A.10}$$

This integrability condition implies that the entire coupling matrix is locally generated by a single holomorphic function. For fixed i, introduce,

$$\omega_i = \tau_{ij}\, da^j. \tag{A.11}$$

Then

$$d\omega_i = \frac{1}{2}\left(\partial_k \tau_{ij} - \partial_j \tau_{ik}\right) da^k \wedge da^j = 0. \tag{A.12}$$

On a contractible patch, the Poincaré lemma therefore gives,

$$\omega_i = d\mathcal{F}_i, \qquad \tau_{ij} = \partial_j \mathcal{F}_i. \tag{A.13}$$

Symmetry of τ_{ij} then implies,

$$\partial_j \mathcal{F}_i = \partial_i \mathcal{F}_j. \tag{A.14}$$

Hence the one-form,

$$\vartheta = \mathcal{F}_i\, da^i \tag{A.15}$$

is also closed,

$$d\vartheta = \frac{1}{2}\left(\partial_j \mathcal{F}_i - \partial_i \mathcal{F}_j\right) da^j \wedge da^i = 0, \tag{A.16}$$

and locally there exists a holomorphic function $\mathcal{F}(a)$ such that,

$$\mathcal{F}_i = \partial_i \mathcal{F}, \qquad \tau_{ij} = \frac{\partial^2 \mathcal{F}}{\partial a^i \partial a^j}. \tag{A.17}$$

Thus the coupling matrix is the Hessian of the four-dimensional $\mathcal{N} = 2$ prepotential. The action can consequently be written entirely in terms of, $\mathcal{F}$,

$$S_{\mathcal{N}=2} = \frac{1}{4\pi}\,\text{Im}\left[\int d^4x\, d^4\theta\, \bar{\Phi}^i \frac{\partial \mathcal{F}}{\partial \Phi^i} + \frac{1}{2}\int d^4x\, d^2\theta\, \frac{\partial^2 \mathcal{F}}{\partial \Phi^i \partial \Phi^j} W^{i\alpha} W^j_\alpha\right], \tag{A.18}$$

up to the chosen normalization. Its bosonic terms include,

$$\begin{aligned}\mathcal{L}^{(4d)}_{\rm bos} \supset\ & -\frac{1}{4\pi}\,\mathrm{Im}\,\mathcal{F}_{ij}\,\partial_\mu a^i \partial^\mu \bar{a}^j - \frac{1}{16\pi}\,\mathrm{Im}\,\mathcal{F}_{ij}\,F^i_{\mu\nu}F^{j\,\mu\nu} \\ & + \frac{1}{16\pi}\,\mathrm{Re}\,\mathcal{F}_{ij}\,F^i_{\mu\nu}\widetilde{F}^{j\,\mu\nu} + \cdots\,. \end{aligned} \tag{A.19}$$

A.2 Five-dimensional prepotential and cubic restriction

A five-dimensional $\mathcal{N}=1$ vector multiplet contains,

$$(A^i_M, \phi^i, \lambda^{iA}), \qquad M = 0, \ldots, 4. \tag{A.20}$$

For the Abelian low-energy theory, let $t_{ij}(\phi)$ denote the symmetric kinetic matrix. The bosonic two-derivative terms contain,

$$\mathcal{L}_{5d} \supset -\frac{1}{4}t_{ij}(\phi)F^i_{MN}F^{j\,MN} - \frac{1}{2}t_{ij}(\phi)\partial_M\phi^i\partial^M\phi^j + \cdots\,. \tag{A.21}$$

Compactifying the fifth direction and retaining only the zero modes, with $F^i_{\mu 4} = \partial_\mu A^i_4$,

$$\begin{aligned}\mathcal{L}_{4d} \supset\ & -\frac{1}{4}t_{ij}(\phi)F^i_{\mu\nu}F^{j\,\mu\nu} \\ & - \frac{1}{2}t_{ij}(\phi)\left(\partial_\mu\phi^i\partial^\mu\phi^j + \partial_\mu A^i_4\partial^\mu A^j_4\right) + \cdots\,. \end{aligned} \tag{A.22}$$

The fields ϕ^i and A^i_4 combine into the complex scalar of a four-dimensional $\mathcal{N}=2$ vector multiplet,

$$a^i = \phi^i + iA^i_4, \tag{A.23}$$

where a radius-dependent normalization has been absorbed into A^i_4. The reduced theory therefore admits a four-dimensional prepotential. Restricting to $A^i_4 = 0$ and absorbing the conventional normalization difference between the four- and five-dimensional definitions gives a real function $\mathcal{F}_{5d}(\phi)$ satisfying

$$t_{ij}(\phi) = \frac{\partial^2\mathcal{F}_{5d}}{\partial\phi^i\partial\phi^j}. \tag{A.24}$$

This is the dimensional-reduction argument used by Seiberg to introduce the five-dimensional prepotential [1]. Five-dimensional gauge invariance places a further restriction on its form. Supersymmetry completes the scalar-dependent kinetic terms with an Abelian Chern–Simons interaction [1, 2],

$$S_{\rm CS} = \frac{1}{24\pi^2}\int \mathcal{F}_{ijk}(\phi)\,A^i\wedge F^j\wedge F^k, \qquad \mathcal{F}_{ijk} \equiv \partial_i\partial_j\partial_k\mathcal{F}. \tag{A.25}$$

Under $A^i \to A^i + d\lambda^i$,

$$\begin{aligned}\delta S_{\rm CS} &= \frac{1}{24\pi^2}\int \mathcal{F}_{ijk}\,d\lambda^i\wedge F^j\wedge F^k \\ &= \frac{1}{24\pi^2}\int d\Big(\mathcal{F}_{ijk}\lambda^i F^j\wedge F^k\Big) - \frac{1}{24\pi^2}\int \lambda^i\,d\mathcal{F}_{ijk}\wedge F^j\wedge F^k. \end{aligned} \tag{A.26}$$

The first term is a boundary contribution whereas,

$$d\mathcal{F}_{ijk} = \mathcal{F}_{ijkl}\, d\phi^l. \tag{A.27}$$

Gauge invariance of the bulk theory for arbitrary field configurations therefore requires,

$$\mathcal{F}_{ijkl} = 0, \tag{A.28}$$

and hence the prepotential is locally at most cubic,

$$\mathcal{F}(\phi) = c_0 + c_i\phi^i + \frac{1}{2}c_{ij}\phi^i\phi^j + \frac{1}{6}c_{ijk}\phi^i\phi^j\phi^k. \tag{A.29}$$

The constant and linear pieces satisfy,

$$\partial_i\partial_j(c_0 + c_k\phi^k) = 0, \tag{A.30}$$

so they do not enter the local vector-multiplet Lagrangian. The prepotential is therefore defined by this Lagrangian only up to an affine function. [1]

A.3 Classical and one-loop terms

A.3.1 Classical coefficients

The quadratic and cubic coefficients are fixed by the Yang–Mills and Chern–Simons couplings. Write,

$$\phi = \phi^i H_i, \qquad F = F^i H_i, \qquad h_{ij} \equiv \mathrm{Tr}(H_i H_j). \tag{A.31}$$

The Cartan components of the Yang–Mills action are,

$$\mathcal{L}_{\mathrm{YM}} \supset -\frac{1}{4g_0^2}h_{ij}F^i_{\mu\nu}F^{j\,\mu\nu} - \frac{1}{2g_0^2}h_{ij}\partial_\mu\phi^i\partial^\mu\phi^j. \tag{A.32}$$

Comparison with Eq. (A.24) gives,

$$\mathcal{F}^{\mathrm{YM}}_{ij} = \frac{1}{g_0^2}h_{ij}, \qquad \mathcal{F}_{\mathrm{YM}} = \frac{1}{2g_0^2}h_{ij}\phi^i\phi^j. \tag{A.33}$$

For a simple gauge algebra admitting a symmetric cubic invariant define,

$$d_{ijk} = \frac{1}{2}\,\mathrm{Tr}\left(H_i\{H_j, H_k\}\right). \tag{A.34}$$

Gauge invariance fixes the classical cubic tensor to be proportional to this invariant,

$$c^{\mathrm{cl}}_{ijk} = k\, d_{ijk}, \tag{A.35}$$

[1] Equivalently, after reduction to four dimensions, five-dimensional gauge invariance acts as a continuous shift of $\mathrm{Im}\, a^i = A^i_4$. Holomorphy together with this shift symmetry restricts the four-dimensional coupling matrix to be at most linear in a^i, and hence the prepotential to be at most cubic. This is the form of the argument given in [1].

where k is the Chern–Simons level. Thus,

$$\mathcal{F}_{\rm CS} = \frac{k}{6} d_{ijk}\phi^i\phi^j\phi^k, \qquad \partial_i\partial_j\partial_k\mathcal{F}_{\rm CS} = k\, d_{ijk}. \tag{A.36}$$

For a simple gauge algebra, an independent cubic invariant of this form exists only for $\mathfrak{su}(N)$ with $N \geq 3$. The quantization and global interpretation of k are discussed in Sec. 2.7. The classical prepotential is therefore,

$$\mathcal{F}_{\rm cl} = \frac{1}{2g_0^2} h_{ij}\phi^i\phi^j + \frac{k}{6} d_{ijk}\phi^i\phi^j\phi^k. \tag{A.37}$$

A.3.2 One-loop thresholds and the parity anomaly

The remaining perturbative terms arise by integrating out the massive charged multiplets on the Coulomb branch. Their form may be obtained directly from the one-loop correction to the five-dimensional Chern–Simons coefficient. Consider first a hypermultiplet component of Abelian charge q_i. Its fermion has signed mass,

$$\mu(\phi) = q_i\phi^i + m, \qquad M_{\rm phys} = |\mu|. \tag{A.38}$$

The distinction between μ and $|\mu|$ is essential: the spectrum depends on μ^2, whereas the parity-odd part of the fermion determinant depends on the sign of μ. Formally,

$$\Gamma[A] = -\log\det(i\not{D} - \mu). \tag{A.39}$$

The cubic term in the background gauge field gives the diagram that induces $A \wedge F \wedge F$. Its three gauge vertices produce $q_iq_jq_k$, while the parity-odd contribution is selected by,

$$\mathrm{tr}\,(\gamma^{\mu_1}\gamma^{\mu_2}\gamma^{\mu_3}\gamma^{\mu_4}\gamma^{\mu_5}) \propto \epsilon^{\mu_1\mu_2\mu_3\mu_4\mu_5}. \tag{A.40}$$

The relevant mass dependence is,

$$\mu\int \frac{d^5p}{(2\pi)^5}\frac{1}{(p^2+\mu^2)^3} \propto \frac{\mu}{|\mu|} = \mathrm{sign}(\mu), \tag{A.41}$$

so the induced Chern–Simons coefficient must be proportional to $q_iq_jq_k\,\mathrm{sign}(\mu)$. In the conventions of [2], the regulated one-loop result for a hypermultiplet is,

$$\Delta c^H_{ijk} = -\frac{1}{2} q_iq_jq_k\,\mathrm{sign}(\mu). \tag{A.42}$$

The half-integer shift is the five-dimensional parity anomaly: a gauge-invariant definition of the massive fermion determinant induces a parity-odd Chern–Simons term. Its relation to large-gauge invariance is discussed in [2]; an explicit five-dimensional diagrammatic derivation is given in [73]. Supersymmetry then fixes the corresponding contribution to the full prepotential. Since,

$$c^{\rm eff}_{ijk} = \partial_i\partial_j\partial_k\mathcal{F}_{\rm eff}, \tag{A.43}$$

Eq. (A.42) requires,

$$\partial_i\partial_j\partial_k\Delta\mathcal{F}_H = -\frac{1}{2} q_iq_jq_k\,\mathrm{sign}(\mu). \tag{A.44}$$

Using $\partial_i \mu = q_i$, this becomes,

$$\frac{d^3}{d\mu^3}\Delta\mathcal{F}_H(\mu) = -\frac{1}{2}\,\mathrm{sign}(\mu). \tag{A.45}$$

For $\mu \neq 0$,

$$\frac{d}{d\mu}|\mu|^3 = 3\mu|\mu|, \qquad \frac{d^2}{d\mu^2}|\mu|^3 = 6|\mu|, \qquad \frac{d^3}{d\mu^3}|\mu|^3 = 6\,\mathrm{sign}(\mu), \tag{A.46}$$

and hence,

$$\Delta\mathcal{F}_H = -\frac{1}{12}|\mu|^3. \tag{A.47}$$

The integration is fixed up to a polynomial of degree at most two, which simply shifts the lower-degree local terms already present in the classical prepotential. For a hypermultiplet in a representation $\mathcal{R}_f$, the Abelian charges are the weights $w \in \mathcal{R}_f$ with,

$$\mu_{w,f} = w(\phi) + m_f. \tag{A.48}$$

Summing over all weights and flavours gives,

$$\Delta\mathcal{F}_{\text{hyper}} = -\frac{1}{12}\sum_f \sum_{w\in\mathcal{R}_f} |w(\phi) + m_f|^3. \tag{A.49}$$

The massive vector multiplets are labelled by the roots $\alpha \in \Delta$ for which,

$$\mu_\alpha = \alpha(\phi), \qquad M_{W_\alpha} = |\alpha(\phi)|. \tag{A.50}$$

Their complete one-loop determinant contributes with the opposite sign to a hypermultiplet [1, 2],

$$\partial_i\partial_j\partial_k\Delta\mathcal{F}_{V,\alpha} = +\frac{1}{2}\alpha_i\alpha_j\alpha_k\,\mathrm{sign}\big(\alpha(\phi)\big), \tag{A.51}$$

and therefore,

$$\Delta\mathcal{F}_{V,\alpha} = \frac{1}{12}|\alpha(\phi)|^3. \tag{A.52}$$

Summing over the root system,

$$\Delta\mathcal{F}_{\text{vector}} = \frac{1}{12}\sum_{\alpha\in\Delta}|\alpha(\phi)|^3. \tag{A.53}$$

Combining the classical contribution with the vector and hypermultiplet contributions[2] gives the perturbatively exact five-dimensional Coulomb-branch prepotential given in Eq. (2.11).

[2]The same one-loop terms may alternatively be obtained by compactifying the theory on a circle, summing the ordinary four-dimensional $\mathcal{N} = 2$ one-loop prepotential over the Kaluza–Klein tower, and then taking the decompactification limit; see for example [23].

B Non-Linear Sigma Models and Hyperkähler Geometry

This appendix explains why the smooth Higgs branch of a theory with eight supercharges is hyperkähler. The result follows from the geometry required of the low-energy nonlinear sigma model and is independent of the particular gauge-theory realisation. The relation between supersymmetry and target-space geometry was established in the early study of supersymmetric sigma models: four supercharges lead to Kähler geometry [74], while eight supercharges require the stronger hyperkähler structure [8, 9].

On a smooth region of the moduli space, let q^X denote the massless scalar fields. Their two-derivative action takes the form,

$$S_{\rm bos} = -\frac{1}{2}\int d^d x\, g_{XY}(q)\, \partial_\mu q^X \partial^\mu q^Y , \tag{B.1}$$

so the scalar fields define a map into a Riemannian target $(\mathcal{M}, g)$. Their fermionic partners take values in $T\mathcal{M}$ and couple through the Levi–Civita connection.

Additional supersymmetries act schematically as, $\delta_r q^X \sim \bar\epsilon (J_r)^X{}_Y \psi^Y$, where $J_r : T\mathcal{M} \to T\mathcal{M}$. Closure of the supersymmetry algebra on the scalar fields requires,

$$J_r J_s = -\delta_{rs}\mathbf{1} + \epsilon_{rst} J_t . \tag{B.2}$$

Thus, writing $I = J_1$, $J = J_2$ and $K = J_3$, one obtains $I^2 = J^2 = K^2 = -1$ and $IJK = -1$, resulting in the tangent spaces carrying a quaternionic algebra. Invariance of the action imposes two further conditions [8].

First, the metric is Hermitian with respect to each J_r, so $g(J_r X, J_r Y) = g(X, Y)$. Second,

$$\nabla J_r = 0, \qquad r = 1, 2, 3, \tag{B.3}$$

where ∇ is the Levi–Civita connection. Since the connection is torsion-free, covariant constancy implies that the corresponding Nijenhuis tensors vanish, thus each J_r is integrable, and hence defines a genuine complex structure. It also implies that the associated two-forms $\omega_r(X, Y) = g(J_r X, Y)$ are parallel and therefore closed: $d\omega_r = 0$. Consequently (g, J_r, ω_r) is Kähler for each r. A Riemannian manifold carrying three parallel complex structures satisfying the quaternionic algebra is, by definition, hyperkähler. Equivalently, its holonomy satisfies,

$$\mathrm{Hol}(g) \subseteq Sp(n), \qquad \dim_{\mathbb{R}} \mathcal{M} = 4n. \tag{B.4}$$

Choosing one complex structure, say I, the remaining two Kähler forms combine into the holomorphic symplectic form $\Omega_I = \omega_J + i\omega_K$.

C Symplectic Geometry and Moment Maps

This appendix reviews the symplectic origin of the moment maps used in Sec. 2.2.2, their appearance in the supersymmetric gauge-theory Lagrangian, and the resulting hyperkähler quotient.

C.1 Symplectic actions and moment maps

A symplectic manifold is a pair $(\mathcal{X}, \omega)$, where ω is closed, $d\omega = 0$, and non-degenerate. For a vector field X, contraction with ω defines the one-form $(\iota_X \omega)(Y) = \omega(X, Y)$. Non-degeneracy then allows a function $f : \mathcal{X} \to \mathbb{R}$ to define a unique Hamiltonian vector field X_f through,

$$df = \iota_{X_f} \omega. \tag{C.1}$$

Let a Lie group G act on $\mathcal{X}$. For $\xi \in \mathfrak{g} = \mathrm{Lie}(G)$, the corresponding fundamental vector field is,

$$\xi_{\mathcal{X}}(x) = \left. \frac{d}{dt} \right|_{t=0} \exp(t\xi) \cdot x. \tag{C.2}$$

If the action preserves the symplectic form, $\mathcal{L}_{\xi_{\mathcal{X}}} \omega = 0$, Cartan's identity gives,

$$0 = \mathcal{L}_{\xi_{\mathcal{X}}} \omega = d(\iota_{\xi_{\mathcal{X}}} \omega) + \iota_{\xi_{\mathcal{X}}} d\omega = d(\iota_{\xi_{\mathcal{X}}} \omega)\,. \tag{C.3}$$

Hence $\iota_{\xi_{\mathcal{X}}} \omega$ is closed and therefore locally exact. When these local potentials may be chosen globally and compatibly with the group action, the action is Hamiltonian and admits a moment map $\mu : \mathcal{X} \to \mathfrak{g}^*$. Writing $\mu^\xi = \langle \mu, \xi \rangle$, its defining relation is,

$$d\mu^\xi = \iota_{\xi_{\mathcal{X}}} \omega. \tag{C.4}$$

Thus μ^ξ is the Hamiltonian whose flow generates the infinitesimal action of ξ. For non-Abelian G, the moment map is chosen equivariantly, $\mu(g \cdot x) = \mathrm{Ad}^*_g \, \mu(x)$. It is defined up to compatible additive constants; for Abelian gauge factors these shifts become Fayet–Iliopoulos parameters [9].

The moment map is therefore a function on field space rather than an additional dynamical scalar. Once the hypermultiplet scalars define the spacetime-dependent map $q : x \mapsto q(x) \in \mathcal{X}$, the quantity $\mu(q(x))$ is an ordinary composite scalar in the field theory. For a linear action on a flat symplectic vector space, the generating vector fields are linear in the coordinates and the corresponding moment maps are quadratic.

C.2 Hyperkähler moment maps and supersymmetry

A hyperkähler manifold $(\mathcal{X}, g, I, J, K)$ carries three complex structures satisfying,

$$I^2 = J^2 = K^2 = IJK = -1, \tag{C.5}$$

with associated symplectic forms,

$$\omega_I(X,Y) = g(IX,Y), \qquad \omega_J(X,Y) = g(JX,Y), \qquad \omega_K(X,Y) = g(KX,Y). \tag{C.6}$$

A triholomorphic action preserves this hyperkähler structure. If k_a is the vector field generated by $T_a \in \mathfrak{g}$, then $\mathcal{L}_{k_a}\omega_r = 0$ for $r = 1,2,3$, so the symplectic construction applies to each form and gives a triplet of moment maps,

$$d\mu_r^a = \iota_{k_a}\omega_r, \qquad r = 1,2,3. \tag{C.7}$$

Equivalently, $\vec{\mu} : \mathcal{X} \to \mathfrak{g}^* \otimes \mathbb{R}^3$. Choosing the complex structure I, it is useful to combine the remaining two components as,

$$\mu_{\mathbb{C}} = \mu_J + i\mu_K, \qquad \Omega_I = \omega_J + i\omega_K, \qquad d\mu_{\mathbb{C}}^a = \iota_{k_a}\Omega_I. \tag{C.8}$$

The same functions arise from the supersymmetric completion of the gauge interaction. In local coordinates q^X on $\mathcal{X}$, the infinitesimal gauge action and covariant derivative are,

$$\delta q^X = \lambda^a k_a^X(q), \qquad D_\mu q^X = \partial_\mu q^X + A_\mu^a k_a^X(q). \tag{C.9}$$

Thus the same vector fields k_a that describe the geometric action determine the coupling of the hypermultiplets to the gauge fields.

The auxiliary fields of the vector multiplet form an $SU(2)_R$ triplet Y_r^a. Their linear coupling to the hypermultiplet scalars is of the form,

$$Y_r^a M_r^a(q) \subset \mathcal{L}_Y, \tag{C.10}$$

where $M_r^a(q)$ is a corresponding triplet of scalar functions on $\mathcal{X}$. Supersymmetry fixes these functions. Schematically, the relevant transformations contain,

$$\delta q^X \sim \bar{\epsilon}\,\zeta^X, \qquad Y_r^a \sigma^r \epsilon \subset \delta\lambda^a, \tag{C.11}$$

where ζ and λ^a denote the hyperino and gaugino, respectively. Varying the gauging-induced gaugino–hyperino interaction produces terms proportional to $Y_r^a\zeta$, with scalar-space coefficient $(\omega_r)_{XY}k_a^Y$. By contrast, varying the scalar in Eq. (C.10) gives,

$$Y_r^a \partial_X M_r^a\, \delta q^X \subset \delta(Y_r^a M_r^a(q))\,. \tag{C.12}$$

Cancellation of these terms therefore requires,

$$\partial_X M_r^a = (\omega_r)_{XY}k_a^Y, \qquad \Longleftrightarrow \qquad dM_r^a = \iota_{k_a}\omega_r. \tag{C.13}$$

Comparison with Eq. (C.7) gives $M_r^a = \mu_r^a$, up to the allowed additive constants. The same identification is therefore obtained independently from symplectic geometry and supersymmetry.

Its local consistency follows directly from the triholomorphic action. Since $\mathcal{L}_{k_a}\omega_r = 0$ and $d\omega_r = 0$, Cartan's identity implies,

$$d(\iota_{k_a}\omega_r) = 0. \tag{C.14}$$

Thus the scalar functions required by the supersymmetric completion are precisely the Hamiltonians associated with the gauge action. A full component derivation depends on spinor and auxiliary-field conventions, whereas this geometric statement does not.

C.3 Hyperkähler reduction

To see why the quotient given in Eq. (2.17) remains hyperkähler, let $Z = \vec{\mu}^{-1}(0)$, with inclusion $i : Z \hookrightarrow \mathcal{X}$ and quotient map $\pi : Z \to Z/G$. Assume that 0 is a regular value of $\vec{\mu}$ and that the G-action on Z is free and proper.

A differential form on Z descends to the quotient when it is G-invariant and horizontal. Invariance of $i^*\omega_r$ follows from the triholomorphic action. For $v \in T_xZ$, tangency to the moment-map level set gives $d\mu_r^a(v) = 0$, and hence,

$$\omega_r(k_a, v) = d\mu_r^a(v) = 0. \tag{C.15}$$

Thus $i^*\omega_r$ vanishes whenever one argument is tangent to a gauge orbit and therefore descends uniquely to a two-form $\bar{\omega}_r$ on the quotient satisfying,

$$\pi^*\bar{\omega}_r = i^*\omega_r. \tag{C.16}$$

Since $d\omega_r = 0$, the descended forms are also closed.

The metric and complex structures descend similarly. Let $V_x \subset T_xZ$ denote the tangent space to the gauge orbit. Using $d\mu_r^a = \iota_{k_a}\omega_r$ together with $\omega_r(X, Y) = g(J_rX, Y)$, the tangent space to the level set is,

$$T_xZ = (IV_x + JV_x + KV_x)^\perp. \tag{C.17}$$

Choosing representatives orthogonal also to V_x identifies the tangent space of the quotient with,

$$T_{[x]}\mathcal{H}_{\text{cl}} \simeq (V_x \oplus IV_x \oplus JV_x \oplus KV_x)^\perp . \tag{C.18}$$

This horizontal subspace is preserved by I, J and K, so the quotient inherits the metric, the three complex structures and their associated symplectic forms. It is therefore hyperkähler [9].

D Seiberg's Type I′ construction

Type I′ string theory is obtained by T-dualising Type I on a circle, giving $\mathbb{R}^{1,8} \times I$, with $I = S^1/\mathbb{Z}_2$, and orientifold action $\Omega\mathcal{R}_9$. The two endpoints support $O8^-$ planes, each carrying -8 units of D8 brane charge in the physical-interval convention, so the background contains sixteen D8-branes. A wrapped Type I D5-brane becomes a D4-brane probe localised on the interval.

Suppose N_f D8-branes coincide with one orientifold. The configuration preserves eight real supercharges, and the orientifold projection gives an $Sp(1) \simeq SU(2)$ gauge theory on the D4 with N_f fundamental hypermultiplets. The probe also carries a free neutral hypermultiplet describing translations in $x^{5,\dots,8}$, which is removed when considering the interacting sector [1]. Writing $\langle\Phi\rangle = \mathrm{diag}(\phi, -\phi)$, with $\phi \geq 0$, and denoting by y the probe position and by y_f the D8-brane positions measured from the orientifold, the moduli are,

$$\phi = \frac{y}{2\pi\alpha'}, \qquad m_f = \frac{y_f}{2\pi\alpha'}, \qquad m_0 = \frac{8\pi^2}{g_0^2}. \tag{D.1}$$

The first two identifications are directly geometric. Strings between the D4-brane and its orientifold image give $M_W = 2\phi$, while D4–D8 strings give masses $|\phi \pm m_f|$, so the D4 position is the Coulomb-branch modulus and the D8 positions are flavour-mass deformations. The gauge coupling is instead encoded in the Type I′ background. In Type I variables, $g^{-2} \sim RM_s^2/\lambda_I \sim M_s/\lambda_{I'}$, so after T-duality m_0 fixes the inverse coupling at the orientifold, while the D4 position samples the local Type I′ dilaton profile [1]. The D8- and O8-plane sources make this profile piecewise linear, in agreement with the five-dimensional prepotential. Using the $SU(2)$ roots ± 2 and fundamental weights ± 1, the Coulomb-branch metric is,

$$G_{\phi\phi} = \frac{h_{\phi\phi} m_0}{8\pi^2} + 8\phi - \frac{1}{2}\sum_{f=1}^{N_f} \left(|\phi - m_f| + |\phi + m_f|\right). \tag{D.2}$$

Each D8-brane crossed by the probe lowers the slope by one, matching the piecewise-linear dilaton profile. When all local D8-branes lie on the orientifold this reduces to $G_{\phi\phi} = h_{\phi\phi} m_0/(8\pi^2) + (8 - N_f)\phi$. For $N_f < 8$, the limit $m_0 \to 0$ leaves a positive metric for $\phi > 0$ while it degenerates at the origin.[1] At this locus the local Type I′ string coupling diverges, so the perturbative description breaks down. The relevant additional states carry D0-brane charge: the D4 Wess–Zumino coupling $\int_{\mathrm{D4}} C_1 \wedge \mathrm{Tr}(F \wedge F)$ identifies D0 charge with the five-dimensional instanton number I, whose central-charge contribution at the Coulomb and flavour origin is $I\, m_0$. Instanton states may therefore become massless when $m_0 \to 0$.

[1] The numerical coefficients follow the $1/12$ normalisation of the one-loop prepotential used in Chapter 2. Seiberg's original effective-coupling convention differs by an overall factor of two.

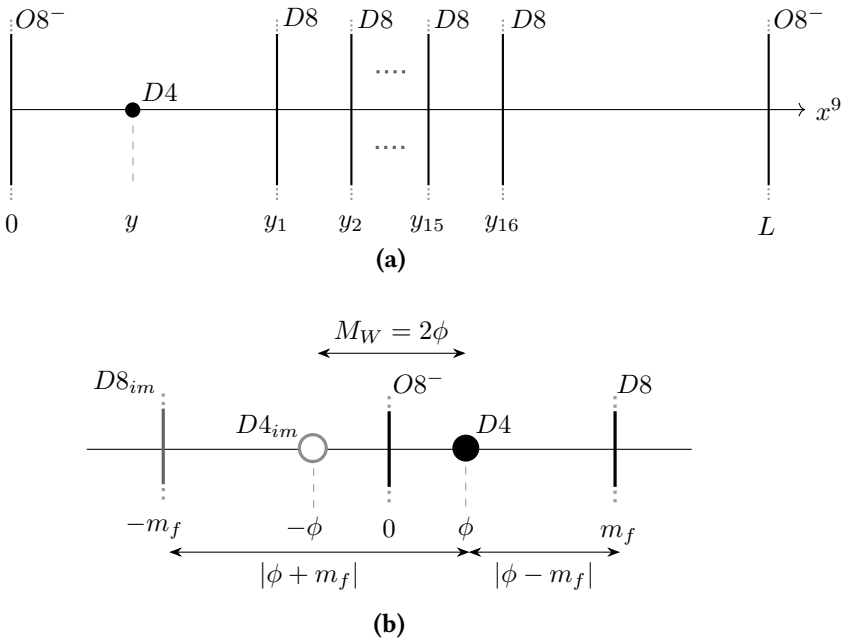


Figure D.1: (a) Type I′ interval with a D4-brane probe and background D8-branes. The probe position determines the Coulomb parameter $\phi = y/(2\pi\alpha')$. (b) Local double-cover description near an orientifold, showing one D8-brane and its image together with the corresponding D4 image. The separations of the branes reproduce the perturbative masses $M_W = 2\phi$ and $|\phi \pm m_f|$.

The enhanced symmetry is fixed by heterotic/Type I/Type I′ duality. For $N_f = 7$, the divergent-coupling locus maps to a heterotic enhancement point at which additional winding states become massless and complete an E_8 gauge symmetry in the ambient theory [1]. On the D4 this appears as flavour symmetry. The seven D8 positions together with m_0 provide the eight Cartan deformation parameters.At vanishing masses and infinite coupling, the finite-coupling symmetry enhances as,

$$\mathfrak{so}(2N_f) \oplus \mathfrak{u}(1)_I \longrightarrow \mathfrak{e}_{N_f+1}, \qquad 1 \leq N_f \leq 7. \tag{D.3}$$

Moving D8-branes away from the orientifold instead turns on quark masses and breaks this enhanced symmetry. For $N_f = 0$, the branch considered here is the E_1 theory with trivial discrete theta angle, for which $\mathfrak{u}(1)_I$ enhances to $\mathfrak{su}(2)$. This is distinct from restoration of the dynamical $SU(2)$ gauge symmetry at $\phi = 0$, and from the separate $\widetilde{E}_1$ endpoint. The Type I′ construction therefore supplies more than the field-theoretic consistency test: the prepotential permits the limit $m_0 \to 0$, while string duality identifies the resulting locus with additional non-perturbative states and enhanced symmetry. With the probe sector decoupled from the bulk, $m_f = m_0 = 0$ at $\phi = 0$ gives the interacting fixed point whose m_0 deformation flows to $SU(2) + N_f F$.

The same setup also explains the ultraviolet Higgs branch. A D4-brane may dissolve into the background gauge system as an instanton on the relative directions $x^{5,\ldots,8}$, as follows from the D8 coupling $\int_{\mathrm{D8}} C_5 \wedge \mathrm{tr}(F_G \wedge F_G)$. Its size and gauge-orientation moduli give Higgs-branch vacuum data, while its position belongs to the decoupled centre-of-mass hypermultiplet. At the strong-coupling locus the ambient symmetry enhances to E_{N_f+1}, and the interacting Higgs branch becomes the reduced moduli space of one E_{N_f+1} instanton [1]. This is distinct from a D0-brane dissolved in the D4, which is an instanton particle of the five-dimensional gauge theory.

E Classical Relations

This appendix collects the classical Higgs-ring relations used in the finite-coupling restrictions of Chapter 6. We consider $SU(N_c)$ SQCD with N_f fundamental hypermultiplets, using the four-supercharge notation adopted throughout the dissertation. Let $Q^i{}_a$ and $\widetilde{Q}^a{}_i$ denote the holomorphic hypermultiplet scalars, with $a = 1, \dots, N_c$ and $i = 1, \dots, N_f$. The flavour-space meson, singlet coordinate and traceless meson are,

$$X^i{}_j = Q^i{}_a \widetilde{Q}^a{}_j, \qquad \eta' = \frac{1}{N_c} \operatorname{tr} X, \qquad M = X - \frac{N_c}{N_f} \eta' \mathbf{1}_{N_f}. \tag{E.1}$$

We use the notation $\mathrm{Comp}_r(X)$ for matrix minors as defined in the main text (Eq.7.16).

Mesonic relations. The hypermultiplet coupling to the adjoint chiral field is $W = \widetilde{Q}\Phi Q$. Since Φ is traceless for $SU(N_c)$, its equation of motion removes only the traceless part of the colour-space matrix $\widetilde{Q}Q$,

$$\widetilde{Q}^a{}_i Q^i{}_b - \frac{1}{N_c} \operatorname{tr}(\widetilde{Q}Q) \delta^a{}_b = 0. \tag{E.2}$$

Using $\operatorname{tr}(\widetilde{Q}Q) = \operatorname{tr} X = N_c \eta'$, the complex moment-map equation and its immediate mesonic consequence are,

$$\widetilde{Q}Q = \eta' \mathbf{1}_{N_c}, \tag{E.3}$$

$$X^2 = Q(\widetilde{Q}Q)\widetilde{Q} = \eta' X. \tag{E.4}$$

Thus, for $Y \equiv X - \eta' \mathbf{1}_{N_f}$, one has $XY = YX = 0$.

Writing $a = N_c/N_f$ and substituting $X = M + a\eta' \mathbf{1}_{N_f}$ gives,

$$M^2 + (2a - 1)\eta' M + (a^2 - a)(\eta')^2 \mathbf{1}_{N_f} = 0, \tag{E.5}$$

$$N_f = 5 : \qquad M^2 + \frac{1}{5}\eta' M - \frac{6}{25}(\eta')^2 \mathbf{1}_5 = 0, \tag{E.6}$$

$$N_f = 6 : \qquad M^2 - \frac{1}{4}(\eta')^2 \mathbf{1}_6 = 0. \tag{E.7}$$

Finally, $X = Q\widetilde{Q}$ factors through an N_c-dimensional colour space, so $\operatorname{rank} X \le N_c$, equivalently $\mathrm{Comp}_{N_c+1}(X) = 0$.

Baryonic relations. For $N_f \ge N_c$, the baryon and antibaryon are,

$$B^{i_1 \cdots i_{N_c}} = \epsilon^{a_1 \cdots a_{N_c}} Q^{i_1}{}_{a_1} \cdots Q^{i_{N_c}}{}_{a_{N_c}}, \tag{E.8}$$

$$\widetilde{B}_{i_1 \cdots i_{N_c}} = \epsilon_{a_1 \cdots a_{N_c}} \widetilde{Q}^{a_1}{}_{i_1} \cdots \widetilde{Q}^{a_{N_c}}{}_{i_{N_c}}. \tag{E.9}$$

Equation (E.3) gives $XQ = \eta' Q$ and $\widetilde{Q}X = \eta'\widetilde{Q}$, so $YQ = \widetilde{Q}Y = 0$. Acting on the baryonic operators gives the incidence relations,

$$Y^{i_1}{}_j B^{ji_2\cdots i_{N_c}} = 0, \tag{E.10}$$

$$\widetilde{B}_{ji_2\cdots i_{N_c}} Y^j{}_{i_1} = 0. \tag{E.11}$$

It is convenient to Hodge dualise the baryons in flavour space,

$$(\star B)_{j_1\cdots j_{N_f-N_c}} = \frac{1}{N_c!}\epsilon_{j_1\cdots j_{N_f-N_c} i_1\cdots i_{N_c}} B^{i_1\cdots i_{N_c}}, \tag{E.12}$$

$$X\cdot(\star B) = 0, \qquad (\star\widetilde{B})\cdot X = 0. \tag{E.13}$$

The second line follows because inserting one additional Q into the antisymmetrised product defining $\star B$ vanishes. For $SU(3)$ with five flavours the Hodge-dual baryons are two-forms, while for six flavours they are three-forms; these are the tensors denoted by B and $\widetilde{B}$ in the main text.

The baryon–antibaryon relation is the corresponding maximal minor of X. Writing $I = (i_1, \ldots, i_{N_c})$ and $J = (j_1, \ldots, j_{N_c})$, the Cauchy–Binet identity gives,

$$B^I \widetilde{B}_J = \mathrm{Comp}_{N_c}(X)^I{}_J. \tag{E.14}$$

This holds up to the normalisation convention associated with antisymmetrisation and Hodge dualisation. After decomposing $X = M + \frac{N_c}{N_f}\eta'\mathbf{1}$, it gives the mixed meson–baryon relations used in the explicit $SU(3)$ rings.

The remaining purely baryonic constraints follow because $B = q_1 \wedge \cdots \wedge q_{N_c}$ is decomposable, and hence,

$$B^{[i_1\cdots i_{N_c}} B^{j_1]j_2\cdots j_{N_c}} = 0, \tag{E.15}$$

$$\widetilde{B}_{[i_1\cdots i_{N_c}} \widetilde{B}_{j_1]j_2\cdots j_{N_c}} = 0. \tag{E.16}$$

For the Hodge-dual two-forms of the five-flavour $SU(3)$ theory these reduce to $B^{[ij}B^{kl]} = 0$ and $\widetilde{B}_{[ij}\widetilde{B}_{kl]} = 0$; for six flavours they are the corresponding Plücker relations for decomposable three-forms.

Trace nilpotency

The behaviour of the singlet coordinate follows from $X^2 = \eta' X$ and $\mathrm{tr}\, X = N_c\eta'$. For $N_f \geq N_c$, η' is an ordinary non-zero coordinate: taking $Q = (v\mathbf{1}_{N_c}, 0)^{\mathsf{T}}$ and $\widetilde{Q} = (v\mathbf{1}_{N_c}, 0)$ gives $X = \mathrm{diag}(v^2\mathbf{1}_{N_c}, 0)$ and $\eta' = v^2$. Since v varies continuously, no relation $(\eta')^p = 0$ can hold identically. This is the regime of the theories studied in the main text, where $N_c = 3$ and $N_f = 5, 6$.

For $N_f < N_c$, suppose instead that $\eta' \neq 0$ at some point. The relation $X(X - \eta'\mathbf{1}) = 0$ has distinct roots, so X is diagonalizable with eigenvalues 0 and η'. If $r = \mathrm{rank}\, X$, then $\mathrm{tr}\, X = r\eta' = N_c\eta'$, requiring $r = N_c > N_f$, a contradiction. Thus η' vanishes at every ordinary point, although it may remain as a non-zero nilpotent element of the chiral ring.

F One-Instanton Zero Modes and Baryon Charge

The baryon charges of the one-instanton operators used in Chapter 6 follow from quantising fermionic zero modes in a charge-one instanton background. We follow Refs. [67, 68].

Instanton background. A charge-one instanton of $SU(N)$ may be represented by embedding the BPST solution into an $SU(2) \subset SU(N)$ subgroup,

$$A_\mu = \begin{pmatrix} A_\mu^{\mathrm{BPST}} & 0 \\ 0 & 0_{N-2} \end{pmatrix}. \tag{F.1}$$

Gauge-orientation collective coordinates recover the full $SU(N)$ one-instanton sector, so the embedding simply provides a convenient representative. Its commutant is $SU(N-2) \times U(1)_G$. Choosing,

$$T_G = \mathrm{diag}(N-2, N-2, -2, \ldots, -2), \qquad \mathbf{N} \longrightarrow (\mathbf{2},\mathbf{1})_{N-2} \oplus (\mathbf{1},\mathbf{N-2})_{-2}, \tag{F.2}$$

reduces the zero-mode calculation to the standard $SU(2)$ instanton problem after decomposing the $SU(N)$ fields under this subgroup.

Zero-mode charges. Gauge invariance of the instanton state is imposed through the unbroken $U(1)_G$. The Chern–Simons term gives a charge,

$$Q_G^{\mathrm{CS}} = (N-2)\kappa. \tag{F.3}$$

The adjoint decomposition relevant for the gaugino zero modes is,

$$\begin{aligned} \mathrm{Adj}_{SU(N)} \longrightarrow\ & (\mathbf{3},\mathbf{1})_0 \oplus (\mathbf{1},\mathrm{Adj}_{SU(N-2)})_0 \oplus (\mathbf{1},\mathbf{1})_0 \\ & \oplus (\mathbf{2},\mathbf{N-2})_{-N} \oplus (\mathbf{2},\overline{\mathbf{N-2}})_{+N}. \end{aligned} \tag{F.4}$$

The $(\mathbf{3},\mathbf{1})_0$ modes are the universal supersymmetry zero modes and do not affect the $U(1)_G$ charge. The off-diagonal terms give $2(N-2)$ complex zero modes. Normal ordering assigns charges $\mp q/2$ to the empty and filled states of a complex fermion of charge q, so the two extreme gaugino sectors, denoted $E_\pm$, have,

$$q_g(E_\pm) = \mp N(N-2). \tag{F.5}$$

Each fundamental hypermultiplet contributes one complex fermionic zero mode with $U(1)_G$ charge $N-2$. Temporarily using the literature convention $B_{\rm lit}(Q)=+1$, a state obtained by occupying ℓ of the N_f matter zero modes transforms in $\wedge^{\ell}\mathbf{N_f}$ and carries,

$$B_{\rm lit}=\ell-\frac{N_f}{2},\qquad Q_G^{(h)}=(N-2)\left(\ell-\frac{N_f}{2}\right). \tag{F.6}$$

The shift by $-N_f/2$ is the normal-ordering charge of the matter Fock vacuum.

Gauge neutrality. Combining the Chern–Simons, gaugino and matter contributions, a physical one-instanton state must satisfy,

$$0=Q_G=(N-2)\kappa+q_g+(N-2)\left(\ell-\frac{N_f}{2}\right). \tag{F.7}$$

For the two extreme gaugino sectors this gives,

$$B_{\rm lit}(E_\pm)=\pm N-\kappa,\qquad \ell_\pm=\frac{N_f}{2}\pm N-\kappa,\qquad 0\le\ell_\pm\le N_f. \tag{F.8}$$

The main text uses the opposite baryon convention, $Q_B(Q)=-1$, so $Q_B=-B_{\rm lit}$ and therefore,

$$Q_B(E_\pm)=\kappa\mp N. \tag{F.9}$$

Application to $SU(3)_{\frac{1}{2}}+5F$. For $N=3$, $N_f=5$ and $\kappa=\frac{1}{2}$, the two extreme sectors require,

$$\ell_+=5,\qquad Q_B(E_+)=-\frac{5}{2},\qquad \ell_-=-1. \tag{F.10}$$

The E_+ state therefore occupies all five matter zero modes and transforms as $\wedge^5\mathbf{5}=\mathbf{1}$ of $SU(5)$. The second extreme sector is absent because ℓ counts occupied fermionic zero modes and must lie between 0 and N_f; the value $\ell_-=-1$ cannot occur in the Fock space. Both E_+ and E_- refer here to possible states in the charge-one sector $Q_I=+1$, so the absence of E_- should not be confused with the anti-instanton. The latter belongs to the separate $Q_I=-1$ sector. The surviving current operator and its anti-instanton conjugate consequently carry,

$$(Q_B,Q_I)=\left(-\frac{5}{2},1\right),\qquad \left(\frac{5}{2},-1\right), \tag{F.11}$$

which are the charges assigned to I_+ and I_- in Chapter 6.

Equation (F.9) concerns only the two extreme gaugino sectors. The remaining gaugino sector of the $SU(3)$ instanton has $q_g=0$, for which gauge neutrality gives $\ell=2$ and $Q_B=\frac{1}{2}$. It therefore produces a one-instanton operator in $\wedge^2\mathbf{5}=\mathbf{10}$ with $(Q_B,Q_I)=(\frac{1}{2},1)$. Its anti-instanton conjugate lies in $\overline{\mathbf{10}}$ with $(Q_B,Q_I)=(-\frac{1}{2},-1)$. These are the non-perturbative components that pair with the perturbative antibaryon and baryon, respectively, in the degree-three ultraviolet multiplets.

G Glueball Nilpotency and $SU(4)$ Relations

G.1 Glueball nilpotency in the $SU(3)$ theories

G.1.1 Five flavours, $|k| = \frac{1}{2}$

The finite-coupling degree-four scalar and neutral degree-six relations give,

$$S^2 = 6\eta' S, \qquad S\left(\eta' + \frac{5}{6}S\right)^2 = 0, \tag{G.1}$$

$$0 = S\left(\eta' + \frac{5}{6}S\right)^2 = \left(\eta' + \frac{5}{6}S\right)S^2 = S^3. \tag{G.2}$$

Here the first line was used in the second through $(\eta' + \frac{5}{6}S)S = S^2$.

G.1.2 Five flavours, $|k| = \frac{3}{2}$

The relevant finite-coupling relations and trace are,

$$S^2 = 2\eta' S, \qquad S\left(\eta' + \frac{S}{2}\right)\left(M - \frac{2\eta'}{5}\mathbf{1}_5\right) = 0, \qquad \mathrm{tr}\left(M - \frac{2\eta'}{5}\mathbf{1}_5\right) = -2\eta'. \tag{G.3}$$

Since $S(\eta' + S/2) = S^2$, the matrix relation gives $S^2(M - \frac{2\eta'}{5}\mathbf{1}_5) = 0$. Taking its trace yields $\eta' S^2 = 0$, and therefore,

$$S^3 = 2\eta' S^2 = 0. \tag{G.4}$$

G.1.3 Five flavours, $|k| = \frac{5}{2}$:

The relevant scalar consequences of the degree-six mixed-bispinor relations may be normalised as,

$$R_1 := S\left(\eta' - \frac{S}{6}\right)^2 = 0, \qquad R_3 := S\left(\eta' - \frac{7S}{6}\right)\left(\eta' + \frac{5S}{6}\right) = 0, \tag{G.5}$$

$$R_1 - R_3 = S^3 = 0. \tag{G.6}$$

Thus cubic nilpotency follows directly by combining the two scalar contractions of the $\mathrm{Spin}(10)$ spinor-bilinear relations. For the six-flavour theories, finite coupling is again obtained by setting all operators with non-zero instanton charge to zero. We write $e_r(A)$ for the r-th elementary symmetric polynomial of a matrix A.

G.1.4 Six flavours, $k = 0$

The finite-coupling relations give,

$$H^2 = \Sigma^2 = 6\ \mathrm{tr}\, M^2, \qquad H = -3(\eta' + S), \qquad \Sigma = -3(\eta' - S), \qquad S\ \mathrm{tr}\, M^2 = 0. \tag{G.7}$$

Since $H^2 - \Sigma^2 = 36\eta' S$, the first equality implies $\eta' S = 0$. Hence,

$$0 = 6S\ \mathrm{tr}\, M^2 = SH^2 = 9S(\eta' + S)^2 = 9S^3. \tag{G.8}$$

G.1.5 Six flavours, $|k| = 1$

Set $A := M + (-\eta'/2 + S/3)\mathbf{1}_6$. The finite-coupling relations give,

$$\eta' S = 0, \qquad A^2 = (S - \eta')A, \qquad \mathrm{tr}\, A = -3\eta' + 2S, \qquad S\, e_2(A) = 0. \tag{G.9}$$

Using $e_2(A) = \frac{1}{2}[(\mathrm{tr}\, A)^2 - \mathrm{tr}\, A^2]$ together with $\eta' S = 0$ gives,

$$e_2(A) = 3(\eta')^2 + S^2, \qquad 0 = S\, e_2(A) = 3(\eta')^2 S + S^3 = S^3. \tag{G.10}$$

G.1.6 Six flavours, $|k| = 2$:

Set $A := M + S\mathbf{1}_6$. The finite-coupling relations are,

$$A^2 = \frac{(\eta')^2}{4}\mathbf{1}_6, \qquad (\eta')^2 S = 0, \qquad e_3(A) = 0. \tag{G.11}$$

Since $\mathrm{tr}\, A = 6S$, the quadratic relation gives $\mathrm{tr}\, A^2 = \frac{3}{2}(\eta')^2$ and $\mathrm{tr}\, A^3 = \frac{3}{2}(\eta')^2 S$. Newton's identity therefore gives,

$$0 = e_3(A) = \frac{1}{6}\left[(\mathrm{tr}\, A)^3 - 3\,\mathrm{tr}\, A\ \mathrm{tr}\, A^2 + 2\,\mathrm{tr}\, A^3\right] = 36S^3 - 4(\eta')^2 S = 36S^3. \tag{G.12}$$

Thus all six $SU(3)$ theories obey the same cubic glueball nilpotency relation,

$$S^3 = 0. \tag{G.13}$$

In each case, S^2 lies outside the linear span of the degree-four singlet relations at finite coupling, while homogeneity prevents higher-degree relations from setting it to zero. Consequently, S and S^2 remain nonzero elements of the chiral ring, so the glueball has nilpotency order exactly three.

G.2 $\mathrm{SU}(4)_{\frac{1}{2}} + 7F$

G.2.1 Glueball nilpotency

At finite coupling, the degree-four scalar relation gives,

$$\Sigma^2 - H^2 = 0, \qquad H = 4\eta' - S, \qquad \Sigma = 4\eta' \qquad \Longrightarrow \qquad S^2 = 8\eta' S. \tag{G.14}$$

Let $Y := M - \frac{3\eta'}{7}\mathbf{1}_7$. The degree-four matrix relation and neutral degree-eight relation give,

$$Y^2 = -\eta' Y, \qquad \operatorname{tr} Y = -3\eta', \qquad S\, e_3(Y) = 0. \tag{G.15}$$

The first two relations imply $e_3(Y) = -(\eta')^3$, so $(\eta')^3 S = 0$. Repeated use of Eq. (G.14) then gives,

$$S^4 = 512(\eta')^3 S = 0. \tag{G.16}$$

Thus the $SU(4)_{\frac{1}{2}} + 7F$ example exhibits quartic glueball nilpotency. The degree-four and degree-six relations leave $S^3 = 64(\eta')^2 S$ nonzero, while homogeneity prevents higher-degree relations from setting it to zero, so S, S^2 and S^3 survive in the finite-coupling chiral ring and the nilpotency order is exactly four.

G.2.2 Relations for $\mathrm{SU}(4)$ with seven flavours, $|k| = \frac{1}{2}$

The ultraviolet symmetry is $\mathrm{SU}(7) \times \mathrm{SU}(2) \times \mathrm{U}(1)_q$. Restricting the enhanced $\mathrm{SU}(2)$ to its Cartan gives,

$$\mathbf{3} \longrightarrow \mathbf{1}_{+2} \oplus \mathbf{1}_0 \oplus \mathbf{1}_{-2}, \qquad \mathbf{2} \longrightarrow \mathbf{1}_{+1} \oplus \mathbf{1}_{-1}. \tag{G.17}$$

The relevant generator terms are,

$$\begin{aligned} \mathrm{PL}[t^2] &= 1 + \nu^2 + \mu_1\mu_6, \\ \mathrm{PL}[t^4] &= \nu\mu_3 q + \nu\mu_4 q^{-1}. \end{aligned} \tag{G.18}$$

The corresponding generators decompose into the physical operators,

d	Ultraviolet multiplet	Branched components
2	$(\mathbf{48}, \mathbf{1})_0$	M
2	$(\mathbf{1}, \mathbf{1})_0$	Σ
2	$(\mathbf{1}, \mathbf{3})_0$	$H_{(0,0)},\ I_{(-7/2,\,1)},\ \widetilde{I}_{(7/2,\,-1)}$
4	$(\mathbf{35}, \mathbf{2})_{+1}$	$B_{(-4,\,0)},\ (B_{-1})_{(-1/2,\,-1)}$
4	$(\overline{\mathbf{35}}, \mathbf{2})_{-1}$	$(\widetilde{B}_{+1})_{(1/2,\,1)},\ \widetilde{B}_{(4,\,0)}$

(G.19)

$$Q_B = -\frac{7x + 9q}{4}, \qquad Q_I = \frac{x - q}{2}. \tag{G.20}$$

With the corresponding normalisation, the neutral ultraviolet scalars and traceful meson are,

$$H = 4\eta' - S, \qquad \Sigma = 4\eta', \qquad X = M + \frac{\Sigma}{7}\mathbf{1}_7. \tag{G.21}$$

Thus Σ is the mesonic trace direction, while H is the neutral component contained in the enhanced $\mathrm{SU}(2)$ current. The enhanced current and degree-four baryonic generators are,

$$\Phi = \begin{pmatrix} H & I \\ \widetilde{I} & -H \end{pmatrix}, \qquad \mathcal{B} = \begin{pmatrix} B \\ B_{-1} \end{pmatrix}, \qquad \widetilde{\mathcal{B}} = \begin{pmatrix} \widetilde{B}_{+1} \\ \widetilde{B} \end{pmatrix}. \tag{G.22}$$

Here $\mathcal{B}$ and $\widetilde{\mathcal{B}}$ transform as $\mathrm{SU}(2)$ doublets and carry three upper and three lower $\mathrm{SU}(7)$ flavour indices, respectively.

Degree four

At degree four,

$$\mathrm{PL}[t^4] = q^{-1}\nu\mu_4 - 2 - \mu_1\mu_6 + q\nu\mu_3. \tag{G.23}$$

The adjoint and two singlet relations are,

$$XY = 0, \qquad \Sigma^2 - H^2 = I\widetilde{I}, \tag{G.24}$$

where

$$Y \equiv X - \frac{\Sigma}{4}\mathbf{1}_7. \tag{G.25}$$

Thus, for $\Sigma \neq 0$, X has eigenvalues 0 and $\Sigma/4$, with multiplicities three and four fixed by $\mathrm{tr}\, X = \Sigma$. The fundamental therefore decomposes as $\mathbb{C}^7 = \ker X \oplus \ker Y$, with dimensions three and four respectively. The second equation directly relates the two neutral components of the enhanced current to the instanton–anti-instanton bilinear.

Degree six

The degree-six terms are,

$$\begin{aligned} \mathrm{PL}[t^6] = {} & \mu_1\mu_6 - q^{-1}\nu(\mu_5\mu_6 + 2\mu_4 + \mu_1\mu_3) \\ & - q\nu(\mu_4\mu_6 + 2\mu_3 + \mu_1\mu_2). \end{aligned} \tag{G.26}$$

The incidence channels are,

$$X^i{}_j\mathcal{B}^{jkl}_\alpha = 0, \qquad \widetilde{\mathcal{B}}_{\alpha,jkl}X^j{}_i = 0, \tag{G.27}$$

together with the hook relations,

$$P_{210}(M \otimes \mathcal{B}_\alpha) = 0, \qquad P_{\overline{210}}\Big(M^{\mathsf{T}} \otimes \widetilde{\mathcal{B}}_\alpha\Big) = 0. \tag{G.28}$$

Since $\ker X$ is the three-dimensional eigenspace selected by the quadratic relation, the first incidence equation restricts the baryonic tensor to $\mathcal{B}_\alpha \in \Lambda^3 \ker X$, with the conjugate equation imposing the corresponding condition on $\widetilde{\mathcal{B}}_\alpha$. The $\mathbf{210}$ is the trace-free hook in $\Lambda^4\mathbf{7} \otimes \overline{\mathbf{7}}$. The enhanced-$\mathrm{SU}(2)$ incidence equations are

$$(\Sigma\mathbf{1}_2 - \Phi)\mathcal{B} = 0, \qquad (\Sigma\mathbf{1}_2 + \Phi)\widetilde{\mathcal{B}} = 0. \tag{G.29}$$

In branched components the first equation contains,

$$(\Sigma - H)B = IB_{-1}, \qquad (\Sigma + H)B_{-1} = \widetilde{I}B. \tag{G.30}$$

Degree eight

At degree eight the terms are,

$$\begin{aligned}\mathrm{PL}[t^8] = \; & -q^{-2}(\mu_3\mu_5 + \nu^2\mu_2\mu_6 + \mu_1) \\ & + q^{-1}\nu(\mu_6^3 + 3\mu_5\mu_6 + 4\mu_4 + \mu_1\mu_4\mu_6 + 3\mu_1\mu_3 + \mu_1^2\mu_2) \\ & - 2 - \nu^2 - (1+\nu^2)(\mu_3\mu_4 + \mu_2\mu_5) - (2+\nu^2)\mu_1\mu_6 \\ & + q\nu(\mu_5\mu_6^2 + 3\mu_4\mu_6 + 4\mu_3 + \mu_1\mu_3\mu_6 + 3\mu_1\mu_2 + \mu_1^3) \\ & - q^2(\mu_6 + \mu_2\mu_4 + \nu^2\mu_1\mu_5).\end{aligned} \tag{G.31}$$

The same-charge products obey,

$$P_{(\mathbf{140},\,\mathbf{3})}\Big(\mathrm{Sym}^2\,\mathcal{B}\Big) = 0, \qquad P_{(\overline{\mathbf{7}}\oplus\mathbf{588},\,\mathbf{1})}\Big(\mathrm{Sym}^2\,\mathcal{B}\Big) = 0, \tag{G.32}$$

together with their conjugates for $\widetilde{\mathcal{B}}$. The first is the polarised three-form Plücker relation, while the second removes the remaining flavour-antisymmetric $\mathrm{SU}(2)$-singlet channel. Together with the degree-six incidence equations, these identify the baryonic generators as decomposable three-forms supported on $\ker X$. For the mixed products define,

$$K_{\mathrm{s}} = B\widetilde{B} - B_{-1}\widetilde{B}_{+1}, \tag{G.33}$$

and,

$$(K_+, K_0, K_-) = \left(-B\widetilde{B}_{+1}, \frac{1}{2}\left(B\widetilde{B} + B_{-1}\widetilde{B}_{+1}\right), B_{-1}\widetilde{B}\right), \qquad (h_+, h_0, h_-) = (I, H, \widetilde{I}). \tag{G.34}$$

These are uncontracted flavour tensors transforming as the singlet and triplet of the enhanced $\mathrm{SU}(2)$. Since,

$$\mathbf{35} \otimes \overline{\mathbf{35}} = \mathbf{1} \oplus \mathbf{48} \oplus \mathbf{392} \oplus \mathbf{784}, \tag{G.35}$$

the mixed relations are compactly

$$P_R K_{\mathrm{s}} = -\frac{\Sigma}{4} P_R\,\mathrm{Comp}_3(Y), \qquad P_R K_a = -\frac{h_a}{8} P_R\,\mathrm{Comp}_3(Y), \tag{G.36}$$

for

$$R = \mathbf{1}, \mathbf{48}, \mathbf{392}, \mathbf{784}. \tag{G.37}$$

Here $\mathrm{Comp}_2(M)$ lies in the $\mathbf{1} \oplus \mathbf{48} \oplus \mathbf{392}$ flavour channels, while the $\mathbf{784}$ appears only at the level of the third compound. The form above follows polynomially from the quadratic relation $XY = 0$.

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